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Dynamical response to local perturbations of an active matter system with polar order

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Dynamical response to local perturbations of an active matter system with polar order

Master thesis. Sapienza – University of Rome

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Introduction

Due to the strongly interacting nature of their individuals, several animal groups exhibit nontrivial large scale properties analogous to the ones found in condensed matter systems: this suggests they can be described through the same statistico-mechanical approach. Biological systems are the archetype of Active Matter, where the effects of interaction combine with nonequilibrium aspects due to the self-propelled motion of the individuals.

Experiments have recently been performed on birds flocks and midges swarms which showed that the statistical laws they obey cannot be correctly reproduced by the main models used to describe collective motion in active matter [3]. In particular:

- perturbations propagate in flocks of birds following a linear dispersion law, in sharp contrast with the diffusive propagation predicted by standard models (such as Vicsek's);
- swarms of insects obey dynamical scaling laws whose dynamical critical exponent differs by the one found in standard models.

This lead to the introduction of the Inertial Spin Model (ISM), which seems to correctly account for such observations. Like other models, the ISM includes the alignment interaction among individuals (*follow your neighbours*); what's new is the presence, in the dynamical update rule for velocities, of inertial terms (which alone would give rise to a time reversible evolution) together with the usual overdamped terms. Numerical simulations have shown that ISM correctly reproduces the linear dispersion law in its ordered phase (*flock*, [11]), while a renormalization group analysis of its small diffusion limit gave a dynamical critical exponent close to the one we observe in the critical disordered phase (*swarm*, [10]).

It is important, at this point, to explore the predictions of the model in some of its biologically relevant aspects, such as the response of the system to an external perturbation: this is the main manifestation of its collective nature and arguably a crucial feature for its survival in the presence of predators. From a Physical point of view, on the other end, it is an interesting problem *per se* to understand how a local perturbation can affect a finite size system like the ones we encounter within the field of active matter.

The purpose of this thesis is an in-depth study of the response properties of ISM in its ordered, low temperature phase. Here the system exhibits spin waves, *i.e.* directional waves which propagate in a linear fashion. The response function can be found analytically, by means of a low temperature linear expansion of ISM, for a

system sitting on a lattice: indeed, it is possible in this case to explicitly compute the propagator of the theory. The real situation (*off lattice*) is far more challenging, and far less amenable to an analytical approach. Our goal is thus to study several perturbation-response protocols in numerical simulations of ISM, with the purpose of understanding how the mutual diffusion of the individuals can affect the propagation of information throughout the network.

In practice, we choose the velocity of the individuals as our order parameter and define the spin to be its canonical conjugated momentum, that is the local generator of velocity rotations; then we couple a time-dependent external field to one of the two variables and we calculate the dynamical response of the system. We expect it to depend in a non trivial way upon the ratio between the scale of variation of the perturbation and the typical time scales of the system: they are the characteristic frequency of the response at the spatial scale of the perturbation (as it would also happen on a fixed lattice) and the microscopic time scale of mutual diffusion of the individuals (which instead is due to the non-equilibrium nature of the system).

This work is organized as follows. In Chapter 1 we present the main model which has been adopted to describe collective motion in biological systems ever since the 90's, which is the Vicsek model [38], and we show how it can be improved to correctly account for the dynamical properties of such systems: this will lead us to the Inertial Spin Model. In Chapter 2 we specialize the ISM to the planar case and we perform a numerical study of its static properties on a regular lattice. In Chapter 3 we introduce some external, local perturbing fields into our description by coupling them to the canonical variables which we use to describe the system on-lattice; hence, we apply Linear Response Theory to derive expressions for the dynamical response of the system. In Chapter 4, we perform numerical simulations to test the results of Chapter 3, and we define perturbation protocols with the aim of controlling the global direction of motion of the simulated flock by tuning the parameters of the external fields. Finally, in Chapter 5 we apply those very same protocols to the full, off-lattice system, and we measure to which extent the dynamical response can still be compared to that of the on-lattice case.

Chapter 1

Models of flocking

1.1 Physical motivation

Active Matter Systems

We use the term *active matter* to speak of condensed systems whose constituents are self-propelled units (active particles), each capable of converting stored or ambient free energy into actual movement. What makes active matter different from other nonequilibrium systems in the usual sense, is that the energy input that drives the system out of equilibrium is local, at the level of each particle, and not at the system's boundaries (as it happens, for instance, in a turbulent liquid flow). This gives rise to many interesting properties, among which emergent structures with collective behavior qualitatively different from that of the individual constituents, non equilibrium order-disorder transitions, pattern formation on mesoscopic scales, and wave propagation.

Although a good deal of non-living systems have successfully been described by models of self-driven particles (granular matter, colloids, complex fluids, just to name a few [27]), active matter finds its natural application to the description of living systems, both at the microscopic (*e.g.* cells, bacteria) and macroscopic level (terrestrial, aquatic and aerials flocks of animals such as insects, birds and mammals). A key property shared by many such biological groups is indeed that of collective behaviour: this occurs when large numbers of units interact locally in space and time giving rise to macroscopic patterns.

Why a Statistical Physics approach?

The aim of Statistical Physics is the description of systems where the large number of units justifies the adoption of a probabilistic approach: one gives up from the start the hope to track individual particles, and focuses instead on a restricted set of collective properties which can thus be predicted and experimentally measured. This proved to be an effective way to extract information from an otherwise overwhelmingly complex system.

In Critical Phenomena, an even more dramatic simplification applies: systems whose fluctuations are correlated over long ranges display in fact *universality*. This means that their large scale properties do not depend on the details of the microscopic

interactions, but rather only upon the dimensions and symmetries of the system; this is also the reason why the behaviour of a wide variety of systems can be effectively described by a few very simple models. Take the case, for instance, of the good old Heisenberg model:

$$\mathcal{H} = - \sum_{i,j} J_{ij} \mathbf{v}_i \cdot \mathbf{v}_j . \quad (1.1)$$

When we adopt it to describe a magnet, we are not implying that a vector variable $\mathbf{v}_i$ should actually be attached to each of the real degrees of freedom of the system; we rather regard it as an effective model, where each *spin* variable represents the averaged value of many real degrees of freedom (this is the essence of the so called *coarse graining*). The avail of such a model lies in its capacity to capture all the relevant features that govern the behaviour of the real system, at least in the vicinity of a critical region. It was a major success of the Renormalization Group to explain why systems that are so different at the microscopic level can show similar macroscopic behaviours (*universality*), and to make predictions about the value of experimentally accessible observables [39].

In the same way, the hope behind a Statistical Physics approach to living systems is that the actual details of the interaction between the individuals become hardly relevant, provided that the correlations are sufficiently strong. As soon as we face the description of a biological system, though, we soon discover how utterly complicated a task this can be: indeed, we have to deal with systems that are complex, strongly interacting and out of equilibrium. On top of that, we know that the tools of Statistical Physics work pretty well on systems of $\mathcal{O}(10^{23})$ particles, and that scaling laws apply at length scales of more than hundreds of interparticle distances. But what happens when we deal with finite groups of a few thousands of individuals?

1.1.1 Experimental evidence

From a theoretical point of view, the question of whether and under what dynamical conditions a statistico-mechanical approach can be successfully applied to out of equilibrium systems is still largely open. However, some experimental observations seem to suggest this to be the right path.

A major outcome of experiments conducted on biological systems during the last decade was the discovery that such groups are always strongly correlated: their correlation length ξ is always much larger than the microscopic length scales of the system. In some cases, ξ has been found to scale with the system's size, which is the landmark (within the scope of finite size systems) of scale-free correlations [13]. What's more, static and dynamical scaling hypotheses (which make predictions about the shape of the correlation functions, both in space and time domains) were found to be verified in systems such as bird flocks and midges swarms, and this can hardly be regarded as a coincidence.

Another aspect of collective behavior - indeed, the one we are primarily interested in within this project - is the existence of some mechanism of *information propagation* throughout animal groups (like mammal herds, fish schools, bird flocks), which enables them to quickly coordinate and maintain their coherence in response to some external stimulus. For example, in the case of a starling flock performing a

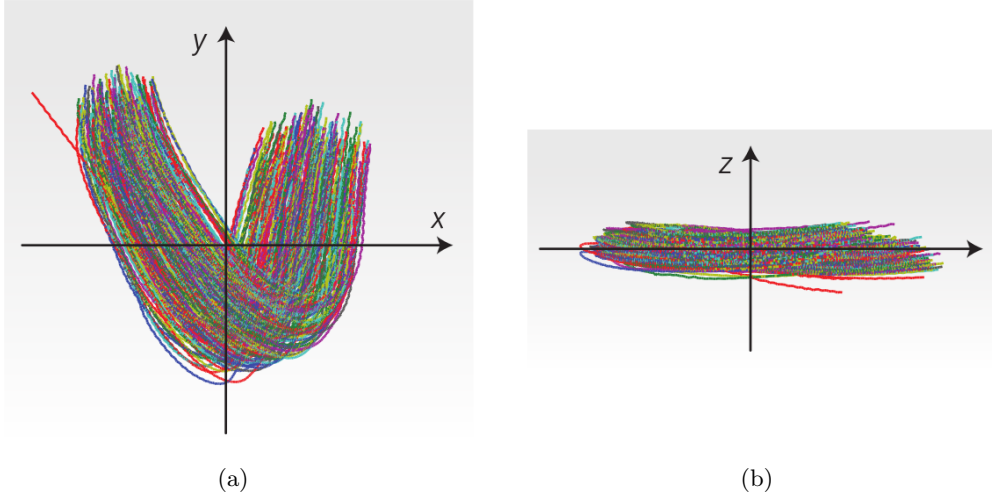


Figure 1.1. Reconstructed 3D trajectories of a starling flock performing a collective turn. Each trajectory lies approximately on a plane, which justifies the planar assumption we will make throughout the rest of this work. Images taken from [3].

collective turn [11], a single individual is always observed to start the turn and to be followed suit by the rest of the flock; the information which propagates in this case is a variation of the orientational angle (with respect to each single bird's moving reference frame) and this was measured to reach the other end of the group with negligible attenuation and following a linear dispersion law [3].

This is how it is achieved. The trajectory of each bird during a collective turn is reconstructed using a 3D camera setup (Figure 1.1) and its radial acceleration profile is plotted as a function of time: its peak can be used to locate the turning time of the corresponding bird (Figure 1.2). Measuring the temporal distance τ_{ij} between any two peaks, one obtains a set of mutual delays $\{\tau_{ij}\}$ which can be used to rank the birds in the flock from 1 (*i.e.* the starter of the turn) to N (the last bird to turn). The *ranking curve* is then obtained by plotting the rank r_i as a function of the absolute turning time t_i for each i -th bird (Figure 1.3(a)). Viewed as a function of t only (*i.e.* dropping the information about the specific bird i whence it came from), the function $r(t)$ tells us how many individuals were reached by the perturbation at time t . If we imagine the signal to propagate spherically in $3d$ space, we expect to find

$$r(t) = \rho \cdot \frac{4}{3}\pi x^3(t), \quad (1.2)$$

where ρ is the number density (the number of individuals per unit volume) and $x(t)$ is the distance travelled by the information at time t : this would give us an operative way to measure $x(t)$ from raw data. In practice, this is only true within the *bulk* of the system, because as soon as the border effects start to kick in, then the hypothesis of spherical propagation fails. However, we do observe a linear regime at least for short times $t < \frac{L}{c_s}$, where c_s is indeed the *sound speed* which we can retrieve from the slope of $x(t)$ (Figure 1.3(b)). In real starling flocks, c_s was measured to fall within the range 20-40 m/s in the center of mass reference frame (notice that the

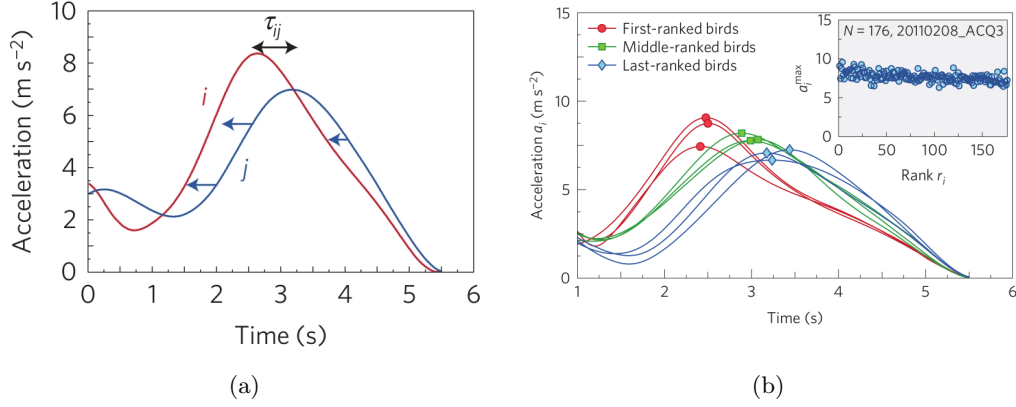


Figure 1.2. (a) The radial acceleration profile can be reconstructed from the 3D trajectory of each individual in the flock. The peak of the curve is taken as the turning time of the i -th bird, and the temporal displacement τ_{ij} represents the mutual turning delay between a given pair of birds. (b) Radial acceleration curves for many individuals in the flock during a collective turn. The peak intensity does not sensibly depend on the distance from the point where the turn was started, meaning a negligible attenuation of the signal. Images taken from [3].

speed of the individual birds, as measured from a fixed reference frame, is instead of the order of $v_0 \sim 7\text{-}12\text{ m/s}$.

It is worth noticing that the peak of the acceleration profile has pretty much the same intensity from the first to the last ranked bird in the flock (Figure 1.2(b)), meaning a minor attenuation of the propagating signal; indeed, a signal which got damped before it managed to reach the other end of the flock would result in its disruption, because only part of the flock would change its flight direction. Moreover, the mere fact that we can identify the starter of the turn is telling us that the signal has a well localized spatial origin.

Understanding what generates such changes in the behavior of a group and what is the mechanism behind their propagation from one individual to the others is at the core of the Physical motivation behind the study of biological systems. One wishes to find a way to model both the perturbation and the active matter system which is observed to respond to it, and then to thoroughly study these models both analytically and numerically.

1.1.2 Models of collective motion

In the effort to build a theory of active matter, we follow the usual procedure of Statistical Mechanics, where a coarse grained description of the large-scale, long-time behavior of the system is given in terms of a small number of continuum, macroscopic fields (such as the velocity field, $\mathbf{v}(\mathbf{x}, t)$, or the density field, $\rho(\mathbf{x}, t)$). In order to write down the evolution equations for such fields, two possible approaches may be adopted.

The first, more practical, is to include in the equations of motion all the possible terms allowed by symmetry. This leads to hydrodynamic laws similar in form to

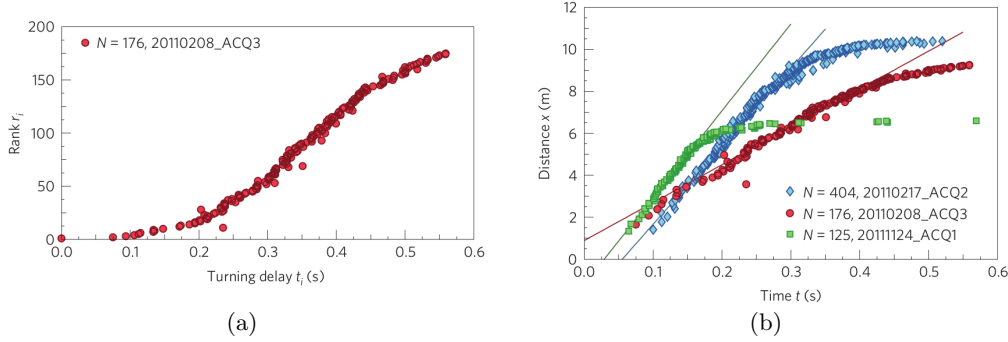


Figure 1.3. (a) Ranking curve. The rank r_i of each bird (*i.e.* its order in the turning sequence) is plotted as a function of its absolute turning time t_i . Under the assumption of spherical propagation of the signal, the quantity $x(t) \sim r(t)^{\frac{1}{3}}$ represents the distance travelled by the information at time t . (b) Dispersion curves $x(t)$ for different flocks. Their linear region can be fitted to extract the propagation speed c_s . Images taken from [3].

Navier-Stokes equations, but where new terms appear (which are usually ruled out when describing thermal equilibrium) so as to account for the non equilibrium nature of the system. This is the essence of Toner and Tu's theory [35], which in a highly simplified version reads

$$\begin{cases} \mathcal{D}_t \mathbf{v} = J \nabla^2 \mathbf{v} - \nabla P - \frac{\partial V}{\partial \mathbf{v}} \\ \partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \end{cases} \quad (1.3)$$

where $\mathcal{D}_t$ is the material derivative, and $V(\mathbf{v})$ is a confining potential in the form

$$V(\mathbf{v}) = \int d\mathbf{r} \left[-\frac{\alpha}{2} \|\mathbf{v}\|^2 + \frac{\beta}{2} (\mathbf{v} \cdot \mathbf{v})^2 \right]. \quad (1.4)$$

The idea of extending continuous field theory to the description of active matter systems is groundbreaking; here we only mention its greatest achievement, which is the proof that long range order is still possible in 2 dimensions. However, we will not indulge on this hydrodynamic theory primarily because some of the predictions it makes (the very existence of sound waves, to name one) are only valid, indeed, in the hydrodynamic limit (*i.e.* for wavenumbers $k \rightarrow 0$). When we deal with finite size systems, as we will systematically do throughout the rest of this work, this long wavelength limit is often practically unattainable. Moreover, as we know well from Fluid dynamics, velocity waves in the longitudinal direction are coupled to density fluctuations, and the speed at which they propagate is a function of the density ρ : it turns out that none of these has ever been experimentally observed in real active matter systems.

A second possible approach prescribes instead to formulate a microscopic model (*i.e.* to write down a microscopic Hamiltonian), and then coarse grain it to lose track of the short scale details; it is analogous to stepping from an Eulerian to a Lagrangian description of the system. This is indeed the type of approach we are going to adopt in the following pages. We will start by presenting the early Vicsek model and its limits of applicability; then we will introduce the Inertial Spin Model.

1.2 Vicsek model

When we deal with active matter systems where some kind of alignment interaction is present, a good starting point is represented by the *Vicsek model*. A system of N point-like particles, each labeled by an index i , move freely in a space of dimension d with constant speed $v_0 = |\mathbf{v}_i|$; the *direction* of motion, on the other end, does evolve depending on the average velocity of the particles surrounding the i -th. The way this neighborhood can be defined is not unique, but once this is done we can encode our choice in a connectivity matrix n_{ij} : its elements are zero everywhere but for the interacting pairs, for which $n_{ij} = 1$. The model is hence defined by the dynamical update rule for the velocities,

$$\mathbf{v}_i(t+1) = v_0(\mathcal{R}_\beta \circ \Theta) \left[\mathbf{v}_i(t) + \sum_j n_{ij} \mathbf{v}_j(t) \right], \quad (1.5)$$

and for the positions,

$$\mathbf{r}_i(t+1) = \mathbf{r}_i(t) + \mathbf{v}_i(t+1). \quad (1.6)$$

In (1.5), we called Θ the normalization operator

$$\Theta \mathbf{x} = \frac{\mathbf{x}}{|\mathbf{x}|} = \hat{\mathbf{x}}, \quad (1.7)$$

while $\mathcal{R}_\beta$ performs a random rotation uniformly distributed around its argument vector: if $d = 3$, this means $\mathcal{R}_\beta \mathbf{v}$ lies within the solid angle subtended by a spherical cap of amplitude $4\pi\beta$ and centered around $\mathbf{v}$.

Two main kinds of interactions are generally considered when constructing the connectivity matrix n_{ij} :

- (i) *Metric*: each particle interacts with those who lie within an interaction radius r_c . Calling $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ the mutual distance between a given pair of particles, what we are saying is that interaction is possible if $r_{ij} < r_c$. This has been found to be the case for living systems like swarms of insects [2].
- (ii) *Topological*: each particle interacts with its n_c nearest neighbours, ranked according to r_{ij} . This is what happens in systems like flocks of birds ([4],[13]), where moreover the number n_c of average nearest neighbours has been found to be independent of the flock's density.

In both cases, the connectivity matrix is easily seen to depend on time through the positions of the particles in the system, $n_{ij} = n_{ij}(\{\mathbf{r}_i(t)\})$: it is their relative motion which gives rise to the non equilibrium features which distinguish the Vicsek model from a simple spin dynamics model. The most spectacular consequence of the introduction of a coupling between local order and local density fluctuations is the fact that this model displays *spontaneous symmetry breaking* even in $d = 2$ (see Section 1.4), which would be otherwise forbidden by the Mermin-Wagner theorem [28]. Indeed, the theorem states that a continuous symmetry cannot be spontaneously broken at finite temperature and in an isotropic system in $d \leq 2$. However, this is no longer true for an *off-lattice* model like Vicsek's. In fact, the VM is known to

display a transition from an ordered phase (*flock*), where the individual velocity vectors align and add up to give a net non-zero velocity of the whole group

$$\mathbf{V} \equiv \frac{1}{N} \sum_{i=1}^N \mathbf{v}_i, \quad (1.8)$$

to a disordered phase (*swarm*), where the order parameter (the total velocity) is null.

1.2.1 Continuous time version of the VM

The evolution equation (1.5) defined in the original Vicsek model is difficult to handle, mainly because of its discrete time nature. Here we introduce a slightly different version of the model which is continuous in time, and is thus easier to treat mathematically:

$$\eta \frac{d\mathbf{v}_i}{dt} = J \sum_j n_{ij} \mathbf{v}_j(t) + \lambda_i \mathbf{v}_i + \boldsymbol{\xi}_i, \quad (1.9)$$

where the positions evolve according to

$$\frac{d\mathbf{r}_i}{dt} = \mathbf{v}_i. \quad (1.10)$$

A few changes were made with respect to (1.5) and (1.6), on top of taking their limit for $\Delta t \rightarrow 0$:

- (i) The parameter J was introduced in order to tune the strength of the *social force* at the right hand side of (1.9), which is responsible for the alignment of the individual velocities. While in (1.5) the term with $j = i$ in the social force has the same weight as all the others, it is useful to write instead

$$\begin{aligned} & \mathbf{v}_i + \sum_{j \neq i} n_{ij} \mathbf{v}_j(t) \\ & \rightarrow \mathbf{v}_i + J dt \sum_{j \neq i} n_{ij} \mathbf{v}_j(t), \end{aligned} \quad (1.11)$$

so that the self contribution remains finite in the limit $dt \rightarrow 0$.

- (ii) The normalization operator Θ has disappeared and a Lagrange multiplier λ_i was introduced in its place: this can be used to enforce the constraint on the speed of the particles, $|\mathbf{v}_i| = v_0$.
- (iii) The operator $\mathcal{R}_\beta$ is replaced by a simple Gaussian noise $\boldsymbol{\xi}_i$, with zero average and variance given by

$$\langle \boldsymbol{\xi}_i(t) \cdot \boldsymbol{\xi}_j(t') \rangle = 2d\eta T \delta_{ij} \delta(t - t'). \quad (1.12)$$

This is analogous to Einstein's relation within the scope of thermal equilibrium systems; here the *temperature* T is a measure of the intensity of the noise, while η will be referred to as the *viscosity*.

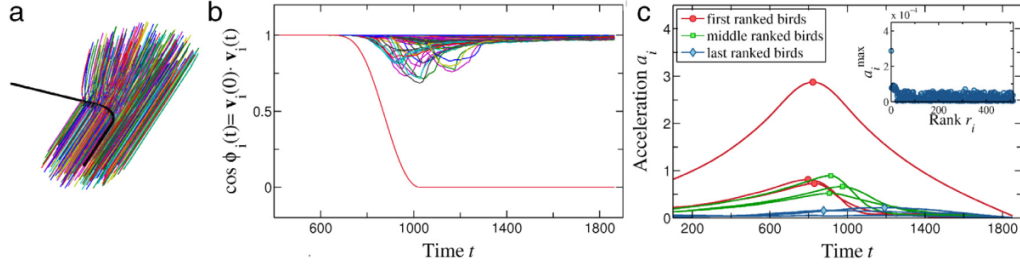


Figure 1.4. Numerical simulations of the Vicsek model. A single individual is manually forced to change its direction of motion, but it is not followed by the rest of the flock. An inspection of the radial acceleration profiles (c) shows that the turning signal is heavily attenuated; this should be compared with the real data in Figure 1.2(b), where a similar attenuation is not observed. Images taken from [13].

By defining the pseudo-Hamiltonian

$$\mathcal{H} = -\frac{J}{2} \sum_{ij} n_{ij} \mathbf{v}_i \cdot \mathbf{v}_j, \quad (1.13)$$

we notice that the update rule (1.9) may be rewritten as

$$\eta \frac{d\mathbf{v}_i}{dt} = -\frac{\partial \mathcal{H}}{\partial \mathbf{v}_i} + \lambda_i \mathbf{v}_i + \boldsymbol{\xi}_i. \quad (1.14)$$

This makes clear the similarity with a Langevin spin dynamics in the presence of ferromagnetic interactions: indeed, the Hamiltonian (1.13) is the same as that of a Heisenberg model built on a random Euclidean lattice, that is the one realized by the particles' positions (where however the connectivity matrix n_{ij} now depends on time). It is not superfluous to remark that (1.13) is only being used as a *pseudo*-Hamiltonian, in the sense that the deterministic part of (1.14) is *not* a canonical equation of motion.

Finally, we notice that we can get rid of the Lagrange multiplier λ_i in (1.9) by simply multiplying both sides of the equation by $\mathbf{v}_i$ and imposing the fixed speed condition

$$\frac{d|\mathbf{v}_i|^2}{dt} = 0; \quad (1.15)$$

this way we obtain

$$\eta \frac{d\mathbf{v}_i}{dt} = \left[J \sum_j n_{ij} \mathbf{v}_j(t) \right]^\perp + \boldsymbol{\xi}_i^\perp, \quad (1.16)$$

where the apex $\perp$ applied to a vector stands for its component perpendicular to $\mathbf{v}_i$, that is

$$\mathbf{x}^\perp = \mathbf{x} - \frac{(\mathbf{x} \cdot \mathbf{v}_i)}{v_0^2} \mathbf{v}_i. \quad (1.17)$$

1.2.2 What is missing in the VM?

We have mentioned that the VM correctly predicts a phase transition at finite T_c from a disordered to an ordered phase, which can be observed by gradually decreasing

the noise intensity T ; in the low temperature phase, long range order sets in and scale free correlations can be observed. However, the Vicsek model fails to describe the collective turns which are observed in real flocks, as it can clearly be seen in the numerical simulations presented in Figure 1.4.

From a theoretical point of view, it is straightforward to find in the *dispersion relation* of the VM the fundamental reason that makes it unable to propagate information, hence to sustain a collective turn. To lighten the notation, let's consider the case where all the velocity vectors $\mathbf{v}_i$ lie on a plane (which is in fact what happens when a flock of birds performs a real collective turn - see [3] and Figure 1.1): then each of them can be specified by its polar angle φ_i via

$$\mathbf{v}_i(t) \equiv v_0 e^{i\varphi_i(t)} . \quad (1.18)$$

Now let's take $v_0 = 1$ for simplicity and consider a system in its highly ordered regime, where the individual velocities deviate little from the flock's one, so that practically speaking $\varphi_i \ll 1$. We can then rewrite the pseudo-Hamiltonian (1.13) as

$$\begin{aligned} \mathcal{H} &\simeq \frac{J}{2} \sum_{ij} n_{ij} (\varphi_i - \varphi_j)^2 \\ &\rightarrow \frac{1}{2} a^2 J n_c \int \frac{d^3x}{a^3} |\nabla \varphi(\mathbf{x}, t)|^2 , \end{aligned} \quad (1.19)$$

where in the last passage we stepped to a field description for the angle φ , useful when the average nearest-neighbour distance a is small enough, and we called n_c the average number of nearest neighbours.

Since the constraint on the speed is automatically enforced, the evolution equation (1.14) of the VM becomes

$$\eta \frac{d\mathbf{v}_i}{dt} = - \frac{\partial \mathcal{H}}{\partial \mathbf{v}_i} + \boldsymbol{\xi}_i , \quad (1.20)$$

whose deterministic part reads, in terms of the field φ ,

$$\eta \frac{\partial \varphi}{\partial t} = - \frac{\delta \mathcal{H}}{\delta \varphi} = a^2 J n_c \nabla^2 \varphi . \quad (1.21)$$

But this is nothing else than a diffusion equation for the angle φ : it is no wonder that information cannot propagate. The dispersion relation $\omega \sim ik^2$ is purely imaginary, which means that the amplitude of any solution decays in time (here, $\varphi \sim \frac{1}{\sqrt{t}}$) and a speed of propagation c_s cannot even be defined. This is in striking contrast with experimental observations.

1.3 Inertial Spin Model (ISM)

The Inertial Spin Model is a new model of flocking which has the advantage, compared to the Vicsek model, of being able to represent the *dynamics* of flocking and not only the static properties of the system [3][11][13]. Here we will start from the VM and see how it can be improved to take *behavioural inertia* into account.

1.3.1 From VM to ISM

Consider again the evolution equation for the polar angle φ in the Vicsek model, which we rewrite here as

$$\eta\dot{\varphi} = a^2 J n_c \nabla^2 \varphi \equiv \mathcal{F} \quad (1.22)$$

in order to stress that the social force $\mathcal{F}$ is acting on the first derivative of φ with respect to time. How is that? Newton's equation would require the force $\mathcal{F}$ to act on the *acceleration*,

$$\chi\ddot{\varphi} = \mathcal{F} , \quad (1.23)$$

where we called χ some kind of generalized *inertia*.

In fact, what we really mean whenever we hypothesize a first order dynamics as we did in (1.22) is the following: we start from a second order dynamics like that of equation (1.23), which describes the motion of a particle subject to a deterministic force $\mathcal{F}$, and then we add friction and a noise term *a posteriori*:

$$\chi\ddot{\varphi} = \mathcal{F} - \eta\dot{\varphi} + \xi . \quad (1.24)$$

Friction originates from all the forces we are not including into $\mathcal{F}$. Take the case, for instance, of a colloid immersed in a thermal bath, whose interactions with the many smaller particles which compose the bath are both uninteresting and utterly complicated to track; however, physical observation tells us that their overall effect is that of damping any initial impulse of the colloid and bringing it to perform *Brownian motion* around its equilibrium position (see, for instance, the seminal work of P. Langevin [26]). The noise term ξ prevents the colloid from simply stopping after its initial momentum has been consumed by the friction term.

By inspecting (1.24), we notice a natural time scale emerges which is given by the ratio of viscosity over inertia:

$$\tau = \frac{\eta}{\chi} . \quad (1.25)$$

We can assume that any initial momentum will be dissipated in a time $t \sim \tau$, after which the inertial term $\chi\ddot{\varphi}$ becomes in fact negligible. For $t \gg \tau$, then, equation (1.23) actually reduces to a first order dynamics:

$$\eta\dot{\varphi} = \mathcal{F} + \xi . \quad (1.26)$$

This is the *overdamped* regime: all the information about the initial transient $t < \tau$ is lost. But it might well be that the initial regime we are discarding is in fact the most relevant, as far as fast information propagation is concerned!

There is another good reason to believe that an inertial term may be missing in the Vicsek model. The Hamiltonian (1.19) clearly presents a global gauge symmetry, namely a rotation of all the velocities $\mathbf{v}_i$,

$$\varphi_i \rightarrow \varphi_i + \delta\varphi , \quad \forall i , \quad (1.27)$$

which according to Noether's theorem would give rise to a conserved quantity under canonical temporal evolution. In general, an excitation of a conserved field cannot

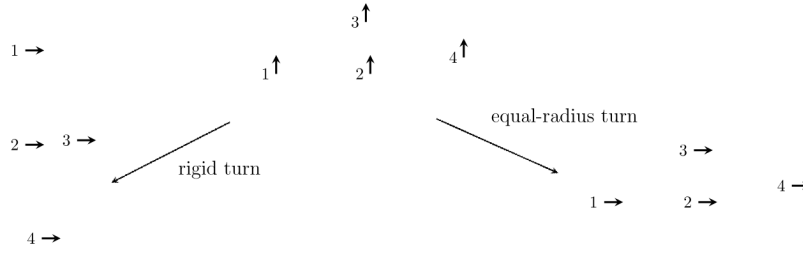


Figure 1.5. Comparison of the effects of a rigid-body turn *versus* an equal radius turn on a small flock. Notice in the latter case the individual trajectories can cross, which is forbidden in a rigid-body scenario (but which is indeed the case in real flocks). Image taken from [13].

just be relaxed locally, but it must be transported away: this is why the presence of an underlying symmetry can deeply affect the dispersion relation. However, no quantity seems to be conserved in the evolution equation (1.21), which in fact is far from being canonical.

Reinstating the inertia into the model

How can we go beyond the overdamped approximation and study the initial, inertial regime $t < \tau$? Indeed, there is no trace of an inertial term in the pseudo-Hamiltonian (1.19) of the planar Vicsek model.

First, notice the kind of inertia we are looking for is not a *mass*: the relevant quantity involved in our dynamics is the angle φ_i *measured in the reference frame of the individual sitting in position $\mathbf{r}_i$* . A change in φ_i means an on-spot rotation of the velocity vector $\mathbf{v}_i$, and a global change $\varphi_i \rightarrow \varphi_i + \delta\varphi \forall i$ encodes a collective turn. Notice the latter is an equal radius turn, which is quite different from the rigid body rotation we would obtain if the angles were measured with respect to a common, fixed pole (Figure 1.5).

The inertia χ should be a measure of the resistance opposed by each individual to changing its flight direction, hence it is really a *moment of inertia*. Since the Hamiltonian (1.19) already contains an angle φ_i , it is quite natural to introduce its conjugated canonical momentum s_i : we call it *spin*¹. The new Hamiltonian will contain the action-angle couple (φ, s) and will read, in analogy with a planar rotator,

$$\mathcal{H} = \int \frac{d^3x}{a^3} \left\{ \frac{1}{2} a^2 J n_c |\nabla \varphi(\mathbf{x}, t)|^2 + \frac{s^2}{2\chi} \right\}. \quad (1.28)$$

From (1.28), the canonical equations of motion can be promptly derived:

$$\begin{cases} \dot{\varphi} = \frac{\delta \mathcal{H}}{\delta s} = \frac{s}{\chi} \\ \dot{s} = -\frac{\delta \mathcal{H}}{\delta \varphi} = a^2 J n_c \nabla^2 \varphi, \end{cases} \quad (1.29)$$

¹Indeed, we will soon find out (by analogy with circular motion) that this is really a spin in strict canonical sense.

where δ denotes a functional variation. We observe that now the *social force* $a^2 J n_c \nabla^2 \varphi$ is no longer acting on $\dot{\varphi}$, but it can only modify the angle φ through the spin s . This fact has deep implications, among which the emergence of a conservation law which was absent in the Vicsek model: indeed, by defining the *current*

$$\mathbf{j}(\mathbf{x}, t) \equiv -a^2 J n_c \nabla \varphi(\mathbf{x}, t) , \quad (1.30)$$

it becomes evident that the second equation in (1.29) takes the form of a continuity equation

$$\frac{ds}{dt} + \nabla \cdot \mathbf{j} = 0 . \quad (1.31)$$

Integrating (1.31) over the whole space $\mathcal{V}$ and using Green's theorem, one finds that the total spin $\mathbf{S}$ of a finite system must be conserved:

$$\frac{dS}{dt} = \frac{d}{dt} \int_{\mathcal{V}} d\mathbf{x} s(\mathbf{x}, t) \equiv \int_{\partial\mathcal{V}} \mathbf{j} \cdot \hat{\mathbf{n}} dA = 0 . \quad (1.32)$$

This, in turn, prevents any local modification of the spin field from being simply dissipated locally: on the contrary, it must be transported away. It is then no surprise that, by deriving again the first equation in (1.29) with respect to time, we find a wave equation,

$$\ddot{\varphi} = \frac{a^2 J n_c}{\chi} \nabla^2 \varphi , \quad (1.33)$$

where the sound speed c_s is given by²

$$c_s = \sqrt{\frac{J n_c}{\chi}} a . \quad (1.34)$$

There remains to include the effects of damping into our equations of motion. Indeed, strict spin conservation would imply for a single, non-interacting bird to maintain its turning state forever, which is bizarre. A way to do so is to introduce a friction-noise pair into the dynamical equation for the spin s [11]. This leads to

$$\begin{cases} \dot{\varphi} = \frac{\delta \mathcal{H}}{\delta s} = \frac{s}{\chi} \\ \dot{s} = -\frac{\delta \mathcal{H}}{\delta \varphi} - \eta \frac{\delta \mathcal{H}}{\delta s} + \xi = J n_c a^2 \nabla^2 \varphi - \frac{\eta}{\chi} s + \xi , \end{cases} \quad (1.35)$$

where the stochastic noise ξ satisfies the same Einstein's relation (1.12) as in the Vicsek model. Even though the spin will no longer be conserved, it is intuitive to realize that as long as the viscosity η is small enough, then a propagating phenomenon will still take place within the system: we will calculate and widely use the new dispersion relation of ISM throughout the rest of this work (see, for instance, Section 3.2).

²Interestingly, this is really the *second sound speed*, that is the speed at which a change in the angle φ propagates. Indeed, it is also possible to define a *first sound* in analogy with ordinary fluids, where the velocity fluctuations are coupled to those of the density ρ . See for instance Toner and Tu [35].

Generalization to the full vectorial case

The canonical equations of motion (1.29) are field equations for $\varphi(\mathbf{x}, t)$ and $s(\mathbf{x}, t)$, useful when the system is big enough that its discrete nature can be disregarded; these two scalar fields are sufficient for completely characterizing the planar model. Moreover, we derived them under the assumption of high order (low temperature): this allowed us to turn the Heisenberg-like potential which appears in (1.19) into a gradient ($\nabla\varphi$), thanks to the small angle condition $\varphi_i \ll 1 \ \forall i$.

Clearly, none of these assumptions is essential to the derivation we carried out above. First, we can write a microscopic Hamiltonian which is the discrete version of (1.28) as

$$\mathcal{H} = V(\{\varphi_i\}) + \sum_i \frac{s_i^2}{2\chi}, \quad (1.36)$$

where we take as $V(\{\varphi_i\})$ the full Heisenberg-like potential

$$V(\{\varphi_i\}) = -\frac{J}{2v_0^2} \sum_{ij} n_{ij} \mathbf{v}_i \cdot \mathbf{v}_j, \quad (1.37)$$

prior to the introduction of the small angle approximation. The Hamilton equations for each φ_i and s_i now read

$$\begin{cases} \frac{d\varphi_i}{dt} = \frac{\partial \mathcal{H}}{\partial s_i} = \frac{s_i}{\chi} \\ \frac{ds_i}{dt} = -\frac{\partial \mathcal{H}}{\partial \varphi_i} = -\frac{\partial V}{\partial \varphi_i}. \end{cases} \quad (1.38)$$

Again, the potential (1.37) is rotationally invariant: calling $\delta\varphi$ a constant vector,

$$V(\varphi_i + \delta\varphi) \simeq V(\varphi_i) + \underbrace{\nabla_{\varphi} V \cdot \delta\varphi}_{=0}, \quad (1.39)$$

which directly leads to the conservation of the total spin $S = \sum_i s_i$,

$$\frac{dS}{dt} = -\sum_i \frac{\partial V}{\partial \varphi_i} = 0. \quad (1.40)$$

Equation (1.40) is nothing else than a discrete version of the continuity equation.

Going beyond the planar model is now straightforward: the Hamiltonian (1.36) admits a direct generalization to 3-dimensional vectors $\mathbf{v}_i$ and $\mathbf{s}_i$ by simply identifying $s_i \equiv \|\mathbf{s}_i\|$. Instead of going through the full derivation of the equations of motion, here we only observe that (1.36) is exactly analogous to the Hamiltonian of a 3-dimensional rotator, after we have made the identifications

$$\hat{\mathbf{v}}_i \leftrightarrow \mathbf{r} \quad (1.41)$$

$$\mathbf{s}_i \leftrightarrow \mathbf{L} \quad (1.42)$$

$$\mathbf{F}_i \equiv -\frac{\partial \mathcal{H}}{\partial \mathbf{v}_i} = -\nabla_{(\mathbf{v}_i)} V, \quad (1.43)$$

where $\mathbf{r}$ is the position vector, $\mathbf{L}$ is the angular momentum and $\mathbf{F}_i$ is a generalized force; as usual, we called

$$\hat{\mathbf{v}}_i = \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|} = \frac{\mathbf{v}_i}{v_0}. \quad (1.44)$$

In analogy with the cardinal equations of Classical Mechanics

$$\begin{cases} \frac{d\mathbf{r}}{dt} = \boldsymbol{\omega} \times \mathbf{r} = \frac{1}{I} \mathbf{L} \times \mathbf{r} \\ \frac{d\mathbf{L}}{dt} = \mathbf{r} \times \mathbf{F}, \end{cases} \quad (1.45)$$

where $\boldsymbol{\omega}$ is the angular velocity and I is the moment of inertia, we can write the deterministic equations

$$\begin{cases} \frac{d\mathbf{v}_i}{dt} = \frac{1}{\chi} \mathbf{s}_i \times \mathbf{v}_i \\ \frac{d\mathbf{s}_i}{dt} = \hat{\mathbf{v}}_i \times \mathbf{F}_i \\ \frac{d\mathbf{r}_i}{dt} = \mathbf{v}_i. \end{cases} \quad (1.46)$$

We conclude by introducing noise and friction into the equation of motion for $\mathbf{s}_i$, alongside with the force term:

$$\frac{d\mathbf{s}_i}{dt} = \hat{\mathbf{v}}_i \times \left[\mathbf{F}_i - \eta \frac{d\hat{\mathbf{v}}_i}{dt} + \boldsymbol{\xi}_i \right]. \quad (1.47)$$

This defines the *Inertial Spin Model* (ISM), whose equations of motion are

$$\begin{cases} \frac{d\mathbf{v}_i}{dt} = \frac{1}{\chi} \mathbf{s}_i \times \mathbf{v}_i \\ \frac{d\mathbf{s}_i}{dt} = \hat{\mathbf{v}}_i \times \left[J \sum_j n_{ij} \hat{\mathbf{v}}_j - \eta \frac{d\hat{\mathbf{v}}_i}{dt} + \boldsymbol{\xi}_i \right] \\ \frac{d\mathbf{r}_i}{dt} = \mathbf{v}_i, \end{cases} \quad (1.48)$$

where as usual

$$\langle \boldsymbol{\xi}_i \rangle = 0 \quad (1.49)$$

$$\langle \boldsymbol{\xi}_i(t) \cdot \boldsymbol{\xi}_j(t') \rangle = 2d\eta T \delta_{ij} \delta(t - t'). \quad (1.50)$$

Since the quantity $\mathbf{s}_i \cdot \mathbf{v}_i$ is preserved by the dynamics defined in (1.48), without loss of generality we can set from the start $\mathbf{s}_i \cdot \mathbf{v}_i = 0$.

1.3.2 Steady state distribution

First of all, does it make any sense to search for the steady state distribution of a system that is inherently out of equilibrium? To see this, let's focus on the variables $\mathbf{v}_i$ and $\mathbf{s}_i$ only. The equations of motion (1.48) tell us that the positions $\mathbf{r}_i$ enter the description of the velocity and spin variables only through the connectivity

matrix $n_{ij}(\{\mathbf{r}_i(t)\})$: as the individuals move and change their relative positions, the interacting pairs specified by n_{ij} have to change as well.

Luckily for us, it has been shown in [29] using inference techniques on real data that local bird alignment happens on a much faster timescale than neighbour rearrangement. If we look at the system at timescales much smaller than the typical swapping time between the individuals, then it makes sense to imagine $\mathbf{v}_i$ and $\mathbf{s}_i$ were sitting on the vertices of a *fixed lattice*; indeed, one could write a Fokker-Planck equation corresponding to the Langevin equation (1.48) and show that (at least in the fixed symmetric network case) it admits a unique, stationary solution in the form of a Boltzmann distribution:

$$\begin{aligned} \mathcal{P}(\{\mathbf{v}_i\}, \{\mathbf{s}_i\}) &\sim e^{-\beta \mathcal{H}(\{\mathbf{v}_i\}, \{\mathbf{s}_i\})} \\ &= e^{-\beta \left(-\frac{J}{2v_0^2} \sum_{ij} n_{ij} \mathbf{v}_i \cdot \mathbf{v}_j + \sum_i \frac{s_i^2}{2\chi} \right)} . \end{aligned} \quad (1.51)$$

What's even more interesting is that this probability distribution factorizes between velocity and spin variables,

$$\mathcal{P}(\{\mathbf{v}_i\}, \{\mathbf{s}_i\}) \sim e^{-\beta \mathcal{H}(\{\mathbf{v}_i\}, \{\mathbf{s}_i\})} = e^{-\beta [\mathcal{H}_v(\{\mathbf{v}_i\}) + \mathcal{H}_s(\{\mathbf{s}_i\})]} . \quad (1.52)$$

In particular, the inertia χ doesn't enter at all the velocity distribution, so that for all practical purposes *the static properties of the ISM on a fixed lattice are the same as those of the Vicsek model* (which, in turn, are the same as those of the Heisenberg model): these include the ordering properties and the velocity correlation functions.

In the following pages, we will briefly summarize the role of the symmetry in this fixed network approximation; however, it is worth mentioning that the same conclusions relatively hold in the off-lattice case as well (see Toner and Tu, [35]).

1.4 Quick recap of the Landau-Ginzburg model

Both the Vicsek model and the Inertial Spin Model present a clear rotational symmetry in the velocities $\mathbf{v}_i$ of the individuals which compose the system. Here we explore the role of this symmetry and we study one of its important consequences, namely the emergence of *Goldstone bosons*, using the language of Field Theory.

We have seen with equation (1.51) that, as long as the positions of the individuals in the flock are kept fixed in time, then the velocity variables $\mathbf{v}_i$ are distributed according to an Heisenberg-like Hamiltonian in the form of equation (1.13). To make our discussion more general, we wish to step to a continuous description of such a model, with the desirable side effect of making calculations far easier. A way to do so is to define a *field* $\varphi(\mathbf{x})$ by averaging over many microscopic degrees of freedom. Let $\delta_{\mathbf{x}}$ denote a small volume centered around position $\mathbf{x}$ in space; $\delta_{\mathbf{x}}$ should be chosen large enough so that many microscopic $\mathbf{v}_i$'s are included within the volume, hence making $\varphi(\mathbf{x})$ a sufficiently smooth function of $\mathbf{x}$, but also much smaller than the correlation length ξ and the size L of the system. Then we define

$$\varphi(\mathbf{x}) = \frac{1}{\delta_{\mathbf{x}}} \sum_{i \in \delta_{\mathbf{x}}} \mathbf{v}_i , \quad (1.53)$$

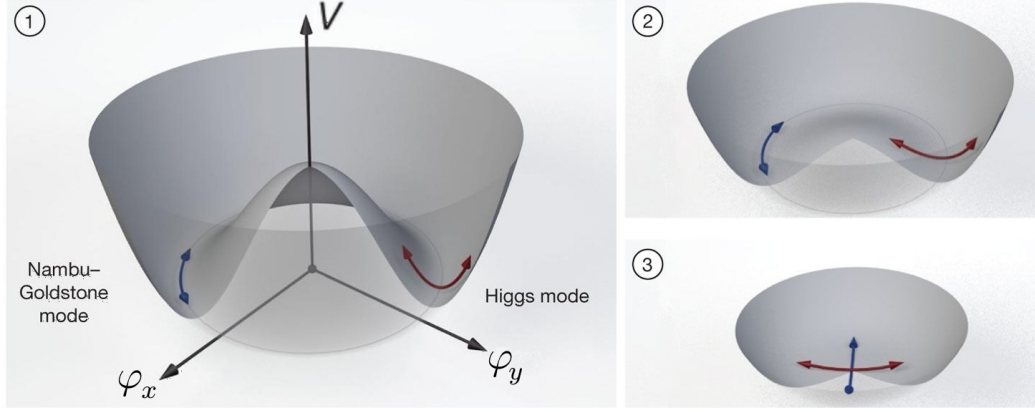


Figure 1.6. Pictorial scheme of the potential $V(\varphi)$ to which φ (here represented as a 2-components field) is subject in the Landau-Ginzburg Hamiltonian (1.56). For $\mu^2 > 0$ it has the form of a simple quadratic potential (3), while it takes that of a *Mexican hat* when μ^2 becomes negative (1 and 2). In the latter case, *soft* fluctuations develop along the bottom of the broken symmetry manifold: they correspond to the so called Nambu-Goldstone mode.

which reads, component by component,

$$\varphi^\alpha(\mathbf{x}) = \frac{1}{\delta_{\mathbf{x}}} \sum_{i \in \delta_{\mathbf{x}}} v_i^\alpha. \quad (1.54)$$

Next, let's introduce the Landau-Ginzburg Hamiltonian [19], an effective model which was devised to represent the most important properties of a spin system (such as its second order phase transition): in the vectorial case, it reads

$$\mathcal{H}_{LG} = \int d^d x \left\{ \frac{1}{2} \partial_\nu \varphi^\alpha \partial^\nu \varphi_\alpha + \frac{1}{2} \mu^2 \varphi_\alpha \varphi^\alpha + \frac{\lambda}{4!} (\varphi_\alpha \varphi^\alpha)^2 \right\}, \quad (1.55)$$

where repeated indices are summed over. The index $\nu = 1, \dots, d$ refers to different directions in physical space, while $\alpha = 1, \dots, D$ runs within the internal space of the components of φ . While the first term in (1.55) is a kinetic term, the second and third can be viewed as an effective potential for the field φ , so that $\mathcal{H}_{LG}$ may be rewritten as

$$\mathcal{H}_{LG} = \int d^d x \left\{ \frac{1}{2} \|\nabla \varphi\|^2 + V(\varphi) \right\}, \quad (1.56)$$

where

$$V(\varphi) \equiv \frac{1}{2} \mu^2 \|\varphi\|^2 + \frac{\lambda}{4!} \|\varphi\|^4. \quad (1.57)$$

The shape of $V(\varphi)$ is represented in Figure 1.6: as we vary the parameter μ^2 , which is called the *mass* of the theory, we step from a paraboloid ($\mu^2 > 0$) to the typical *Mexican hat* potential ($\mu^2 < 0$). Finally, the presence of the coupling term in λ makes Landau-Ginzburg a self-interacting field theory.

1.4.1 Spontaneous Symmetry Breaking

In most of this work we will take $D = 2$, so that φ is really a 2-components vector lying on a plane. Calling $\mathcal{R}$ the operator which enforces a rotation in such plane, we notice that the Landau-Ginzburg model enjoys the obvious symmetry

$$\mathcal{H}(\mathcal{R}\varphi) = \mathcal{H}(\varphi) . \quad (1.58)$$

Such a symmetry is said to be *continuous*, as it is described by a continuous parameter: calling θ the polar angle as in Figure 1.7, the operator $\mathcal{R}$ is in fact represented in matrix form by

$$\mathcal{R}(\theta) \leftrightarrow \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} . \quad (1.59)$$

Now let φ_0 denote the ground state of $\mathcal{H}_{LG}$. Figure 1.6 shows that, when $\mu^2 < 0$, there is in fact a whole manifold of points which minimize the energy: it is enough for them to satisfy

$$\begin{aligned} \nabla V(\varphi) \Big|_{\varphi=\varphi_0} &= 0 \\ \rightarrow \|\varphi_0\|^2 &= -\frac{6\mu^2}{\lambda} , \end{aligned} \quad (1.60)$$

and they are all connected to each other through rotations enforced by $\mathcal{R}$,

$$\begin{aligned} \varphi'_0 &= \mathcal{R}\varphi_0 \neq \varphi_0 \\ \mathcal{H}(\mathcal{R}\varphi_0) &= \mathcal{H}(\varphi_0) . \end{aligned} \quad (1.61)$$

In general, we say that we are in the presence of *Spontaneous Symmetry Breaking* (SSB) whenever the ground state φ_0 does not share the same symmetry $\mathcal{R}$ of the Hamiltonian, *i.e.* when

$$\begin{aligned} \mathcal{H}(\mathcal{R}\varphi) &= \mathcal{H}(\varphi) \\ \mathcal{R}\varphi_0 &\neq \varphi_0 . \end{aligned} \quad (1.62)$$

This is the case whenever a spin system in its low temperature phase develops a nonzero magnetization along a specific direction in space, even in the absence of an external directional magnetic field which would break the rotational symmetry *explicitly*. It is also the case for our Inertial Spin Model, which develops a nonzero polarization $\mathbf{V}$ in its ordered phase.

1.4.2 Goldstone modes

According to Nambu-Goldstone's theorem, *whenever a n -parameters continuous symmetry is spontaneously broken, then n massless fields emerge*. Massless fields

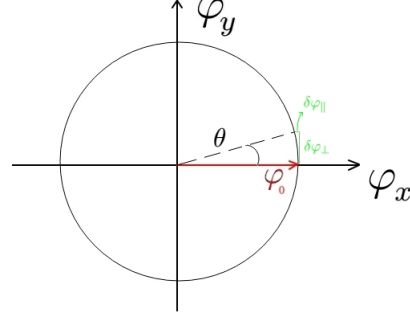


Figure 1.7. Scheme of a 2D field.

correspond to infinite susceptibilities, being χ in general proportional to the inverse of the squared (renormalized) mass³.

A way to see this is to study the Helmotz' free energy

$$f(\mathbf{h}) \equiv -\frac{1}{N} \ln \int \mathcal{D}\varphi e^{-\mathcal{H}(\varphi) - (\mathbf{h}, \varphi)}, \quad (1.63)$$

given that χ can be obtained from $-f(\mathbf{h})$ by deriving it twice with respect to its argument. Here we called $(\cdot, \cdot)$ the scalar product, and $\mathcal{D}\varphi$ denotes a *functional integral* over all possible configurations of the field $\varphi(\mathbf{x})$, which allows to evaluate the *partition function*

$$\mathcal{Z} = \int \mathcal{D}\varphi e^{-\mathcal{H}(\varphi) - (\mathbf{h}, \varphi)}. \quad (1.64)$$

Without loss of generality, let's take the ground state φ_0 to be aligned along the $\hat{x}$ axis as in Figure 1.7; then any external field $\mathbf{h}$ can be decomposed into its components parallel and perpendicular to the order parameter,

$$\mathbf{h} \equiv \begin{pmatrix} h_{\parallel} \\ h_{\perp} \end{pmatrix}. \quad (1.65)$$

Exploiting the symmetry (1.62) of the Hamiltonian, we can apply the unitary change of variable $\varphi \rightarrow \mathcal{R}\varphi$ to $f(\mathbf{h})$ and obtain

$$\begin{aligned} f(\mathbf{h}) &= -\frac{1}{N} \ln \int \mathcal{D}\varphi e^{-\mathcal{H}(\varphi) - (\mathbf{h}, \varphi)} \\ &= -\frac{1}{N} \ln \int \mathcal{D}\varphi e^{-\mathcal{H}(\mathcal{R}\varphi) - (\mathbf{h}, \varphi)} \\ &= -\frac{1}{N} \ln \int \mathcal{D}\varphi e^{-\mathcal{H}(\varphi) - (\mathbf{h}, \mathcal{R}^\dagger \varphi)} \\ &= -\frac{1}{N} \ln \int \mathcal{D}\varphi e^{-\mathcal{H}(\varphi) - (\mathcal{R}\mathbf{h}, \varphi)} \\ &= f(\mathcal{R}\mathbf{h}). \end{aligned} \quad (1.66)$$

Since for small angles θ the action of the rotation matrix (1.59) on the vector $\mathbf{h}$ can be expanded as

$$\mathcal{R}\mathbf{h} \simeq \begin{pmatrix} h_{\parallel} - \theta h_{\perp} \\ h_{\perp} + \theta h_{\parallel} \end{pmatrix}, \quad (1.67)$$

then equation (1.66) can be expressed in components as

$$\begin{aligned} f(h_{\parallel}, h_{\perp}) &= f(h_{\parallel} - \theta h_{\perp}, h_{\perp} + \theta h_{\parallel}) \\ &\simeq f(h_{\parallel}, h_{\perp}) - \theta h_{\perp} \frac{\partial f}{\partial h_{\parallel}} + \theta h_{\parallel} \frac{\partial f}{\partial h_{\perp}}, \end{aligned} \quad (1.68)$$

which leads to the *Ward identity*

$$h_{\perp} \frac{\partial f}{\partial h_{\parallel}} = h_{\parallel} \frac{\partial f}{\partial h_{\perp}}. \quad (1.69)$$

³This is quite standard in Statistical Field Theory, and it finds its most profound motivation in the Källén-Lehmann spectral representation of the two-points correlation function (see for instance [34]).

To obtain an expression for the susceptibility $\chi_\perp$ in the direction orthogonal to that of the order parameter, we differentiate relation (1.69) once again by $h_\perp$:

$$\left(\frac{\partial f}{\partial h_\parallel} + h_\perp \frac{\partial^2 f}{\partial h_\perp \partial h_\parallel} \right) \Big|_{h_\perp=0} = h_\parallel \frac{\partial^2 f}{\partial h_\perp^2} \Big|_{h_\perp=0}, \quad (1.70)$$

whence

$$\frac{\partial f}{\partial h_\parallel} = h_\parallel \frac{\partial^2 f}{\partial h_\perp^2}, \quad (1.71)$$

where it is understood that the two sides of the equation are to be evaluated at $h_\perp = 0$. Since the magnetization m and the susceptibility are given, by definition, by

$$m \equiv - \frac{\partial f}{\partial h_\parallel} \quad (1.72)$$

$$\chi_\perp \equiv - \frac{\partial^2 f}{\partial h_\perp^2}, \quad (1.73)$$

then equation (1.71) can be rephrased as

$$m = h_\parallel \cdot \chi_\perp. \quad (1.74)$$

In the disordered phase of the model, where the symmetry is unbroken, relation (1.74) is trivially saying that $m \rightarrow 0$ when $h_\parallel \rightarrow 0$. But things change in the broken symmetry phase, where a nonzero residual magnetization m can be obtained even when the field $\mathbf{h} \rightarrow \mathbf{0}$. Then relation (1.74) expresses the fact that

$$h_\parallel \rightarrow 0 \quad (1.75)$$

$$m \neq 0 \quad (1.76)$$

$$\chi_\perp = \frac{m}{h_\parallel} \xrightarrow{h_\parallel \rightarrow 0} \infty. \quad (1.77)$$

This is the so called *Nambu-Goldstone mode*: fluctuations of the order parameter in the direction of the broken symmetry manifold spanned by $\mathcal{R}$ are massless, which means that the susceptibility in the direction transverse to that of the order parameter is infinite. In Figure 1.6, they correspond to fluctuations within the bottom of the Mexican hat potential. We will make wide use of this result to justify the emergence of long range correlations between velocities in the ordered phase of the Inertial Spin Model.

Chapter 2

The planar Inertial Spin Model

In this Chapter we specialize the Inertial Spin Model presented in Chapter 1 to the case of a 2-dimensional order parameter immersed in 3-dimensional physical space: this simplifies its description to a great extent, by reducing the number of degrees of freedom. Once we have derived the full set of equations of motion, we will build an integration scheme which allows to solve them numerically; we will then employ it to study the static properties of the system in its *on lattice* approximation.

2.1 Equations of motion

The Hamiltonian (1.36) for the microscopic Inertial Spin Model on a fixed lattice can be generalized by including some external fields $\mathbf{H}_i$ and $\mathbf{h}_i$ coupled to the velocity and to the spin respectively: then it would read

$$\mathcal{H} = -\frac{J}{2v_0^2} \sum_{ij} n_{ij} \mathbf{v}_i \cdot \mathbf{v}_j + \sum_i \frac{s_i^2}{2\chi} - \frac{1}{v_0} \sum_i \mathbf{H}_i \cdot \mathbf{v}_i - \sum_i \mathbf{h}_i \cdot \mathbf{s}_i. \quad (2.1)$$

Let's specialize it for 3-dimensional space ($d = 3$), but choosing the order parameter to be 2-dimensional ($D = 2$). Since the speed is taken to be the same for all the individuals in the flock, namely $\|\mathbf{v}_i\| = v_0$, then it is enough to give the polar angle φ_i lying in the xy plane of the internal space of velocity to completely specify each vector

$$\mathbf{v}_i = v_0 \begin{pmatrix} \cos \varphi_i \\ \sin \varphi_i \end{pmatrix}. \quad (2.2)$$

With this choice of the reference frame, all the spin vectors $\mathbf{s}_i$ lie along the $\hat{\mathbf{z}}$ axis,

$$\mathbf{s}_i = s_i \hat{\mathbf{z}}. \quad (2.3)$$

Moreover, let's choose

$$\begin{aligned} \mathbf{H}_i &= H_i(t) \hat{\mathbf{x}} \\ \mathbf{h}_i &= h_i(t) \hat{\mathbf{z}} \end{aligned} \quad (2.4)$$

so that the Hamiltonian (2.1) takes the form

$$\mathcal{H} = -\frac{J}{2} \sum_{ij} n_{ij} \cos(\varphi_i - \varphi_j) + \sum_i \frac{s_i^2}{2\chi} - \sum_i H_i(t) \cos \varphi_i - \sum_i h_i(t) s_i \quad (2.5)$$

whence we can derive the canonical equations of motion for $\varphi_i(t)$ and $s_i(t)$,

$$\begin{cases} \dot{\varphi}_i(t) = \frac{\partial \mathcal{H}}{\partial s_i} = \frac{s_i}{\chi} - h_i \\ \dot{s}_i(t) = -\frac{\partial \mathcal{H}}{\partial \varphi_i} = J \sum_j n_{ij} \sin(\varphi_j - \varphi_i) - H_i(t) \sin \varphi_i(t) . \end{cases} \quad (2.6)$$

The derivation we carried out so far accounts for a completely reversible dynamics, and in fact only deterministic terms appear into (2.6). Indeed, several kinds of dynamics may be associated to the same given Hamiltonian; for the sake of generality, and as we did in Section 1.3.1 when we introduced the Inertial Spin Model, here we add a dissipating term and a stochastic noise in the equation which governs the evolution of $s_i(t)$:

$$\dot{s}_i(t) = -\frac{\partial \mathcal{H}}{\partial \varphi_i} - \eta \frac{\partial \mathcal{H}}{\partial s_i} + \xi_i \quad (2.7)$$

where η is a damping coefficient and ξ is a white uncorrelated noise,

$$\begin{aligned} \langle \xi \rangle &= \mathbf{0} \\ \langle \xi_i(t) \xi_j(t') \rangle &= 2T\eta \delta_{ij} \delta(t - t') . \end{aligned} \quad (2.8)$$

This way the equations of motion become

$$\begin{cases} \dot{\varphi}_i(t) = \frac{s_i}{\chi} - h_i \\ \dot{s}_i(t) = J \sum_j n_{ij} \sin(\varphi_j - \varphi_i) - H_i(t) \sin \varphi_i - \frac{\eta}{\chi} s_i + \eta h_i(t) + \xi_i(t) . \end{cases} \quad (2.9)$$

Finally, if we allow the individual positions to evolve in time as well (a situation we will refer to as *off lattice*), then the update rule for the positions should be included:

$$\begin{cases} \frac{dx_i}{dt} = v_0 \cos \varphi_i \\ \frac{dy_i}{dt} = v_0 \sin \varphi_i ; \end{cases} \quad (2.10)$$

as a result, the connectivity matrix $n_{ij} = n_{ij}(t)$ becomes a function of time.

2.2 Numerical integration of the equations of motion

Most of our work will be devoted to the study of the response of the system when an external perturbation is applied. In Chapter 3 we will be concerned with the analytic description of such response and the derivation of the exact propagators in the case where the individuals which constitute the flock are kept fixed on the vertices of a regular lattice; however, a full analytic treatment will no longer be possible as soon as we allow the individual positions $\mathbf{r}_i$ to evolve according to their own respective velocity $\mathbf{v}_i$, hence going *off lattice*. Even in the cases when we *do* have the propagators, nonetheless, it will reveal useful to check one's findings against some kind of empirical evidence. We thus devote some efforts to the derivation of

a numerical integration scheme that will allow us to build a Langevin dynamics computer simulation of the planar ISM.

Let's start by defining

$$\begin{aligned} x_i &\equiv \varphi_i, \quad u_i \equiv \frac{s_i}{\chi} \\ \gamma &\equiv \frac{\eta}{2\chi}, \quad \sigma \equiv \frac{\sqrt{2T\eta}}{\chi}, \quad \beta \equiv \frac{1}{\chi} \\ f_i(\mathbf{x}(t), t) &\equiv \frac{J}{\chi} \sum_j n_{ij}(t) \sin(x_j(t) - x_i(t)), \end{aligned} \quad (2.11)$$

so that the system (2.9) becomes equivalent to

$$\begin{cases} dx_i(t) = [u_i(t) - h_i(t)] dt \\ du_i(t) = \{f_i(\mathbf{x}(t), t) - 2\gamma[u_i(t) - h_i(t)] - \beta H_i(t) \sin x_i(t)\} dt + \sigma dW_i(t) \end{cases} \quad (2.12)$$

where $W_i(t)$ is a Wiener process satisfying

$$\langle \mathbf{W}(t) \rangle = \mathbf{0} \quad (2.13)$$

$$\langle W_i(t) W_j(t') \rangle = \delta_{ij} \min\{t, t'\}. \quad (2.14)$$

We aim at discretizing these differential equations in order to obtain a numerical integration scheme which is at least of the second order in Δt . To do so, we will follow a procedure similar to that presented by Ciccotti and Vanden-Eijnden in [37].

First, notice the system (2.12) can be rewritten as

$$\begin{cases} x_i(t + \Delta t) = x_i(t) + \int_t^{t+\Delta t} [u_i(s) - h_i(s)] ds \\ u_i(t + \Delta t) = u_i(t) + \int_t^{t+\Delta t} \underbrace{[f_i(\mathbf{x}(s), s)]}_A - 2\gamma \underbrace{[u_i(s) - h_i(s)]}_B - \underbrace{\beta H_i(s) \sin x_i(s)}_D ds \\ \quad + \underbrace{\sigma [W_i(t + \Delta t) - W_i(t)]}_E \end{cases} \quad (2.15)$$

We will treat each of the $A - E$ terms separately.

Stochastic terms

Let's start from the stochastic term E ; by definition, a Wiener process satisfies the distribution equality

$$W_i(t + \Delta t) - W_i(t) \equiv \sqrt{\Delta t} \xi_i \quad (2.16)$$

where ξ_i is a Gaussian stochastic variable completely characterized by

$$\langle \xi \rangle = \mathbf{0} \quad (2.17)$$

$$\langle \xi_i \xi_j \rangle = \delta_{ij}. \quad (2.18)$$

We will also need the integrated version of (2.16), namely

$$c_i \equiv \int_t^{t+\Delta t} [W_i(s) - W_i(t)] ds . \quad (2.19)$$

Using the defining property (2.14), it is straightforward to prove that

$$\begin{aligned} \langle c_i c_j \rangle &= \left\langle \left(\int_t^{t+\Delta t} [W_i(s_1) - W_i(t)] ds_1 \right) \left(\int_t^{t+\Delta t} [W_j(s_2) - W_j(t)] ds_2 \right) \right\rangle \\ &= \delta_{ij} \frac{\Delta t^3}{3} \end{aligned} \quad (2.20)$$

and that

$$\left\langle (W_i(t + \Delta t) - W_i(t)) \left(\int_t^{t+\Delta t} [W_j(s) - W_j(t)] ds \right) \right\rangle = \delta_{ij} \frac{\Delta t^2}{2} . \quad (2.21)$$

Relations (2.20) and (2.21) cannot possibly be satisfied by a single Gaussian variable; however, inspecting (2.19) we notice that

$$c_i = \left(\int_t^{t+\Delta t} W_i(s) ds \right) - W_i(t) \Delta t . \quad (2.22)$$

The first addendum is surely independent from the second, as it is evaluated within the interval $[t, t + \Delta t]$; on the other hand, the second addendum must still be related in some way to the variable ξ defined in (2.18). We formulate the *ansatz*

$$\mathbf{c} = a \xi + b \zeta \quad (2.23)$$

where ζ is another Gaussian stochastic variable,

$$\begin{aligned} \langle \zeta \rangle &= \mathbf{0} \\ \langle \zeta_i \zeta_j \rangle &= \delta_{ij} \\ \langle \zeta_i \xi_j \rangle &= 0 . \end{aligned} \quad (2.24)$$

It is enough to impose (2.20) and (2.21) to fix the value of a and b in (2.23) and obtain

$$\mathbf{c} = \frac{\Delta t^{\frac{3}{2}}}{2} \left(\xi + \frac{1}{\sqrt{3}} \zeta \right) . \quad (2.25)$$

Deterministic terms

We can now use these results to integrate the other deterministic terms. Using the second equation in (2.15) and choosing $s \in [t, t + \Delta t]$, so that $(s - t) = \mathcal{O}(\Delta t)$, we get to

$$\begin{aligned} u_i(s) &= u_i(t) + (s - t) \{ f_i(\mathbf{x}(t), t) - 2\gamma [u_i(t) - h_i(t)] - \beta H_i(t) \sin x_i(t) \} \\ &\quad + \sigma [W_i(s) - W_i(t)] + \mathcal{O}(\Delta t^{\frac{3}{2}}) \end{aligned} \quad (2.26)$$

and then we iterate, inserting the expression we have just derived back again into (2.15):

$$\begin{aligned} \int_t^{t+\Delta t} u_i(s) ds &= \Delta t u_i(t) + \frac{\Delta t^2}{2} \{f_i(\mathbf{x}(t), t) - 2\gamma [u_i(t) - h_i(t)] - \beta H_i(t) \sin x_i(t)\} \\ &\quad + \sigma c_i + \mathcal{O}(\Delta t^{\frac{5}{2}}) \\ &\equiv \Delta t u_i(t) + A_i(\mathbf{x}(t), t) + \mathcal{O}(\Delta t^{\frac{5}{2}}), \end{aligned} \quad (2.27)$$

where we defined the variable

$$A_i(\mathbf{x}(t), t) \equiv \frac{\Delta t^2}{2} \{f_i(\mathbf{x}(t), t) - 2\gamma [u_i(t) - h_i(t)] - \beta H_i(t) \sin x_i(t)\}. \quad (2.28)$$

This concludes the treatment of the term B ; let's step to term A , for which we first note that

$$\begin{aligned} \mathbf{f}(\mathbf{x}(s)) &= \mathbf{f}(\mathbf{x}(t)) + \int_t^s \mathbf{u}(q) \cdot [\nabla \otimes \mathbf{f}(\mathbf{x}(q))] dq \\ &= \mathbf{f}(\mathbf{x}(t)) + (s-t)\mathbf{u}(t) \cdot [\nabla \otimes \mathbf{f}(\mathbf{x}(t))] + \mathcal{O}(\Delta t^{\frac{3}{2}}), \end{aligned} \quad (2.29)$$

hence

$$\int_t^{t+\Delta t} \mathbf{f}(\mathbf{x}(s)) ds = \Delta t \mathbf{f}(\mathbf{x}(t)) + \frac{\Delta t^2}{2} \mathbf{u}(t) \cdot [\nabla \otimes \mathbf{f}(\mathbf{x}(t))] + \mathcal{O}(\Delta t^{\frac{5}{2}}). \quad (2.30)$$

We can get rid of the Jacobian simply by noticing that

$$\mathbf{f}(\mathbf{x}(t+\Delta t)) = \mathbf{f}(\mathbf{x}(t)) + \Delta t \mathbf{u}(t) \cdot [\nabla \otimes \mathbf{f}(\mathbf{x}(t))] + \mathcal{O}(\Delta t^{\frac{3}{2}}) \quad (2.31)$$

whence

$$\frac{\Delta t^2}{2} \mathbf{u}(t) \cdot [\nabla \otimes \mathbf{f}(\mathbf{x}(t))] = \frac{\Delta t}{2} [\mathbf{f}(\mathbf{x}(t+\Delta t)) - \mathbf{f}(\mathbf{x}(t))] + \mathcal{O}(\Delta t^{\frac{5}{2}}). \quad (2.32)$$

Finally, the terms C and D can be treated in a similar way as follows. Define

$$F(y) \equiv \int_t^{t+y} \mathbf{g}(\mathbf{x}(s), s) ds = \int_0^y \mathbf{g}(\mathbf{x}(t+s), t+s) ds \quad (2.33)$$

and expand it in MacLaurin series,

$$\begin{aligned} F(\Delta t) &\simeq F(0) + F'(0)\Delta t + F''(0)\frac{\Delta t^2}{2} + \dots \\ &= \Delta t \mathbf{g}(\mathbf{x}(t), t) + \frac{\Delta t^2}{2} \left\{ \frac{\partial}{\partial s} \mathbf{g}(\mathbf{x}(s), s) + \mathbf{u}(s) \cdot [\nabla \otimes \mathbf{g}(\mathbf{x}(s))] \right\} \Big|_{s=t} + \mathcal{O}(\Delta t^{\frac{5}{2}}). \end{aligned} \quad (2.34)$$

By noticing that

$$\begin{aligned} \mathbf{g}(\mathbf{x}(t+\Delta t), t+\Delta t) &= \mathbf{g}(\mathbf{x}(t), t) + \Delta t \left\{ \frac{\partial}{\partial s} \mathbf{g}(\mathbf{x}(s), s) + \mathbf{u}(s) \cdot [\nabla \otimes \mathbf{g}(\mathbf{x}(s))] \right\} \Big|_{s=t} \\ &\quad + \mathcal{O}(\Delta t^{\frac{3}{2}}), \end{aligned} \quad (2.35)$$

one obtains

$$\int_t^{t+\Delta t} \mathbf{g}(\mathbf{x}(s), s) ds = \frac{\Delta t}{2} [\mathbf{g}(\mathbf{x}(t+\Delta t), t+\Delta t) + \mathbf{g}(\mathbf{x}(t), t)] + \mathcal{O}(\Delta t^{\frac{5}{2}}). \quad (2.36)$$

Integration scheme

Putting all these integrated terms back together into (2.15) finally yields the integration scheme

$$\begin{cases} x_i(t + \Delta t) = x_i(t) + \Delta t [u_i(t) - \bar{h}_i] + A_i(t) + \mathcal{O}(\Delta t^{\frac{5}{2}}) \\ u_i(t + \Delta t) = u_i(t) + \sigma \sqrt{\Delta t} \xi_i + \Delta t \left\{ \bar{f}_i - 2\gamma [u_i(t) - \bar{h}_i] - \beta \overline{H_i \sin x_i} \right\} - 2\gamma A_i(t) \\ \quad + \mathcal{O}(\Delta t^{\frac{5}{2}}) \\ A_i(t) = \frac{\sigma}{2} \Delta t^{\frac{3}{2}} \left(\xi_i + \frac{\zeta_i}{\sqrt{3}} \right) + \frac{\Delta t^2}{2} \{ f_i(\mathbf{x}(t), t) - 2\gamma [u_i(t) - h_i(t)] - \beta H_i(t) \sin x_i(t) \} \end{cases} \quad (2.37)$$

where ξ_i, ζ_i are white uncorrelated stochastic noises, with the same properties as in (2.24), and the bar over a variable stands for its midpoint value, for instance

$$\bar{f}_i = \frac{1}{2} [f_i(\mathbf{x}(t + \Delta t), t + \Delta t) + f_i(\mathbf{x}(t), t)] . \quad (2.38)$$

There remains to specify how the positions $\mathbf{r}_i(t)$ evolve according to their corresponding velocity $\mathbf{v}_i(t)$. This can be done by a simple Euler-Cromer update rule,

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t + \Delta t) \Delta t , \quad (2.39)$$

or by the Midpoint method,

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \frac{\mathbf{v}(t) + \mathbf{v}(t + \Delta t)}{2} \Delta t . \quad (2.40)$$

Clearly the connectivity matrix $n_{ij}(t)$ (here contained in the force term $\mathbf{f}(\mathbf{x}(t), t)$) should be updated regularly in order to take the relative motion of the individual positions into account; the way this can be achieved is described in the final Chapter, which is devoted to the study of the off-lattice system.

2.3 Static properties of ISM on a fixed lattice

We begin by studying the static properties of the model, namely its static correlations and the way the polarization and the susceptibility vary with the temperature T , which controls the amplitude of the stochastic noise in the equations of motion (2.9). We will first consider the fixed network case, for which reference resources already exist in literature for the static equilibrium properties of the system. This will mainly serve as a check on the correct operation of the numerical simulation, and will guide us in the choice of the parameters for the future measurements we will take.

In the next Chapter, we will move to investigate the dynamical properties of the system. However, it will not be until Chapter 5 that we will reinstate the evolution of the positions $\mathbf{r}_i(t)$ back into our description; from now on and up until then, they will be taken to be fixed in time. As we explained in Paragraph 1.3.2, this corresponds to observing the system on timescales which are much longer than the typical reshuffling time of the connectivity matrix n_{ij} .

2.3.1 Polarization and susceptibility

The order parameter of the model is the polarization Φ , which is defined as the average velocity of the flock normalized to unity:

$$\Phi = \frac{\mathbf{V}}{v_0} = \frac{1}{v_0 N} \sum_i \mathbf{v}_i ; \quad (2.41)$$

we will call its modulus

$$\phi \equiv |\Phi| \in [0, 1] , \quad (2.42)$$

and use it as a measure of the global alignment of the system. Just like the Heisenberg model, the ISM too is known to display a second order transition from an ordered to a disordered phase at finite temperature: the order parameter, which is equal to $\phi = 1$ at $T = 0$, gradually decreases to zero in correspondence of $T = T_c$, a value known as the *critical temperature* of the model.

A solid way to locate T_c is by studying the susceptibility χ of the system. For a system on the lattice, the static fluctuation-dissipation theorem (FDT) simply relates χ to the connected correlation of the polarization via

$$\chi = \beta N \left[\langle \phi^2 \rangle - \langle \phi \rangle^2 \right] , \quad (2.43)$$

where β is the inverse of T ; for a system in its thermodynamic limit, $N \rightarrow \infty$, the susceptibility χ is expected to diverge at $T = T_c$. In a finite size system, χ still reaches its maximum in correspondence of $T = T_c^*(N)$, a value which depends slightly on N and is generally bigger than its asymptotic limit

$$T_c = \lim_{N \rightarrow \infty} T_c^*(N) ; \quad (2.44)$$

the peak value χ_{max} also grows with N .

2.3.2 Correlation functions

Using the discretized equations of motion (2.37), the evolution of the system can be simulated numerically and the value assumed by all the degrees of freedom $\mathbf{v}_i(t)$, $\mathbf{r}_i(t)$ and $s_i(t)$, $\forall i$ can be measured for every time t along the trajectory in phasespace. Here we choose to keep the positions $\mathbf{r}_i(t)$ fixed on the vertices of a regular cubic lattice, which further decreases the number of degrees of freedom.

Notice that, when we wrote equation (2.43), we did not specify what the brackets $\langle \cdot \rangle$ referred to. Let's then assume that the system explores its whole phasespace if granted a sufficient amount of time (*i.e.* let's assume the system to be *ergodic*), and define $\langle \cdot \rangle$ to be the time average computed along the trajectory. When we measure it, different samples should be chosen sufficiently far from one another so that the state of the system can safely be assumed to be decorrelated from the previous: but how long should we wait in between two distinct measurements? An estimate of the decorrelation time τ_{decorr} can be obtained by computing the autocorrelation function, which we are about to define.

Autocorrelation function

Consider at first that, for a planar model like ours ($D = 2$), the polarization at any time t along the trajectory simply reduces to

$$\Phi(t) = \frac{1}{N} \sum_i \begin{pmatrix} \cos \varphi_i(t) \\ \sin \varphi_i(t) \end{pmatrix} \equiv \begin{pmatrix} \phi_x(t) \\ \phi_y(t) \end{pmatrix}. \quad (2.45)$$

We define the autocorrelation function of the polarization vector as

$$C(t) = \frac{1}{C(t=0)} \left\{ \frac{1}{T_{max} - t} \sum_{t_0=1}^{T_{max}-t} \Phi(t_0) \cdot \Phi(t_0 + t) - |\langle \Phi \rangle|^2 \right\}, \quad (2.46)$$

where

$$C(t=0) = \langle |\Phi|^2 \rangle - |\langle \Phi \rangle|^2 \quad (2.47)$$

$$\Phi(t_0) \cdot \Phi(t_0 + t) = \phi_x(t_0)\phi_x(t_0 + t) + \phi_y(t_0)\phi_y(t_0 + t) \quad (2.48)$$

$$|\langle \Phi \rangle|^2 = \langle \phi_x \rangle^2 + \langle \phi_y \rangle^2, \quad (2.49)$$

and the brackets denote the time average along the whole trajectory. The meaning of the sum in (2.46) is this: given a trajectory whose total duration is T_{max} , we first sum over all pairs at time distance t and finally divide by the number of such pairs, which is $(T_{max} - t)$.

We can similarly define an autocorrelation function for the polarization modulus as

$$C^{scalar}(t) = \frac{\langle \phi(t_0)\phi(t_0 + t) \rangle_{t_0} - \langle \phi(t_0) \rangle_{t_0}^2}{\langle \phi^2(t_0) \rangle_{t_0} - \langle \phi(t_0) \rangle_{t_0}^2}, \quad (2.50)$$

where we called $\phi = |\Phi|$ and, as in equation (2.46),

$$\langle f(t; t_0) \rangle_{t_0} \equiv \frac{1}{T_{max} - t} \sum_{t_0=1}^{T_{max}-t} f(t; t_0). \quad (2.51)$$

In all the cases we will consider, the shape of $C(t)$ presents an exponential decay (see for instance Figure 2.3(a)); the quantity τ_{decorr} is defined as the typical time scale of this decay.

Spatial correlations

The deviations of the velocity $\mathbf{v}_i$ of the single individual from the global mean $\mathbf{v}$ are described by the fluctuation

$$\delta \mathbf{v}_i \equiv \mathbf{v}_i - \mathbf{V} = \mathbf{v}_i - \langle \mathbf{v}_i \rangle. \quad (2.52)$$

We can measure the degree of similarity between two individual deviations i and j using $\langle \delta \mathbf{v}_i \cdot \delta \mathbf{v}_j \rangle$; again, though, it is not completely clear what the brackets should refer to, at least in the general off-lattice case. Indeed, in equilibrium Statistical Physics they would denote an average over some statistical ensemble, but we do

not have a well defined such ensemble for an active matter system (or, in other words, the single $\delta \mathbf{v}_i$'s do not admit a stationary distribution). The best we can do is to assume the interactions to be isotropic and imagine the correlations to depend only on the mutual distance between each pair of individuals. Hence, we define the spatial connected correlation function $C(r)$ through the spatial average

$$C(r) = \frac{\sum_{ij} \delta \mathbf{v}_i \cdot \delta \mathbf{v}_j \delta(r - r_{ij})}{\sum_{ij} \delta(r - r_{ij})}, \quad (2.53)$$

where $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$. The sum runs over all the distinct pairs at distance r from one another, and the factor at the denominator is the number of such pairs: this way the density fluctuations are decoupled from those of the order parameter. Notably, $C(r)$ is always well defined because it *does* admit a stationary distribution.

Scaling laws

According to Widom and Kadanoff's scaling hypothesis (initially based on experimental evidence and later justified through the theory of Critical Phenomena - see for instance [25]), the general form of the spatial correlation function in real space is

$$C(r) = \frac{1}{r^{d-2+\eta}} f\left(\frac{r}{\xi}\right), \quad (2.54)$$

where d is the space dimension, ξ is the correlation length of the system, η is its anomalous dimension and f is a scaling function (for example $f(x) \sim e^{-x}$). In the vicinity of the critical point of a second order phase transition, ξ is thought to be the only relevant scale for the system; it depends on the temperature via

$$\xi \sim (T - T_c)^{-\nu}, \quad (2.55)$$

ν being its critical exponent. As ξ diverges, $f(\frac{r}{\xi} \rightarrow 0)$ simply tends to a constant, and (2.54) predicts a power law (*scale free*) decay for $C(r)$. The velocity of the decay is basically controlled by the dimension d of the space, since η generally consists in a small correction to its behaviour.

The divergence of ξ also gives rise to that of the susceptibility χ in the critical region, being

$$\begin{aligned} \chi(T) &= \beta \hat{C}(\mathbf{k} = \mathbf{0}) = \beta \int d^d x C(\mathbf{x}) \\ &\simeq \int_{|\mathbf{x}| < \xi(T)} \frac{d^d x}{|\mathbf{x}|^{d-2+\eta}} \sim \xi(T)^{2-\eta} \sim (T - T_c)^{-\nu(2-\eta)} \equiv (T - T_c)^{-\gamma}, \end{aligned} \quad (2.56)$$

where we called γ the critical exponent of χ . This justifies the claim we made in Section 2.3.1 that it is possible to locate T_c by studying the behaviour of $\chi(T)$.

Finally, a scaling law also holds for the autocorrelation function $C(t)$, whose general shape is

$$C(t) \sim e^{-\frac{t}{\tau}}; \quad (2.57)$$

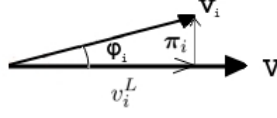


Figure 2.1. Expansion of a generic velocity vector $\mathbf{v}_i$ into its component parallel (v_i^L) and perpendicular (π_i) to the direction of the polarization $\mathbf{V}$.

the decorrelation time τ ($= \tau_{decorr}$) depends on the correlation length ξ via

$$\tau \sim \xi^z, \quad (2.58)$$

where z is the dynamical critical exponent¹.

Correlations in the broken symmetry phase

As we have summarily revised in Section 1.4, in the broken symmetry phase of a model whose order parameter is continuous (which is the case for the Heisenberg model and the Inertial Spin Model with $D = 2$) we see the emergence of *Goldstone modes*. Fluctuations of the order parameter parallel and transverse to the direction of the symmetry manifold (see Figure 1.6) contribute differently to the correlation function $C(r)$, so that it is useful to write them separately as

$$C(r) \sim C^{\parallel}(r) + C^{\perp}(r). \quad (2.59)$$

The Goldstone mode is massless, which means that $C^{\perp}(r)$ decays as a power law,

$$C^{\perp}(r) \sim \frac{1}{r^{d-2}}; \quad (2.60)$$

moreover, since the modulus of the order parameter is a constrained (hard) variable, longitudinal and perpendicular fluctuations are not independent [33] and longitudinal correlations are scale free too². For example, close to the critical temperature one finds [9]

$$\chi_{\parallel} \simeq (\chi_{\perp})^{\frac{\epsilon}{2}}, \quad (2.61)$$

where $\epsilon = 4 - d$. All in all, due to the soft Goldstone mode, we expect a slow decay of the correlation function $C(r)$ like that of (2.60) in the whole ordered phase of the model.

A simple argument can give us an idea of how relevant the two contributions in $C(r)$ are with respect to one another. As we will often do in the rest of this work whenever we consider a system into its deeply polarized phase, we now define $\hat{\mathbf{n}}$ to

¹For the Heisenberg model in $d = 3$, for instance, it is known that $z = 2$ for a purely dissipative dynamics.

²The proof is subtle, and it makes use of the fact that an infinitesimal displacement of the order parameter along its broken symmetry manifold (see Figure 1.6) does produce a small change in the *parallel* component of the field as well, as it can be clearly seen in Figure 1.7. A possible workaround is to adopt a polar rather than a Cartesian parameterization for the two components of the field: this would clearly show that the only massless component is the one in the *angular* direction, while correlations decay exponentially along the *radial* direction.

be the direction of the global velocity $\mathbf{V}$ of the flock and we decompose each velocity vector $\mathbf{v}_i$ into its components parallel and perpendicular to $\hat{\mathbf{n}}$:

$$\mathbf{v}_i = v_i^L \hat{\mathbf{n}} + \boldsymbol{\pi}_i, \quad (2.62)$$

where $\boldsymbol{\pi}_i$ is a $(D - 1)$ -dimensional vector; in our planar model, v_i^L and π_i are both scalar (see Figure 2.1) and it is clearly, by construction,

$$\sum_i \pi_i = 0. \quad (2.63)$$

Due to the constraint on the modulus of the velocity (here we take $v_0 = 1$ for brevity),

$$|\mathbf{v}_i|^2 = \left(v_i^L\right)^2 + \pi_i^2 = 1 \quad (2.64)$$

$$\rightarrow v_i^L \simeq 1 - \frac{\pi_i^2}{2}. \quad (2.65)$$

Since π_i represents the fluctuation of $\mathbf{v}_i$ from the average in the direction perpendicular to that of the order parameter, *i.e.*

$$\delta \mathbf{v}_i = \delta v_i^L \hat{\mathbf{n}} + \boldsymbol{\pi}_i, \quad (2.66)$$

then we conclude that, given two variables sitting on positions $\mathbf{r}_i$ and $\mathbf{r}_j$, their spacial correlation is made up of the two contributions

$$\begin{aligned} C_{ij}^\perp &\sim \langle \pi_i \pi_j \rangle \sim \mathcal{O}(\pi^2) \\ C_{ij}^\parallel &\sim \langle \delta v_i^L \delta v_j^L \rangle = \left\langle \frac{\pi_i^2 \pi_j^2}{4} \right\rangle \sim \mathcal{O}(\pi^4). \end{aligned} \quad (2.67)$$

This is telling us that, if the system is highly polarized, then $C^\parallel(r)$ can safely be discarded and we expect

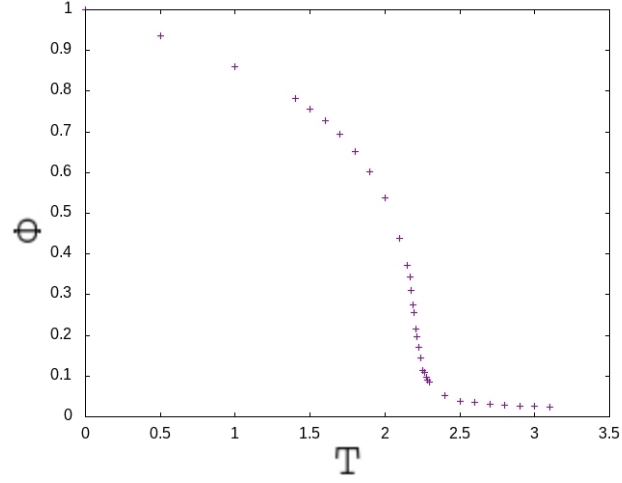
$$C^{(T \ll T_c)}(r) \sim \frac{1}{r^{d-2}}. \quad (2.68)$$

2.3.3 Numerical simulations

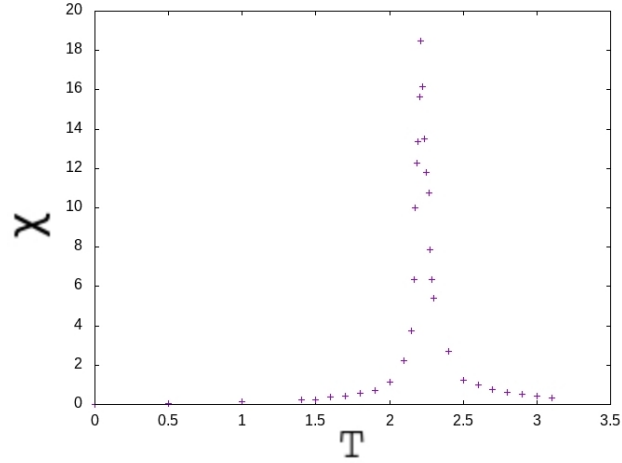
We chose for our numerical simulations the set of parameters

$$\begin{aligned} J &= \chi = \eta = 1 \\ L &= 10 \\ T_{max} &\sim 10^6 \\ \Delta t &= 0.01, \end{aligned} \quad (2.69)$$

and we varied the temperature T to explore the phase diagram of the system. Indeed, χ and η are dynamical parameters which will prove relevant in our study of the information transfer throughout the flock, but they have no influence on the static properties of the system: we have seen in equation (1.51) that the steady state distribution of velocities in the ISM is controlled by the combination βJ , hence



(a)



(b)

Figure 2.2. On top, polarization of the system across the second order transition driven by the parameter T . On the bottom, the susceptibility χ as a function of T .

setting $J = 1$ and varying β is enough to completely characterize the phase diagram. Finally, the choice of T_{max} and Δt is dictated by a compromise between an accurate simulation and a reasonable computation time.

Periodic boundary conditions were applied, so that the bulk properties are similar to those of an infinite size system.

Polarization and susceptibility

In the case $D = 2$, the modulus of the polarization can be evaluated as

$$\phi = \left\langle \frac{1}{N} \left[\left(\sum_i \cos \varphi_i(t) \right)^2 + \left(\sum_i \sin \varphi_i(t) \right)^2 \right]^{\frac{1}{2}} \right\rangle_t, \quad (2.70)$$

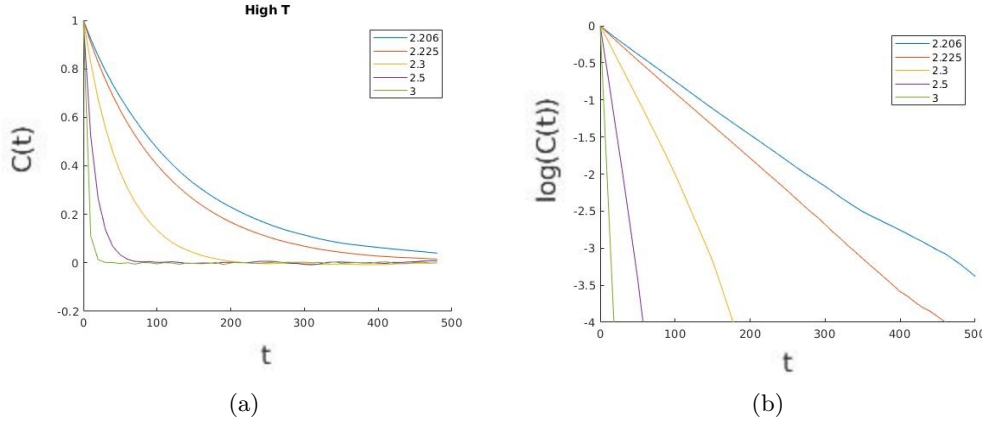


Figure 2.3. (a) Autocorrelation function $C(t)$ in the high temperature, disordered phase of the model, for some selected temperatures $T > T_c$. (b) The same curves plotted in semilog scale; the slope of each curve is proportional to the inverse of $\tau_{decorr}(T)$.

where $\langle \cdot \rangle_t$ denotes a time average over those configurations separated by the decorrelation time τ_{decorr} .

Figure 2.2(a) shows how the polarization drops to zero around the critical temperature $T_c \simeq 2.2125$. This result is only slightly larger than the one found in literature for the Heisenberg model on a $3d$ cubic lattice and with a 2-dimensional order parameter, which is $T_c = 2.2018$ (see for instance [22], where the value $\beta_c = 0.454170(7)$ was extrapolated numerically by studying cubic lattices up to $L = 32$). This discrepancy is totally imputable to finite size effects, as we are studying a lattice of $N = 1000$ individuals only.

Similarly, the susceptibility of the system across the transition can be computed using equation (2.43); Figure 2.2(b) shows that χ is peaked around $T_c \simeq 2.2125$ as well.

Correlations in the disordered phase

The autocorrelation function in the high temperature phase can be computed along the trajectory using equation (2.46); the result for some selected temperatures $T > T_c$ is shown in Figure 2.3(a). Since $C(t)$ decays exponentially as predicted by equation (2.57), it is useful to plot $\log C(t)$ vs t : this is done in Figure 2.3(b). A rough estimate of the decorrelation time $\tau_{decorr}(T)$ can directly be read from the graphs; this has been done for all the temperatures involved in the realization of the curves $\phi(T)$ and $\chi(T)$ in Figure 2.2(a) and 2.2(b), where the averages were performed over decorrelated configurations selected at intervals of $\tau_{decorr}(T)$ along each trajectory. The increasing steepness of the straight lines is telling us that $\tau_{decorr}(T)$ gets smaller and smaller as we move away from the critical temperature T_c , as predicted by (2.58).

To measure the spatial correlation function $C(r)$, we specialize equation (2.53)

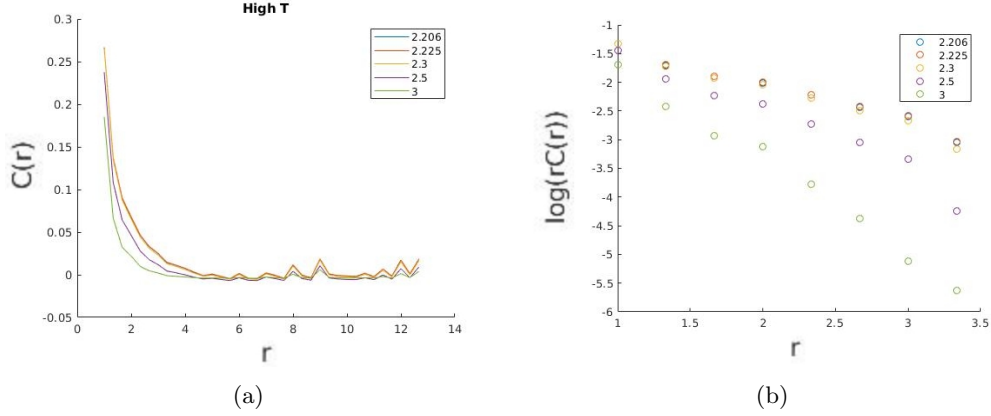


Figure 2.4. (a) Spatial correlation function $C(r)$ in the high temperature, disordered phase of the model, for some selected temperatures $T > T_c$. (b) Plots of $\log rC(r) \sim r$; the slope of each curve is proportional to the inverse of the correlation length $\xi(T)$.

for our planar $D = 2$ model and we find

$$C(r) = \frac{1}{C_0} \left\{ \frac{\sum_{ij} \delta(r - r_{ij}) \langle \cos[\varphi_i(t) - \varphi_j(t)] \rangle_t}{\sum_{ij} \delta(r - r_{ij})} - |\langle \Phi \rangle|^2 \right\}, \quad (2.71)$$

where again $\langle \cdot \rangle_t$ denotes a time average over uncorrelated configurations, and the explicit expression of $|\langle \Phi \rangle|^2$ is given in (2.49). The expression above of $C(r)$ is divided by the value $C_0 = C(r = 0)$ it assumes in zero, so as to better compare correlations at different temperatures. In the high temperature phase, $\langle \Phi \rangle = \mathbf{0}$ and the connected and unconnected correlation functions coincide.

Plots of $C(r)$ for some selected temperatures $T > T_c$ are shown in Figure 2.4(a). In Figure 2.4(b) we plotted $\log rC(r)$ vs r for the same data points: if $C(r)$ behaves as predicted by the scaling law (2.54), and being $d = 3$, $\eta \ll 1$ for our system, then we should observe straight lines whose slope is proportional to the inverse of the correlation length $\xi(T)$. It is important to bear in mind that periodic boundary conditions were adopted, hence our measured $C(r)$ is only significant up to $r \sim \frac{L}{2} = 5$ (after which we expect it to start growing again).

Correlations in the ordered phase

Measurements in the broken symmetry phase of the system are made more complicated by the so called *wandering of the order parameter*. To get a quick grasp of what this means, let's consider first a d -dimensional Ising model in its low temperature phase; the order parameter, *i.e.* the magnetization m , can assume the two opposite values $m = \pm m_0$ which are equally energetic, thus leading to a situation of *bistability* (see Figure 2.5(a)). If the system were to perform a transition from one state to the other, it would have to win first a potential energy barrier

$$\Delta G \sim \sigma L^{d-1}, \quad (2.72)$$

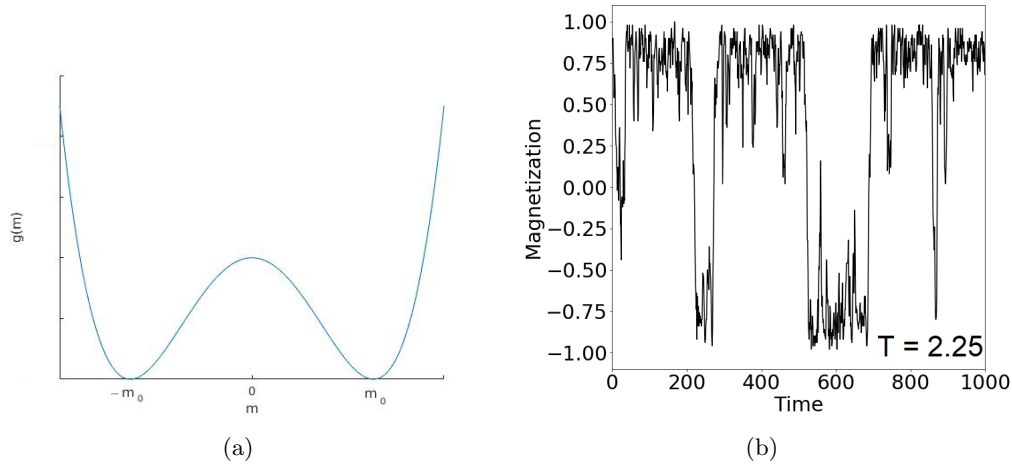


Figure 2.5. (a) Gibbs' free energy for the d -dimensional Ising model plotted as a function of the order parameter, which is the magnetization m . It presents two minima of equal energy at $m = \pm m_0$. (b) The order parameter jumps from one minimum to the other in a finite-size numerical simulation of the Ising model, a phenomenon known as *flip-flop*.

where σ is the *surface tension* and L is the size of the system. For a system in a thermal bath at temperature T , the typical transition time τ_{FF} from one state to the other can be estimated using Arrhenius' law:

$$\tau_{FF} \sim e^{\frac{\Delta G}{k_B T}}, \quad (2.73)$$

where k_B is the Boltzmann constant.

As a consequence, if we simulate numerically a finite size Ising system in the presence of thermal noise, we typically observe *flip-flops*: the order parameter suddenly jumps from one equilibrium state to the other every $t \sim \mathcal{O}(\tau_{FF})$ (see Figure 2.5(b)). Whenever we measure an observable in a numerical simulation, we always have to make sure that the system has remained in the same state for the whole duration of the measurement; if, on the contrary, we exceed the time window where this remains true, we risk obtaining trivial zeros due to symmetry reasons.

When we study the broken symmetry phase of a system whose order parameter is continuous, such as the Heisenberg model or the ISM, then instead of two points alone ($m = \pm m_0$) we find a whole manifold of points in the space of the order parameter which represent states of equal energy (Figure 1.6). If the system is given a small amount of noise (as we always do in our Langevin dynamics simulations), then the order parameter will eventually migrate from its initial position and explore this manifold, a process that we call *wandering*. In Section 3.1.2 we will be able to give an estimate of the typical wandering time of the planar ISM as

$$\tau_w \sim \frac{\eta N}{2T}; \quad (2.74)$$

we notice that

$$\lim_{N \rightarrow \infty} \tau_w = \infty, \quad (2.75)$$

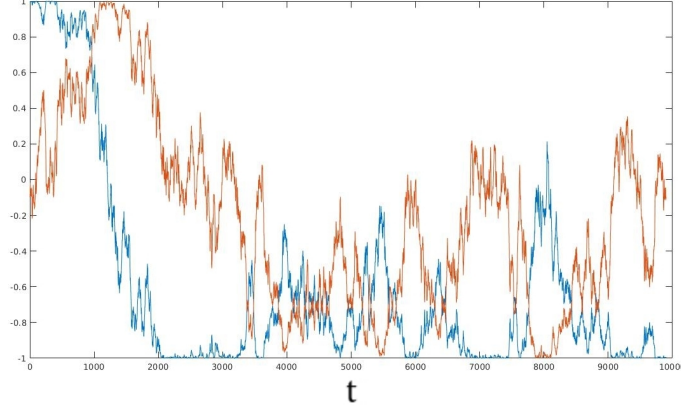


Figure 2.6. Wandering of the order parameter in a typical simulation of ISM with $D = 2$, for a system of size $L = 10$. The two components ϕ_x and ϕ_y of the polarization defined in (2.45) are plotted as a function of time, showing that $\mathbf{V} = v_0 \mathbf{\Phi}$ is rotating.

as it should be: this way a system in its thermodynamic limit can actually exhibit spontaneous symmetry breaking and the ergodicity is lost. For a finite size system, on the other hand, equation (2.75) predicts that the time windows during which the polarization is more or less stable are of order $\tau_w \sim \mathcal{O}(N)$.

Figure 2.6 gives a taste of how this wandering looks like in a typical simulation of ISM with $D = 2$: the polarization $\mathbf{\Phi}(t)$ rotates in time, so that its angle measured with respect to any fixed frame of reference assumes all the values between 0 and 2π .

What does this imply for our correlation functions? Equations (2.46) and (2.71) which we use to calculate $C(t)$ and $C(r)$ in our numerical simulations both contain the quantity $|\langle \mathbf{\Phi} \rangle|$. This is irrelevant in the high temperature phase, where $\langle \mathbf{\Phi} \rangle = \mathbf{0}$ anyways, but it becomes utterly important in the broken symmetry phase, where it renders $C(t)$ and $C(r)$ *connected* correlation functions and allows them to eventually drop to zero, which they would not do otherwise. This is well exemplified by Figure 2.7, where we show the results of measurements taken on a system of $N = 1000$ individuals. In the first graph (Figure 2.7(a)), the spatial correlation function $C(r)$ for temperatures $T < T_c$ simply does not connect and reaches a plateau $C(r \gg 1) > 0$, instead of going to zero.

The situation is more interesting for $C(t)$ (Figure 2.7(b)), which actually *does* go to zero, but only because of the wandering of the order parameter. The decay time τ that we observe in Figure 2.7(c) actually represents an estimate of the wandering time τ_w at that specific temperature.

We must conclude that the only possible way to observe the scaling of $C(r)$ predicted by (2.68) is to manually examine the single trajectories and measure $C(r)$ within a time window where the order parameter does not wander too much; we have seen such windows are of order τ_w . According to equation (2.75), we can make τ_w larger by choosing a bigger system (which we will not do, for consistency with our initial choice of $L = 10$) or else by lowering the temperature; we thus choose $T = 0.5$.

When we use equation (2.71) to evaluate $C(r)$, we need to average over uncorrelated configurations of the system. Our previous estimates of $C(t)$ in the broken symmetry phase cannot be relied upon if we want to determine the decorrelation time $\tau_{decorr}(T = 0.5)$, but we can at least set a lower bound for the real τ_{decorr} by inspecting the scalar autocorrelation function defined in (2.50). Again, this is useful because $C^{scalar}(t)$ is not subject to the wandering of the polarization. A plot of $C^{scalar}(t)$ for several temperatures $T < T_c$ is illustrated in Figure 2.8(b), while Figure 2.8(c) shows the same curves in semi-log scale, whence the value of the lower bound τ_{scalar} can be extracted; we notice that $\tau_{scalar}(T)$ grows as we approach T_c from below, as it should (equation (2.58)).

When we actually compute $C(r)$ via equation (2.71), we choose to work with as big a τ_{decorr} as it is allowed by the dimension of the wandering-free time window, whilst still retaining a decent number of configurations to average over; in other words, we pick $\tau_{scalar} < \tau_{decorr} < \tau_w$. Finally, we plot $C(r)$ in double-log scale in order to extract its slope, which is in reasonable agreement with the prediction $\alpha \simeq -1$ of the scaling law (2.68): this result is shown in Figure 2.8(a).

Feeling satisfied, we decree our simulation to be correct and reliable. Next, we will move on to the study of the effects of external perturbations applied to our system.

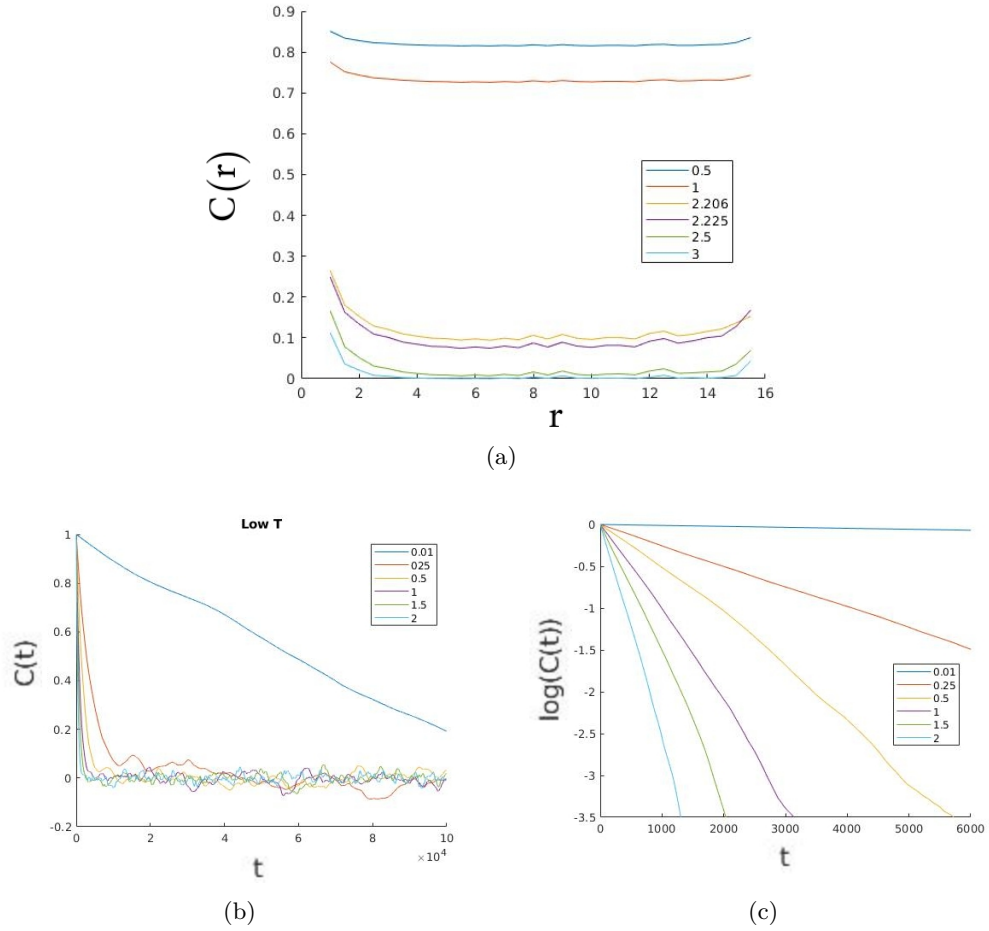


Figure 2.7. (a) Wrong version of the spatial correlation function $C(r)$ in the high and low temperature phases of the model, calculated without taking into account the wandering of the order parameter. The curves for $T < T_c$ never go to zero because they are not connected properly (see text). (b) Wrong version of the autocorrelation functions $C(t)$, which do eventually drop to zero, but only thanks to the wandering of the polarization. (c) A semilog graph of the wrong $C(t)$ allows to estimate (from the inverse of the slopes) the decay time τ , which is in this case nothing else than the wandering time $\tau_w(T)$.

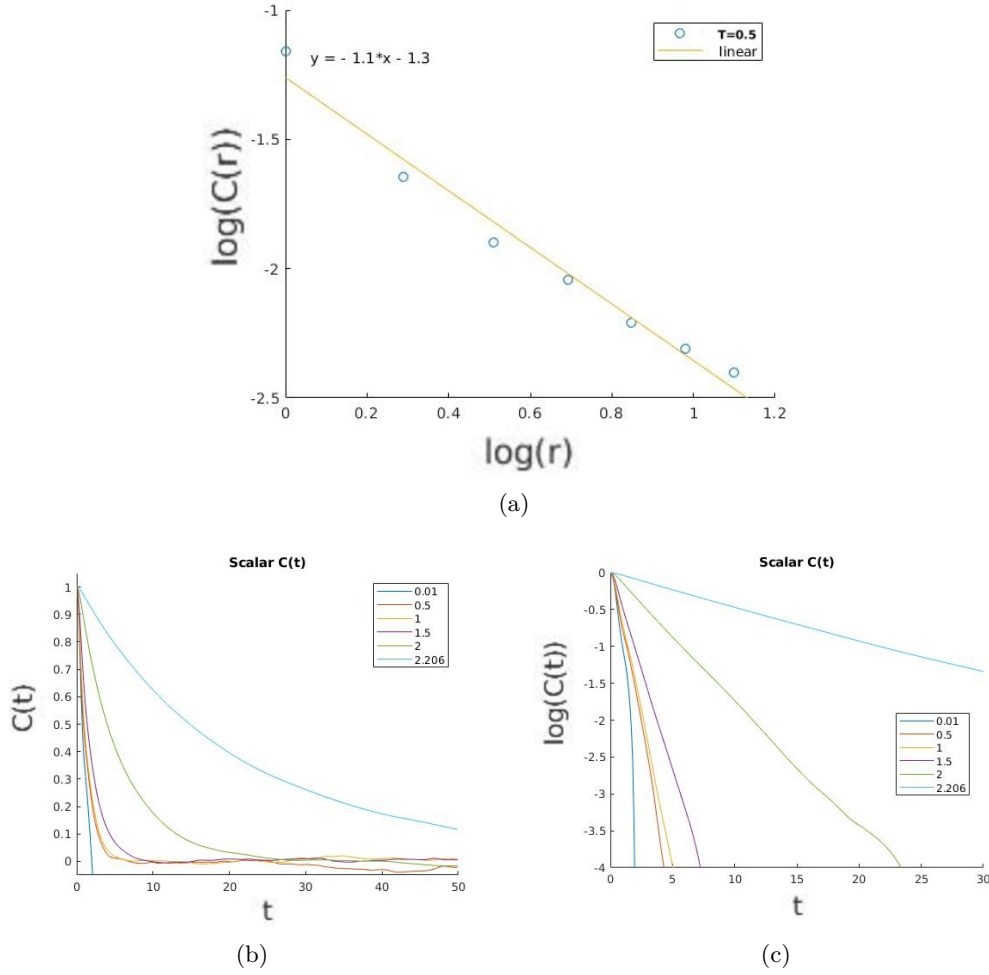


Figure 2.8. (a) Spatial correlation function $C(r)$ measured in a wandering-free time window, and here plotted in double-log scale so as to stress that it correctly reproduces the slope $\alpha \simeq -1$ predicted by the scaling law (2.68). (b) Scalar autocorrelation function, evaluated according to its definition (2.50). (c) A semilog graph of the scalar $C(t)$ allows to estimate (by the inverse of the slopes) the decay time τ_{scalar} , which serves as a lower bound to the real decorrelation time τ_{decorr} of the full vectorial polarization Φ .

Chapter 3

Analytical study of the response on-lattice

This Chapter is devoted to the exact derivation of the propagators in the low temperature phase of the ISM, with the aim of writing an explicit expression for the response of the system to a local perturbing field with a specific time dependence, at first order precision in linear Perturbation Theory. We will consider the positions $\mathbf{r}_i$ to be fixed in time, which represents the *on lattice* approximation, valid for times shorter than the reshuffling time of the connectivity matrix n_{ij} .

First, we will face the problem working directly on the microscopic Hamiltonian and trying to solve the $2N$ coupled dynamical equations of motion which describe the evolution of the $2N$ degrees of freedom (*i.e.* spins and velocities). Next, we will give a coarse grained description of the system in terms of the fields $\varphi(\mathbf{x}, t)$ and $s(\mathbf{x}, t)$, and we will check how the same results can be obtained by solving a single partial differential equation for the evolution of the field $\varphi(\mathbf{x}, t)$.

Why is this an issue?

As we have seen in Section 1.3.1, the ISM is superior to the VM in that it is able to reproduce the propagation of information which is observed in real active matter systems, here paradigmatically represented by the occurrence of coordinate movements such as collective turns. In Section 1.1.1, we noticed that real turns in bird flocks are usually observed to present a well localized origin, as if in response to some punctual external stimulus, and only then to propagate throughout the group. However, nothing is said in the model itself about what it is that generates such collective changes in the first place. A compact and satisfactory description of this point requires the clarification of the following aspects:

- (i) Finding a suitable field to be included in the Hamiltonian (as in equation (2.1)) which represents the external perturbation. Such a field must be *local*, since we know that each collective turn exhibits a well localized origin.
- (ii) Identifying the correct pair of conjugated fields. We have seen that two sets of canonical variables (spins and velocities) appear in the Hamiltonian of ISM

in its on-lattice approximation, so that in principle an external field may be coupled to each of them.

- (iii) Defining some parameters of the external field which can be tuned in order to control the dynamical response of the system, or more specifically the direction assumed by the global polarization $\Phi = \frac{1}{v_0} \mathbf{V}$ in the ordered phase of the model.

In general, when a spatially uniform external field is applied to a spin system in its low temperature phase, whose ground state enjoys some kind of symmetry (think of the Ising or the Heisenberg models, for instance), then the only macroscopic effect we observe is that the order parameter aligns, sooner or later, to the direction selected by the external field. On the other hand, the effect of any local field (*e.g.* a field applied on a single perturbed site, $H \sim \mathcal{O}\left(\frac{1}{N}\right)$) becomes completely negligible in the thermodynamic limit, so that virtually nothing should happen at all.

When we deal with biological systems, where the number of components is certainly far from infinite, then the question arises of *how a system with $N < \infty$ responds to a local perturbation*. The aim of the following pages is to try to understand how a local field can change the direction of the order parameter $\mathbf{V}$ in the broken symmetry phase of a finite size ISM system.

3.1 Microscopic approach

Consider a system in its deeply polarized phase, and let $\hat{\mathbf{n}}$ indicate the direction of the polarization Φ , that is

$$\mathbf{V} = \frac{1}{N} \sum_{i=1}^N \mathbf{v}_i \equiv V \hat{\mathbf{n}} \quad (3.1)$$

(recall that, being $\|\mathbf{v}_i\| = v_0 \forall i$, then it is simply $\mathbf{V} = v_0 \Phi$). Let's write its microscopic Hamiltonian in analogy with the one we defined in equation (2.1), but this time choosing $\hat{\alpha}_i$ to be the direction of the field $\mathbf{H}_i$ coupled to the velocity $\mathbf{v}_i$, as shown in Figure 3.1. Specializing to the planar model as in equation (2.5), we can write

$$\mathcal{H} = -\frac{J}{2} \sum_{ij} n_{ij} \cos(\varphi_i - \varphi_j) + \sum_i \frac{s_i^2}{2\chi} - \sum_i H_i(t) \cos(\varphi_i - \alpha_i) - \sum_i h_i(t) s_i, \quad (3.2)$$

where this time φ_i is measured with respect to the direction assumed by the polarization $\mathbf{V}$ at the initial time taken as $t = 0$.

We have already derived the corresponding equations of motion, which can simply be read in (2.9) up to an angle shift: we get

$$\begin{cases} \dot{\varphi}_i(t) = \frac{s_i}{\chi} - h_i(t) \\ \dot{s}_i(t) = J \sum_j n_{ij} \sin(\varphi_j - \varphi_i) - H_i(t) \sin(\varphi_i - \alpha_i) - \frac{\eta}{\chi} s_i + \eta h_i(t) + \xi_i(t) \end{cases} \quad (3.3)$$

Notice, again, that the connectivity matrix n_{ij} is not yet a function of time, because the whole system is sitting on a lattice and the individuals are not allowed to move and change their relative positions.

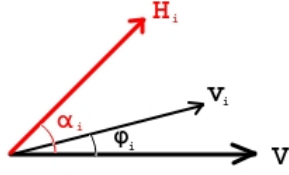


Figure 3.1. Relative positions of the polarization $\mathbf{V}$ at time $t = 0$, a generic velocity vector $\mathbf{v}_i$ and the external field $\mathbf{H}_i$ coupled to it.

The Spin Wave Approximation

When the system is in its ordered phase, then it is worthy decomposing each velocity vector into its components parallel and perpendicular to the global polarization $\mathbf{V}$,

$$\mathbf{v}_i = v_i^L \hat{\mathbf{n}} + \boldsymbol{\pi}_i, \quad (3.4)$$

where $\boldsymbol{\pi}_i$ is a $(D - 1)$ -dimensional vector. In the planar model, π_i is simply a scalar,

$$\pi_i = v_0 \sin \varphi_i. \quad (3.5)$$

The *Spin Wave Approximation* (SWA) consists in assuming all the angles φ_i to be small when the system is strongly polarized. This allows us to expand

$$\begin{aligned} \sin(\varphi_i - \alpha_i) &= \sin \varphi_i \cos \alpha_i - \cos \varphi_i \sin \alpha_i \\ &\simeq \varphi_i \cos \alpha_i - \left(1 - \frac{\varphi_i^2}{2}\right) \sin \alpha_i \\ &\simeq \varphi_i \cos \alpha_i - \sin \alpha_i, \end{aligned} \quad (3.6)$$

and

$$\sin(\varphi_i - \varphi_j) \simeq (\varphi_i - \varphi_j). \quad (3.7)$$

Applying this simplification to the system of equations in (3.3), we see that the factors coming from the expansion of the interactive term $\sum_{ij} n_{ij} \sin(\varphi_i - \varphi_j)$ in the equation for $\dot{s}_i(t)$ build up to give the discrete Laplacian, defined as

$$\begin{aligned} \Lambda_{ij} &= -n_{ij} + \delta_{ij} \sum_k n_{ik} \\ &\simeq -n_{ij} + \delta_{ij} n_c, \end{aligned} \quad (3.8)$$

where the last passage is exact for an homogeneous connectivity matrix n_{ij} , and n_c denotes the average number of nearest neighbours:

$$n_c \equiv \frac{1}{N} \sum_j n_{ij}. \quad (3.9)$$

In this approximation, the equations of motion become

$$\begin{cases} \dot{\varphi}_i(t) = \frac{s_i}{\chi} - h_i(t) \\ \dot{s}_i(t) = -J \sum_j \Lambda_{ij} \varphi_j - \frac{\eta}{\chi} s_i - \varphi_i H_i(t) \cos \alpha_i + H_i(t) \sin \alpha_i + \eta h_i(t) + \xi_i(t). \end{cases} \quad (3.10)$$

It should be born in mind that these equations are only useful for times t such that the SWA still holds or, in other terms, such that $|\varphi_i| \ll 1 \forall i$. We expect them to correctly describe the initial part of a collective turn, but surely not the final regime when the global velocity $\mathbf{V}$ saturates along the direction selected by the external field, should a collective turn occur at all.

3.1.1 Solution with external fields switched off

In the absence of the external fields $\mathbf{h}(t)$ and $\mathbf{H}(t)$ in the Hamiltonian (3.2), the system (3.10) simply reads

$$\begin{cases} \dot{\varphi}_i(t) - \frac{s_i}{\chi} = 0 \\ \dot{s}_i(t) + \frac{\eta}{\chi} s_i + J \sum_j \Lambda_{ij} \varphi_j = \xi_i(t) . \end{cases} \quad (3.11)$$

This can be cast into a more transparent form (as it was first done in [14]) by defining the $2N$ -dimensional vectors

$$\vec{\Psi} = \begin{pmatrix} \varphi \\ s \end{pmatrix}, \quad \vec{\mathbf{F}} = \begin{pmatrix} \mathbf{0} \\ \xi \end{pmatrix} \quad (3.12)$$

and the block matrix

$$\hat{L} = \begin{pmatrix} \mathbb{O} & -\frac{1}{\chi} \mathbb{I} \\ J\Lambda & \frac{\eta}{\chi} \mathbb{I} \end{pmatrix}, \quad (3.13)$$

where $\mathbb{I}$ and $\mathbb{O}$ are the unit matrix and the null matrix in N dimensions. This way the temporal evolution is simply governed by

$$\frac{d}{dt} \vec{\Psi} + \hat{L} \vec{\Psi} = \vec{\mathbf{F}}(t), \quad (3.14)$$

which admits a straightforward solution in terms of the propagators:

$$\vec{\Psi}(t) = e^{-t\hat{L}} \vec{\Psi}(0) + \int_0^t dt' e^{-(t-t')\hat{L}} \vec{\mathbf{F}}(t'). \quad (3.15)$$

Spectrum of $\hat{L}$

Solution (3.15) is simple but only formal: in order to evaluate the action of the exponential operator $e^{-t\hat{L}}$, we need first to find a basis where $\hat{L}$ is diagonal. In order to simplify the notation, we will borrow the *bra-ket* formalism from Quantum Mechanics and write

$$\varphi \rightarrow |\varphi\rangle, \quad s \rightarrow |s\rangle. \quad (3.16)$$

The *state* of the system will thus be described by a vector living in the tensor product space

$$\vec{\Psi} \rightarrow |\Psi\rangle = |\varphi\rangle \otimes |s\rangle, \quad (3.17)$$

which can be represented in the canonical basis

$$\mathcal{B}_{can} : |i\rangle \otimes |i\rangle, \quad (3.18)$$

where $|i\rangle$ is a N -dimensional vector. The components of $|\varphi\rangle$ and $|s\rangle$ along this basis (which will be called the *position basis*) are found by taking the scalar products

$$\begin{aligned}\varphi^i &= \langle i|\varphi\rangle \\ s^i &= \langle i|s\rangle .\end{aligned}\tag{3.19}$$

Looking at the definition (3.13) of the matrix $\hat{L}$, we notice it contains Λ , the discrete Laplacian defined in (3.8): let's thus call

$$\mathcal{B}_a : |a\rangle \otimes |a\rangle \tag{3.20}$$

the basis where Λ is diagonal, so that

$$\Lambda |a\rangle = \lambda_a |a\rangle . \tag{3.21}$$

We will denote the components of $|\varphi\rangle$ and $|s\rangle$ along this basis using the superscript a , as in

$$\varphi^a = \langle a|\varphi\rangle = \langle a|\left(\sum_i |i\rangle\langle i|\right)|\varphi\rangle = \sum_i \langle a|i\rangle \varphi^i \equiv \sum_i (U^{-1})^a_i \varphi^i , \tag{3.22}$$

where we have inserted a completeness sum in the second passage (*i.e.* a decomposition of the identity $\mathbb{I}$ into projectors) and defined the $N \times N$ matrix U , which enforces the change of basis

$$U : \mathcal{B}_a \rightarrow \mathcal{B}_{can} . \tag{3.23}$$

When written in the basis $\mathcal{B}_{can}$, the matrix U contains the eigenvectors of Λ as its *columns*; moreover, U can be used to diagonalize Λ as

$$\begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \end{pmatrix} = U^{-1} \Lambda U . \tag{3.24}$$

In the case of symmetric interactions, the condition $\Lambda_{ij} = \Lambda_{ji}$ makes U unitary ($U^{-1} = U^\dagger$); it is not superfluous to notice that the interaction network does not need to be regular (*e.g.* a cubic lattice) in order for this to occur.

Now we are ready to look for the eigenstates of the operator $\hat{L}$, which we will call

$$\begin{aligned}\hat{L} |l\rangle &= \omega_l |l\rangle \\ |l\rangle &= |l\rangle_\varphi \otimes |l\rangle_s .\end{aligned}\tag{3.25}$$

Let's work in the representation defined by the basis $\mathcal{B}_a$, which means taking both $|l\rangle_\varphi$ and $|l\rangle_s$ to be proportional to $|a\rangle$, the eigenstates of Λ ; without loss of generality, we set

$$|l\rangle = |a\rangle \otimes c(l) |a\rangle \xrightarrow{\mathcal{B}_a} \kappa \begin{pmatrix} 1 \\ c(l) \end{pmatrix} , \tag{3.26}$$

where κ is a normalization constant; the functional form of $c(l)$ is to be found by solving the eigenvalue equation (3.25), which now takes the form of a linear system:

$$\begin{pmatrix} \omega_l & \frac{1}{\chi} \\ -J\lambda_a & \omega_l - \frac{\eta}{\chi} \end{pmatrix} \begin{pmatrix} 1 \\ c(l) \end{pmatrix} = \vec{0}. \quad (3.27)$$

Taking the determinant of a block matrix is not, in general, a good idea; however, the spectrum can be found by eliminating $c(l)$ from the system, which leads to

$$-J\lambda_a - \omega_l^2 \chi + \eta \omega_l = 0, \quad (3.28)$$

solved by

$$\begin{aligned} \omega_l &= \gamma \pm i\omega_a \\ \gamma &\equiv \frac{\eta}{2\chi}, \quad \omega_a \equiv \sqrt{\frac{J\lambda_a}{\chi} - \gamma^2}. \end{aligned} \quad (3.29)$$

By using again (3.27), we finally find

$$c(l) = -\chi \omega_l, \quad (3.30)$$

which gives us the eigenvectors (here represented in the basis $\mathcal{B}_a$)

$$|l\rangle \xrightarrow{\mathcal{B}_a} \kappa \begin{pmatrix} 1 \\ -\chi \omega_l \end{pmatrix}. \quad (3.31)$$

Notice equation (3.29) shows a two-fold degeneracy for each fixed value of a (which was totally to be expected), so that we might as well set

$$\omega_l \equiv \omega_{\pm}(\lambda_a). \quad (3.32)$$

But we can go even further: for each fixed value of a (recall there are N such values), $\hat{L}$ can actually be represented by a 2×2 matrix,

$$\hat{L} \xrightarrow[a \text{ fixed}]{\mathcal{B}_a} L \equiv \begin{pmatrix} 0 & -\frac{1}{\chi} \\ J\lambda_a & \frac{\eta}{\chi} \end{pmatrix}, \quad (3.33)$$

which is diagonalized by

$$\begin{pmatrix} \omega_+ & 0 \\ 0 & \omega_- \end{pmatrix} = M^{-1} L M \quad (3.34)$$

where M is a non-unitary matrix (even when Λ were symmetrical, L would not be so). The matrix M and its inverse are given by

$$M = \kappa \begin{pmatrix} 1 & 1 \\ -\chi \omega_+ & -\chi \omega_- \end{pmatrix}, \quad M^{-1} = \frac{1}{2i\chi\kappa\omega_a} \begin{pmatrix} -\chi \omega_- & -1 \\ \chi \omega_+ & 1 \end{pmatrix}. \quad (3.35)$$

Temporal evolution in the $|l\rangle$ basis

Equation (3.15) takes a simple form when it is projected along the eigenstates $|l\rangle$ of the operator $\hat{L}$. Indeed, by expressing

$$\begin{aligned} |\Psi\rangle &= \left(\sum_l |l\rangle \langle l| \right) |\Psi\rangle = \sum_l \Psi^l |l\rangle \\ |F\rangle &= \sum_l F^l |l\rangle, \end{aligned} \quad (3.36)$$

it is enough to use the completeness relation

$$e^{-t\hat{L}} = \sum_l e^{-t\omega_l} |l\rangle\langle l| \quad (3.37)$$

to obtain the set of $2N$ decoupled equations

$$\Psi^l(t) = e^{-t\omega_l} \Psi^l(0) + \int_0^t dt' e^{-(t-t')\omega_l} F^l(t'). \quad (3.38)$$

In order to use this result, we need to:

- (i) Find the eigenvectors of Λ_{ij} (they depend on the choice of the connectivity matrix n_{ij}) and construct the matrices U (whose columns are the normalized eigenvectors) and U^{-1} ;
- (ii) Use the matrices U and U^{-1} on each of the two N -dimensional subspaces of the source term $\vec{\mathbf{F}}$ defined in equation (3.12), so as to move to the basis where the Laplacian is diagonal;
- (iii) Use the matrices M and M^{-1} on the 2-dimensional space at fixed a , so as to move to the basis where L is diagonal. This gives the source term components $F^l(t)$;
- (iv) Use equation (3.38) to compute the time evolution of each of the components $\Psi^l(t)$;
- (v) Apply the inverse of the two changes of basis used above (in reverse order), so as to translate the result back into the position basis: this gives $\varphi^i(t)$ and $s^i(t)$.

The above procedure is tedious but straightforward. Here we state the result for the angular field:

$$\begin{aligned} \varphi^i(t) = & \sum_{a,j=1}^N U_a^i (U^{-1})_j^a \int_0^t dt' e^{-\gamma(t-t')} \left\{ f_{(\varphi)}^j(t') \left[\frac{\gamma}{\omega_a} \sin \omega_a(t-t') + \cos \omega_a(t-t') \right] \right. \\ & \left. + \frac{1}{\chi \omega_a} f_{(s)}^j(t') \sin \omega_a(t-t') \right\}, \end{aligned} \quad (3.39)$$

where the external source term was set equal to

$$\vec{\mathbf{F}} = \begin{pmatrix} \mathbf{f}_{(\varphi)} \\ \mathbf{f}_{(s)} \end{pmatrix} \quad (3.40)$$

for future reference.

3.1.2 Wandering of the order parameter

The result we have just derived can be used to evaluate the typical time taken by the order parameter $\mathbf{V}$ to explore its broken symmetry manifold, when the system is driven by thermal noise only; we encountered this phenomenon in Section 2.3.3, when we were trying to measure numerically the correlation functions of the velocities

$\mathbf{v}_i(t)$ in the low temperature phase of the system. To this end, we will follow the steps of [14].

Recall every velocity vector $\mathbf{v}_i$ can be expressed, in SWA, as

$$\mathbf{v}_i = v_i^L \hat{\mathbf{n}} + \boldsymbol{\pi}_i , \quad (3.41)$$

where it is always, by definition of the average velocity,

$$\sum_i \boldsymbol{\pi}_i = 0 . \quad (3.42)$$

Now let's keep calling $\hat{\mathbf{n}}$ the direction assumed by the polarization $\mathbf{V}$ at time $t = 0$, but allow $\mathbf{V}(t)$ to change in time, so that it makes sense to separate the two components

$$\mathbf{V} = \frac{1}{N} \sum_i \mathbf{v}_i = (\mathbf{V} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} + \delta \mathbf{V}^\perp , \quad (3.43)$$

where

$$\begin{aligned} (\mathbf{V} \cdot \hat{\mathbf{n}})(t) &= \frac{1}{N} \sum_i v_i^L(t) \\ \delta \mathbf{V}^\perp(t) &= \frac{1}{N} \sum_i \boldsymbol{\pi}_i = \frac{v_0}{N} \sum_i \sin \varphi_i(t) \hat{\pi} , \end{aligned} \quad (3.44)$$

being $\hat{\pi}$ the direction perpendicular to $\hat{\mathbf{n}}$.

We define the wandering time τ_w as the time when

$$\left\langle \left| \delta \mathbf{V}^\perp(t) \right|^2 \right\rangle \sim \mathcal{O}(1) , \quad (3.45)$$

where the brackets denote the average over the noise. Using (3.44), we find

$$\left\langle \left| \delta \mathbf{V}^\perp(t) \right|^2 \right\rangle = \frac{1}{N^2} \sum_{ik} \langle \boldsymbol{\pi}_i \cdot \boldsymbol{\pi}_k \rangle , \quad (3.46)$$

which reduces to

$$\left\langle \left| \delta \mathbf{V}^\perp(t) \right|^2 \right\rangle = \frac{v_0^2}{N^2} \sum_{ik} \left\langle \varphi^i(t) \varphi^k(t) \right\rangle , \quad (3.47)$$

or equivalently, since $\mathbf{V} = v_0 \boldsymbol{\Phi}$,

$$\left\langle \left| \delta \boldsymbol{\Phi}^\perp(t) \right|^2 \right\rangle = \frac{1}{N^2} \sum_{ik} \left\langle \varphi^i(t) \varphi^k(t) \right\rangle \quad (3.48)$$

in the case with $D = 2$ and having applied the SWA for small angles φ^i .

The evolution of $\varphi^i(t)$ is given by (3.39); defining the source term as in (3.12) and choosing Λ_{ij} to be symmetric, so that the matrix U is unitary, we find

$$\varphi^i(t) = \sum_{a,j=1}^N U_a^i (U^\dagger)^a_j \int_0^t dt' \frac{e^{-\gamma(t-t')}}{\chi \omega_a} \xi^j(t') \sin \omega_a(t-t') . \quad (3.49)$$

The last expression can be further simplified by exploiting the properties of the interaction matrix Λ_{ij} . Indeed, using its definition (3.8) we see that by construction

$$\sum_j \Lambda_{ij} = \sum_j \left(-n_{ij} + \delta_{ij} \sum_k n_{ik} \right) = 0, \quad (3.50)$$

which means that the Laplacian matrix always admits a zero mode (*i.e.* a $\lambda_0 = 0$ eigenvalue) with constant right eigenvector. Normalizing it for a N -dimensional vector, we deduce that the matrix U (which contains the eigenvectors as its columns) must be such that

$$U_0^i = \frac{1}{\sqrt{N}} \quad \forall i, \quad (3.51)$$

or equivalently

$$(U^\dagger)_i^0 = \frac{1}{\sqrt{N}} \quad \forall i, \quad (3.52)$$

since the transposed of the matrix U contains the same (right) eigenvectors as its *rows*. Notice in general they do *not* coincide with the left eigenvectors of Λ ; they only do when the matrix Λ_{ij} is symmetric, because then the orthogonality condition reads

$$\mathbb{I} = U^{-1}U = U^\dagger U. \quad (3.53)$$

Then in particular we have

$$\sum_i U_a^i = \delta_{a0} \sqrt{N}. \quad (3.54)$$

When evaluating the sums over i and k prescribed in (3.48), the only surviving terms are those with $a = 0$: let's then isolate the zero mode term in

$$\begin{aligned} \varphi^i(t) &= \frac{1}{N} \sum_{j=1}^N \int_0^t dt' \frac{\xi^j(t')}{\eta} \left[1 - e^{-2\gamma(t-t')} \right] + (a \neq 0) \\ &= \frac{1}{N} \sum_{j=1}^N \int_0^t du \frac{\xi^j(t-u)}{\eta} \left[1 - e^{-2\gamma u} \right] + (a \neq 0), \end{aligned} \quad (3.55)$$

where we have used the expression for ω_a given in (3.29). Using the property

$$\left\langle \xi^j(t-u_1) \xi^k(t-u_2) \right\rangle = 2T\eta \delta_{jk} \delta(u_1 - u_2), \quad (3.56)$$

it is easy to evaluate

$$\left\langle \varphi^i(t) \varphi^k(t) \right\rangle = \frac{2T}{N\eta} \left[t - \frac{1}{\gamma} \left(1 - e^{-2\gamma t} \right) + \frac{1}{4\gamma} \left(1 - e^{-4\gamma t} \right) \right]. \quad (3.57)$$

We notice this result does not depend on the indices i, k ; moreover, the only term growing in time is the first. For $t \gg \frac{1}{\gamma}$ and using (3.48), we can calculate

$$\begin{aligned} \left\langle \left| \delta \Phi^\perp(t) \right|^2 \right\rangle &\sim \frac{2T}{\eta N} t \\ &\equiv \frac{2Dt}{\eta}, \end{aligned} \quad (3.58)$$

where we defined the *diffusion coefficient*

$$D \equiv \frac{T}{N} . \quad (3.59)$$

Setting this quantity of $\mathcal{O}(1)$ finally gives the wandering time

$$\tau_w \sim \frac{\eta N}{2T} . \quad (3.60)$$

3.1.3 Solution in the presence of local external fields

Switching on the external fields $\mathbf{h}(t)$ and $\mathbf{H}(t)$ in the Hamiltonian (3.2) leads us to the full system (3.10), which we will try to solve perturbatively building on the free case.

First we notice that, by defining a modified Laplacian matrix

$$\begin{aligned} \tilde{\Lambda}_{ij} &\equiv \Lambda_{ij} + \delta_{ij} \frac{H_i \cos \alpha_i}{J} \\ &= -n_{ij} + \delta_{ij} \left(n_c + \frac{H_i \cos \alpha_i}{J} \right) , \end{aligned} \quad (3.61)$$

the system (3.10) can be cast in a form which is completely analogous to (3.11),

$$\begin{cases} \dot{\varphi}_i - \frac{s_i}{\chi} = -h_i \\ \dot{s}_i + \frac{\eta}{\chi} s_i + J \sum_j \tilde{\Lambda}_{ij} \varphi_j = \eta h_i + H_i \sin \alpha_i + \xi_i , \end{cases} \quad (3.62)$$

with $\tilde{\Lambda}$ in place of Λ and the source term set equal to

$$\vec{\mathbf{F}} = \begin{pmatrix} -\mathbf{h} \\ \eta \mathbf{h} + \mathbf{H} \sin \alpha + \boldsymbol{\xi} \end{pmatrix} . \quad (3.63)$$

The solution is already known (equation (3.39)), provided that we compute the eigenvalues and eigenvectors of the new Laplacian matrix.

Corrections to the spectrum of Λ

We will start from the unperturbed spectrum of Λ (which in the following discussion is assumed to be symmetric) and use Perturbation Theory to compute its corrections when a small external field is applied. We thus consider

$$\begin{aligned} \tilde{\Lambda} &= \Lambda + \delta\Lambda \\ (\delta\Lambda)_{ij} &= \delta_{ij} \frac{H_i \cos \alpha_i}{J} , \end{aligned} \quad (3.64)$$

where the field $\mathbf{H}$ itself plays the role of the small parameter. Using again the bra-ket notation, the new eigenvalues and eigenvectors are given in Perturbation Theory by

$$\begin{aligned} \tilde{\lambda}_a &= \lambda_a + \langle a | \delta\Lambda | a \rangle \\ |\tilde{a}\rangle &= |a\rangle + \sum_{b \neq a} \frac{\langle b | \delta\Lambda | a \rangle}{\lambda_a - \lambda_b} |b\rangle . \end{aligned} \quad (3.65)$$

When represented in the position basis $\mathcal{B}_{can}$, the first equation becomes

$$\begin{aligned}\tilde{\lambda}_a &= \lambda_a + \sum_{ijk} (U^\dagger)^a_i (\delta\Lambda)^{ik} \underbrace{\delta_{kj}}_{\text{flat metric}} U^j_a \\ &= \lambda_a + \sum_{ij} (U^\dagger)^a_i \frac{H_i \cos \alpha_i}{J} \delta^i_j U^j_a ,\end{aligned}\tag{3.66}$$

while the eigenvectors get corrected as

$$\begin{aligned}\tilde{U}^i_a &= U^i_a + \sum_{b \neq a} \sum_{jkl} \frac{(U^\dagger)^b_j (\delta\Lambda)^{jl} \delta_{lk} U^k_a}{\lambda_a - \lambda_b} U^i_b \\ &= U^i_a + \sum_{b \neq a} \sum_{jk} \frac{(U^\dagger)^b_j (H_j \cos \alpha_j) \delta^j_k U^k_a}{J(\lambda_a - \lambda_b)} U^i_b .\end{aligned}\tag{3.67}$$

A particularly interesting case is when the external field is uniform, $\mathbf{H}_i = \mathbf{H} \forall i$. Then, using $UU^\dagger = \mathbb{I}$, the expression (3.66) above reduces to

$$\tilde{\lambda}_a = \lambda_a + \frac{H \cos \alpha}{J} ,\tag{3.68}$$

which shows that the effect of a uniform external field is to add a *mass* to the Laplacian: this will be seen more clearly in Section 3.2, when we will derive the same result using a field approach, and a mass term will actually appear into the Hamiltonian due to the presence of the external field.

Placing instead a *local* field H_p on site p only, we would find

$$\begin{aligned}\tilde{\lambda}_a &= \lambda_a + (U^\dagger)^a_p U^p_a \frac{H_p \cos \alpha_p}{J} \\ \tilde{U}^i_a &= U^i_a + \sum_{b \neq a} (U^\dagger)^b_p U^p_a \frac{H_p \cos \alpha_p}{J(\lambda_a - \lambda_b)} U^i_b .\end{aligned}\tag{3.69}$$

Notice the correction to the eigenvalues is of $\mathcal{O}\left(\frac{1}{N}\right)$, because the orthogonality condition imposes

$$\sum_{k=1}^N (U^\dagger)^a_k U^k_a = 1 ,\tag{3.70}$$

hence

$$(U^\dagger)^a_p U^p_a \sim \mathcal{O}\left(\frac{1}{N}\right) .\tag{3.71}$$

Time dependence of the polarization angle

Given a polarization vector $\mathbf{V}(t=0) = V\hat{\mathbf{n}}$, we want to use the results we have derived so far to predict the behaviour of the quantity $\Phi(t)$, namely the angle formed by the polarization $\mathbf{V}(t)$ with $\hat{\mathbf{n}}$ as it moves in response to the application of an external field. To do so, we first decompose $\mathbf{V}(t)$ into its components parallel to $\hat{\mathbf{n}}$

and perpendicular to it (*i.e.* $\hat{\boldsymbol{\pi}}$), as we did in (3.44); then we apply the SWA to obtain

$$\begin{aligned}\delta\mathbf{V}^\perp(t) &= \frac{1}{N} \sum_i \boldsymbol{\pi}_i = \frac{v_0}{N} \sum_i \sin \varphi_i(t) \hat{\boldsymbol{\pi}} \\ &\simeq \frac{v_0}{N} \sum_i \varphi_i(t) \hat{\boldsymbol{\pi}} .\end{aligned}\tag{3.72}$$

Now, we define the polarization angle

$$\Phi(t) \equiv \frac{1}{N} \sum_i \langle \varphi_i(t) \rangle ,\tag{3.73}$$

so that averaging over the noise we obtain

$$\langle |\delta\mathbf{V}^\perp(t)| \rangle \simeq v_0 \Phi(t) .\tag{3.74}$$

By construction, such an approach will yield sound results only as long as the angle $\Phi(t)$ is small. In order to evaluate it, we use the full expression (3.39) for $\varphi_i(t)$, where however the matrices U and U^{-1} are to be replaced by $\tilde{U}$ and $\tilde{U}^\dagger$, and the characteristic frequency ω_a by $\tilde{\omega}_a$ (which contains the corrected eigenvalues). The components of the source term $\vec{\mathbf{F}}$ are to be read in (3.63).

Fortunately, when summing over i to compute $\Phi(t)$ some simplifications apply, thanks again to the symmetry of Λ . Indeed, being $\lambda_0 = 0$, the orthogonality condition (3.54) can be used on the corrected eigenvectors (3.67) to give

$$\begin{aligned}\sum_i \tilde{U}_a^i &= \delta_{a0} \sqrt{N} + (1 - \delta_{a0}) \sum_j \frac{H_j \cos \alpha_j}{J \lambda_j} U_a^j \\ &= \delta_{a0} \sqrt{N} + \mathcal{O}(H) ,\end{aligned}\tag{3.75}$$

which clearly shows that the contributions of all the modes $a \neq 0$ are already at least of $\mathcal{O}(H)$. Since the expression (3.39) for $\varphi_i(t)$ contains a convolution with the external fields $\mathbf{h}$ and $\mathbf{H}$, we can tell from the start that only the mode $a = 0$ will determine the dominant contributions for small fields, all the other terms yielding corrections of $\mathcal{O}(H^2)$, $\mathcal{O}(hH)$. Thus, using (3.73), at leading order in H we are left with

$$\begin{aligned}\Phi(t) &= \frac{1}{N} \sum_{j=1}^N \int_0^t dt' e^{-\gamma(t-t')} \left\{ -h^j(t') \left[\frac{\gamma}{\tilde{\omega}_0} \sin \tilde{\omega}_0(t-t') + \cos \tilde{\omega}_0(t-t') \right] \right. \\ &\quad \left. + \frac{\eta h_j(t') + H_j(t') \sin \alpha_j}{\chi \tilde{\omega}_0} \sin \tilde{\omega}_0(t-t') \right\} .\end{aligned}\tag{3.76}$$

We still have to compute the correction to ω_0 . To do so, we first use (3.66), recalling that $\lambda_0 = 0$, to find

$$\tilde{\lambda}_0 = \frac{1}{N} \sum_i \frac{H_i \cos \alpha_i}{J} \sim \mathcal{O}(H) ;\tag{3.77}$$

actually, if we pick a local field H_p on site p , we even have

$$\tilde{\lambda}_0 \sim \mathcal{O}\left(\frac{H}{N}\right). \quad (3.78)$$

Now we apply the definition (3.29) to compute

$$\begin{aligned} \tilde{\omega}_0 &= \sqrt{\frac{J\tilde{\lambda}_0}{\chi} - \gamma^2} = i\gamma\sqrt{1 - \frac{J\tilde{\lambda}_0}{\chi\gamma^2}} \\ &\simeq i\gamma(1 + \mathcal{O}(H)). \end{aligned} \quad (3.79)$$

Finally, we can substitute $\tilde{\omega}_0$ back into (3.76), discarding the correction of $\mathcal{O}(H)$; at leading order and for a local field on site p , we find

$$\Phi(t) = \frac{1}{N} \int_0^t dt' \left\{ \frac{H_p(t')}{\eta} \sin \alpha_p \left[1 - e^{-2\gamma(t-t')} \right] - h_p(t') e^{-2\gamma(t-t')} \right\}. \quad (3.80)$$

Case 1: $\theta(t)$ -like time dependence

We restricted to a single site p the spatial dependence of the fields $\mathbf{h}(t)$, $\mathbf{H}(t)$; we still have to specify their time dependence. We will analyze thoroughly two specific cases (a stepfunction and a narrow impulse) that will be useful in the numerical study of the next Chapter; our current aim is to find an explicit expression for the response of the system to such perturbations, *i.e.* a prediction for the time dependence of the polarization angle $\Phi(t)$.

First, let's choose two external fields of the form

$$\begin{aligned} H_p(t) &= H_p \theta(t) \\ h_p(t) &= h_p \theta(t), \end{aligned} \quad (3.81)$$

where $\theta(t)$ is the Heavyside distribution. Integrating the right hand side of equation (3.80) gives

$$\Phi(t) = \frac{H_p}{\eta} \sin \alpha_p \frac{t}{N} - \frac{1}{2\gamma N} \left[h_p - \frac{H_p}{\eta} \sin \alpha_p \right] (1 - e^{-2\gamma t}). \quad (3.82)$$

For times $t \gg \frac{1}{\gamma}$, the only term that still counts is the first. Notably, equations (3.80) and (3.82) are telling us that *coupling a field $\mathbf{h}$ to the spin $\mathbf{s}$ in the Hamiltonian is not an effective way of inducing a collective turn.*

In our numerical simulations, we will replace the stepfunction with a smooth, albeit steep, ramp-like field of the form

$$H_p(t) = \frac{A_0}{2} \left[1 + \tanh \left(\frac{t - t_0}{\tau_0} \right) \right]. \quad (3.83)$$

Discarding a rapid initial transient, we thus expect to observe a linear regime where the polarization angle grows like

$$\Phi(t) \sim \frac{A_0 \sin \alpha_p}{\eta N} t, \quad (3.84)$$

followed by a saturation, which corresponds to the fact that $\Phi(t)$ is approaching the direction of the external field. Equation (3.80) does not predict this behaviour, as it is far beyond the regime of validity of the Spin Wave Approximation.

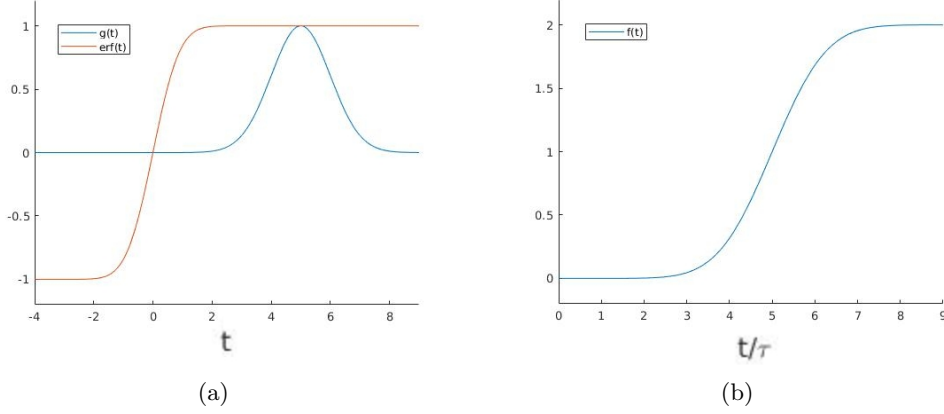


Figure 3.2. (a) Time dependence of the applied external field ($g(t)$ in the legend) and shape of the errorfunction defined in (3.89). By choosing $t_0 = 5\tau$, we are making sure that the Gaussian impulse is wholly contained in the semiaxis $t > 0$. (b) The function $f(t)$ defined in (3.93), plotted against $\frac{t}{\tau}$.

Case 2: $\delta(t)$ -like time dependence

Now that we know better, we set $h_p(t) = 0$ from the start and we choose for the field $H_p(t)$ an impulse-like time dependence in the form

$$H_p(t) = A_0 e^{-\frac{(t-t_0)^2}{2\tau^2}}. \quad (3.85)$$

Calling

$$A'_0 \equiv \frac{A_0 \sin \alpha_p}{\eta N}, \quad (3.86)$$

equation (3.80) reads

$$\Phi(t) = \int_0^t dt' A'_0 e^{-\frac{(t'-t_0)^2}{2\tau^2}} \left[\underbrace{1}_{I_1} - \underbrace{e^{-2\gamma(t-t')}}_{I_2} \right]. \quad (3.87)$$

Let's study the two contributions separately. The first gives

$$\begin{aligned} I_1 &= \int_0^t dt' A'_0 e^{-\frac{(t'-t_0)^2}{2\tau^2}} \\ &= A'_0 \sqrt{\frac{\pi}{2}} \tau \left[\text{erf}\left(\frac{t-t_0}{\sqrt{2}\tau}\right) + \text{erf}\left(\frac{t_0}{\sqrt{2}\tau}\right) \right], \end{aligned} \quad (3.88)$$

where we made use of the errorfunction (Figure 3.2(a)) defined as

$$\text{erf}(t) \equiv \frac{2}{\sqrt{\pi}} \int_0^t e^{-x^2} dx, \quad (3.89)$$

which is the integral of a Gaussian distribution with variance $\sigma^2 = \frac{1}{2}$.

In our numerical simulations, we will choose

$$t_0 \equiv 5\tau \quad (3.90)$$

so as to be able to appreciate the full Gaussian impulse (see Figure 3.2(a)). This means that the second errorfunction in equation (3.88) is basically at its saturation,

$$\operatorname{erf}\left(\frac{t_0}{\sqrt{2}\tau}\right) = \operatorname{erf}(5\sigma) \simeq 1, \quad (3.91)$$

and we can safely approximate

$$I_1 \simeq A'_0 \sqrt{\frac{\pi}{2}} \tau \left\{ 1 + \operatorname{erf} \left[\frac{1}{\sqrt{2}} \left(\frac{t}{\tau} - 5 \right) \right] \right\}. \quad (3.92)$$

The function between braces,

$$f(t; \tau) \equiv 1 + \operatorname{erf} \left[\frac{1}{\sqrt{2}} \left(\frac{t}{\tau} - 5 \right) \right], \quad (3.93)$$

saturates to the value 2 (see Figure 3.2(b)) when

$$\begin{aligned} \frac{t}{\tau} &\sim 5 + 3\sigma \sim 7 \\ &\rightarrow t \sim 7\tau \end{aligned} \quad (3.94)$$

and its slope in the center of the impulse, $t = t_0 = 5\tau$, is

$$\left. \frac{\partial f}{\partial t} \right|_{t=t_0} = \sqrt{\frac{2}{\pi}} \frac{1}{\tau}, \quad (3.95)$$

which makes the slope of the impulse (3.92) (*i.e.* the rate of change of the angle of the polarization $\Phi(t)$) substantially *independent of the choice of τ , and controlled instead by the amplitude A_0 of the signal only.*

There remains to show that the contribution I_2 in equation (3.87) is irrelevant, and can thus be discarded. To do so, notice it can be rewritten as

$$\begin{aligned} I_2 &= -A'_0 \int_0^t dt' e^{-2\gamma(t-t')} e^{-\frac{(t'-t_0)^2}{2\tau^2}} \\ &= -A'_0 e^{-2\gamma t} \underbrace{e^{2\gamma(t_0+\gamma\tau^2)}}_A \underbrace{\int_0^t dt' e^{-\frac{[t'-(t_0+2\gamma\tau^2)]^2}{2\tau^2}}}_B. \end{aligned} \quad (3.96)$$

Factor A is quite big, but factor B is another errorfunction. Having chosen $t_0 = 5\tau$, then it is always

$$(t_0 + 2\gamma\tau^2) \gg 3\tau, \quad (3.97)$$

which means that the Gaussian function which is being integrated is peaked in a region well distant from the origin (*i.e.* it looks like that in Figure 3.2(a)). For $t > 0$, we conclude that factor B only begins to differ substantially from zero when

$$t \sim (t_0 + 2\gamma\tau^2), \quad (3.98)$$

and it stops growing when it saturates to the value $\tau\sqrt{2\pi}$. By the time this happens, the factor $e^{-2\gamma t}$ in front of A and B has long since suppressed anything else.

3.2 Field approach

Following the same steps as in the microscopic approach, here we analyze the response of the system from a different perspective.

To derive a differential equation for the dynamics of ISM in $d = 3$ and with $D = 2$, let's first start from its microscopic Hamiltonian (3.2), which we rewrite here after having set $\alpha_i = \alpha \forall i$ for simplicity:

$$\mathcal{H} = -\frac{J}{2} \sum_{ij} n_{ij} \cos(\varphi_i - \varphi_j) + \sum_i \frac{s_i^2}{2\chi} - \sum_i H_i(t) \cos(\varphi_i - \alpha) - \sum_i h_i(t) s_i. \quad (3.99)$$

We wish to step to a coarse-grained description of the system in terms of the fields $\varphi(\mathbf{x}, t)$ and $s(\mathbf{x}, t)$. Using again the Spin Wave Approximation, we have

$$\begin{aligned} \cos(\varphi_i - \alpha) &\simeq \left(1 - \frac{1}{2}\varphi_i^2\right) \cos \alpha + \varphi_i \sin \alpha \\ \cos(\varphi_i - \varphi_j) &\simeq 1 - \frac{1}{2}(\varphi_i - \varphi_j)^2 \rightarrow 1 - \frac{a^2}{2} |\nabla \varphi|^2, \end{aligned} \quad (3.100)$$

where we called a the lattice spacing. Recalling that we defined n_c to be the average number of interacting neighbours (see equation (3.9)) and applying the above approximations to the Hamiltonian (3.99), we find

$$\mathcal{H} = \int \frac{d^3x}{a^3} \left\{ \frac{Jn_c a^2}{2} |\nabla \varphi|^2 + \frac{1}{2} \cos \alpha H(\mathbf{x}, t) \varphi^2 + \frac{s^2}{2\chi} - \sin \alpha H(\mathbf{x}, t) \varphi - h(\mathbf{x}, t) s \right\}, \quad (3.101)$$

which shows that the component of the external field $\mathbf{H}$ parallel to the global polarization $\mathbf{V}$ (see Figure 3.1) is actually a *mass* for the field φ . The other component, $H \sin \alpha$, and the field h couple linearly to φ and s respectively.

Again, we stress that $\mathcal{H}$ has been derived under the SWA, hence it has to be considered as a low temperature expansion of ISM. In fact, a field theory can still be constructed once we have relaxed this assumption [10], but it is in general more complicated as it would contain new, non-Gaussian terms.

The equations of motion can be derived in the usual way from the Hamiltonian by taking the variations

$$\begin{cases} \dot{\varphi} = \frac{\delta \mathcal{H}}{\delta s} \\ \dot{s} = -\frac{\delta \mathcal{H}}{\delta \varphi} - \eta \frac{\delta \mathcal{H}}{\delta s} + \xi, \end{cases} \quad (3.102)$$

which gives

$$\begin{cases} \dot{\varphi} = \frac{s}{\chi} - h(\mathbf{x}, t) \\ \dot{s} = Jn_c a^2 \nabla^2 \varphi - \cos \alpha H(\mathbf{x}, t) \varphi + \sin \alpha H(\mathbf{x}, t) - \eta \left(\frac{s}{\chi} - h(\mathbf{x}, t) \right) + \xi. \end{cases} \quad (3.103)$$

This can easily be turned into a second order partial differential equation in φ by deriving again the first equation by t :

$$\chi \ddot{\varphi} + \eta \dot{\varphi} - Jn_c a^2 \nabla^2 \varphi + \cos \alpha H(\mathbf{x}, t) \varphi = \sin \alpha H(\mathbf{x}, t) - \chi \dot{h}(\mathbf{x}, t) + \xi(\mathbf{x}, t). \quad (3.104)$$

First of all, we set $h(\mathbf{x}, t) = 0$ as we are interested in studying the collective response induced by the field $H(\mathbf{x}, t)$ alone (indeed, we have seen that h alone is ineffective¹). We thus study the PDE

$$\chi\ddot{\varphi} + \eta\dot{\varphi} - Jn_c a^2 \nabla^2 \varphi + \cos \alpha H(\mathbf{x}, t) \varphi = \sin \alpha H(\mathbf{x}, t) + \xi(\mathbf{x}, t) . \quad (3.105)$$

As a second simplification, let's discard momentarily the term in $H\varphi$, so that our PDE becomes simply

$$\chi\ddot{\varphi} + \eta\dot{\varphi} - Jn_c a^2 \nabla^2 \varphi = \sin \alpha H(\mathbf{x}, t) + \xi(\mathbf{x}, t) , \quad (3.106)$$

which can be cast in the familiar form

$$\begin{aligned} \mathcal{D}_{\mathbf{x},t} \varphi(\mathbf{x}, t) &= f(\mathbf{x}, t) \\ \mathcal{D}_{\mathbf{x},t} &= \chi \partial_{tt}^2 + \eta \partial_t - L_{\mathbf{x}} , \end{aligned} \quad (3.107)$$

where we recognize

$$\begin{aligned} f(\mathbf{x}, t) &= \sin \alpha H(\mathbf{x}, t) + \xi(\mathbf{x}, t) \\ L_{\mathbf{x}} &= Jn_c a^2 \nabla_{\mathbf{x}}^2 . \end{aligned} \quad (3.108)$$

This passage is quite intuitive in the spirit of Linear Response Theory, which we revise in Appendix A: indeed, the response function is in general evaluated with the external field set equal to zero (see equation (A.1)). However, it will be formally justified in Section 3.2.2.

The propagator can be computed with standard methods, as we did in Appendix A: its expression can be read in equation (A.26). The result is

$$G(\mathbf{k}, t) = e^{-\gamma t} \frac{\sin \omega_{\mathbf{k}} t}{\chi \omega_{\mathbf{k}}} , \quad (3.109)$$

where the dispersion relation is given by

$$\begin{aligned} \omega_{\pm} &= -i\gamma \pm \omega_{\mathbf{k}} \\ \omega_{\mathbf{k}} &= \sqrt{\frac{\lambda_{\mathbf{k}}}{\chi} - \gamma^2} \\ &= \sqrt{\frac{Jn_c a^2 k^2}{\chi} - \gamma^2} . \end{aligned} \quad (3.110)$$

By defining

$$\begin{aligned} c_s &\equiv \sqrt{\frac{Jn_c}{\chi}} a \\ k_0 &\equiv \frac{\eta}{2a\sqrt{Jn_c\chi}} = \frac{\gamma}{c_s} , \end{aligned} \quad (3.111)$$

¹Actually, by inspecting the second order PDE (3.104), we notice that h never appears on the right hand side, while only $\dot{h}$ does. We might conjecture that by applying a time varying field $h(t)$ we should be able to produce a more conspicuous effect on the system (but this is out of the scope of this work).

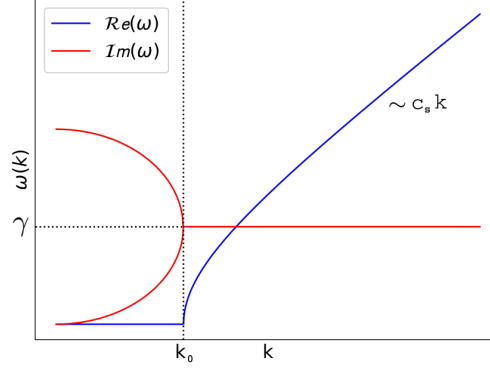


Figure 3.3. Amplitude of the real and imaginary parts of the dispersion relation (3.112).

where c_s is the *sound speed*, we notice that the dispersion relation (3.110) can be rewritten as

$$\omega_{\pm} = -i\gamma \pm c_s \sqrt{k^2 - k_0^2}, \quad (3.112)$$

whose real and imaginary parts are plotted in Figure 3.3. For large k 's (*underdamped regime*), the propagation becomes linear with speed c_s and constant attenuation γ .

3.2.1 Polarization angle

We choose to work in $\mathbf{k}$ space because this makes it easier to evaluate the collective response, which is the polarization angle defined in (3.73), whose continuum space version is

$$\begin{aligned} \Phi(t) &= \frac{1}{\mathcal{V}} \int_{\mathcal{V}} d^d x \langle \varphi(\mathbf{x}, t) \rangle \\ &= \frac{1}{\mathcal{V}} \langle \varphi(\mathbf{k} = \mathbf{0}, t) \rangle. \end{aligned} \quad (3.113)$$

Here $\mathcal{V}$ is the volume occupied by the entire flock, and it is finite. We can still work with continuous wavevectors $\mathbf{k}$ for big enough $\mathcal{V}$ provided that, for any function $f(\mathbf{k})$, we bear in mind the underlying approximation

$$\sum_{\mathbf{k}} f(\mathbf{k}) \simeq \frac{\mathcal{V}}{(2\pi)^d} \int d^d k f(\mathbf{k}). \quad (3.114)$$

Using again the propagator we computed in Appendix A, we can evaluate the average response of the system as

$$\langle \varphi(\mathbf{k}, t) \rangle = \int_0^t dt' \langle G(\mathbf{k}, t - t') f(\mathbf{k}, t') \rangle, \quad (3.115)$$

where the effect of the average is to eliminate the noise term $\xi(\mathbf{x}, t)$, so that

$$\begin{aligned} \Phi(t) &= \frac{1}{\mathcal{V}} \int_0^t dt' G(\mathbf{k} = \mathbf{0}, t - t') \sin \alpha H(\mathbf{k} = \mathbf{0}, t') \\ &= \frac{1}{\mathcal{V}} \int_0^t dt' \frac{H(\mathbf{k} = \mathbf{0}, t')}{\eta} \sin \alpha [1 - e^{-2\gamma(t-t')}] , \end{aligned} \quad (3.116)$$

where we used the fact that $\omega_{\mathbf{k}=0} = i\gamma$. We stress that equation (3.116) is the exact analogous of equation (3.80), which we derived using the microscopic approach; in that case, having placed a single spin variable on each vertex of a regular cubic lattice, it was actually $\mathcal{V} \sim N$. To make contact with expression (3.82), for instance, we can choose a local field and give it a stepfunction-like time dependence,

$$\begin{aligned} H(\mathbf{x}, t) &= H_0 \delta(\mathbf{x} - \mathbf{x}_0) \theta(t) \\ &\rightarrow H(\mathbf{k}, t) = H_0 e^{i\mathbf{k} \cdot \mathbf{x}_0} \theta(t) . \end{aligned} \quad (3.117)$$

Then we can use equation (3.116) to evaluate²

$$\Phi(t) = \frac{H_0}{\eta \mathcal{V}} \sin \alpha \left[t + \frac{1}{2\gamma} (1 - e^{-2\gamma t}) \right] . \quad (3.118)$$

3.2.2 Consistency check

There remains to prove that the term in $H\varphi$ we have neglected in passing from (3.105) to (3.106) only brings corrections of $\mathcal{O}(H^2)$ or higher to the final result. To lighten the notation, consider the 1-dimensional problem

$$\mathcal{D}_x \varphi(x) = f(x) \quad (3.119)$$

where $\mathcal{D}_x$ is a linear differential operator; this will work as a prototype for the full problem (3.107). By definition,

$$\mathcal{D}_x e^{ikx} = G^{-1}(k) e^{ikx} , \quad (3.120)$$

where $G(k)$ is the propagator in k -space: here the solution of (3.119) simply reads

$$\varphi(k) = G(k) f(k) . \quad (3.121)$$

Now imagine we correct (3.119) by adding an $\mathcal{O}(H)$ term to the differential operator on the left hand side:

$$\mathcal{D}_x \varphi(x) + H g(x) \varphi(x) = f(x) . \quad (3.122)$$

Here $g(x)$ can be any function of x ; if we want it to represent a local perturbation, as we do in (3.117), we can choose $g(x)$ to be a finite width approximation of a δ -like distribution, such as a narrow Gaussian impulse.

Let's try to solve (3.122) in Fourier space, where it takes the form

$$\int dk e^{ikx} G^{-1}(k) \varphi(k) + H g(x) \int dk e^{ikx} \varphi(k) = \int dk e^{ikx} f(k) . \quad (3.123)$$

Now, let's apply the inverse Fourier transform $\int dx e^{-iqx}$ to both sides of the equation, so as to move back into real space:

$$\begin{aligned} &\int dk dx e^{i(k-q)x} G^{-1}(k) \varphi(k) + H \int dk \left(\int dx g(x) e^{i(k-q)x} \right) \varphi(k) \\ &= \int dk dx e^{i(k-q)x} f(k) , \end{aligned} \quad (3.124)$$

²Notice that performing the same time convolution in ω space, via (A.29), would not have been as easy: in fact, if $\Phi(t) \sim t$ in real space, then we must conclude that $\hat{\Phi}(\omega)$ diverges in Fourier space.

hence

$$\begin{aligned} G^{-1}(q)\varphi(q) + H \int dk g(k-q)\varphi(k) &= f(q) \\ \varphi(q) &= G(q)f(q) - H \int dk G(q)g(k-q)\varphi(k) . \end{aligned} \tag{3.125}$$

The last expression can be iterated to give

$$\begin{aligned} \varphi(q) &= G(q)f(q) - H \int dk G(q)g(k-q) [G(k)f(k) + \mathcal{O}(H)] \\ &= G(q)f(q) - H \int dk G(q)g(k-q)G(k)f(k) + \mathcal{O}(H^2) . \end{aligned} \tag{3.126}$$

This seems quite tautological, but it becomes useful when we consider an $f(x)$ which is already of $\mathcal{O}(H)$ in itself, as it is the case in the problem (3.105) we are trying to solve. Then we would find

$$\varphi(q) = G(q)f(q) + \mathcal{O}(H^2) , \tag{3.127}$$

which justifies the approximation we made in stepping from equation (3.105) to (3.106).

3.3 Summary

We recap here for future reference the main results of this Chapter, as far as the response of the polarization $\Phi(t)$ to a local external perturbation is concerned.

The first conclusion we reached is that the correct way to include a local external field into the Hamiltonian of the planar ISM, and which is able to induce a collective turn in the system, is to couple it to the velocity (equation (3.82)), so that the new Hamiltonian looks like

$$\mathcal{H} = -\frac{J}{2} \sum_{ij} n_{ij} \cos(\varphi_i - \varphi_j) + \sum_i \frac{s_i^2}{2\chi} - H_p(t) \cos(\varphi_p - \alpha) \quad (3.128)$$

or, in its continuous space version (under the Spin Wave Approximation and for short times t since the external field has been switched on),

$$\mathcal{H} = \int \frac{d^3x}{a^3} \left\{ \frac{Jn_c a^2}{2} |\nabla \varphi|^2 + \frac{1}{2} \cos \alpha H(\mathbf{x}, t) \varphi^2 + \frac{s^2}{2\chi} - \sin \alpha H(\mathbf{x}, t) \varphi \right\}. \quad (3.129)$$

In this latter case, we take the space dependence of the field $H(\mathbf{x}, t)$ to be $\delta(\mathbf{x})$ -like; hence, there remains to control its *amplitude* and its *time dependence*.

In the next Chapter we will systematically study the effects of varying these two parameters separately in numerical simulations of the planar ISM. A way to do so is to adopt ramp-like fields in the form

$$H_p(t) = \frac{A_0}{2} \left[1 + \tanh \left(\frac{t - t_0}{\tau_0} \right) \right], \quad (3.130)$$

for which the time dependence is lost and the amplitude A_0 is the only relevant parameter, or instead impulse-like fields in the form

$$H_p(t) = A_0 e^{-\frac{(t-t_0)^2}{2\tau^2}}, \quad (3.131)$$

whose temporal width τ can be tuned to observe the effect of signals with different durations.

In these two cases, we found the response of the polarization angle of the system to be given respectively by the expressions

$$\Phi(t) \sim \frac{A_0 \sin \alpha}{\eta N} t \quad (3.132)$$

for a ramp-like perturbation, and

$$\Phi(t) \sim \frac{A_0 \sin \alpha}{\eta N} \sqrt{\frac{\pi}{2}} \tau \left[\operatorname{erf} \left(\frac{t - t_0}{\sqrt{2}\tau} \right) + \operatorname{erf} \left(\frac{t_0}{\sqrt{2}\tau} \right) \right] \quad (3.133)$$

for a Gaussian-like perturbation.

Chapter 4

Numerical study of the response on-lattice

The aim of this Chapter is twofold:

- (i) Verifying if and to what extent the predictions we formulated in Chapter 3 about the dynamical response of the system to external perturbations are confirmed by a numerical simulation of ISM;
- (ii) Building a numerical protocol which *induces a collective turn in the simulated flock*, and controlling the speed and the direction of such turns.

4.1 Choice of parameters

We begin by choosing the parameters to feed into our simulation of ISM, that we realize using the integration scheme (2.37) we derived in Section 2.2.

4.1.1 Parameters of the system

In order to observe the propagation of any signal at all, our first concern is to put the system into its deeply underdamped regime. We have seen with equation (3.111) that, when the system is highly polarized, then perturbations propagate in a linear fashion with a speed given by

$$c_s = \sqrt{\frac{Jn_c}{\chi}} a , \quad (4.1)$$

while the amplitude of such perturbations is exponentially attenuated in time (see the dispersion relation (3.112)) with a damping coefficient

$$\gamma = \frac{\eta}{2\chi} . \quad (4.2)$$

Thus, we expect to observe an interplay between two intrinsic time scales of the system: they are that of the sound propagation,

$$\tau_1 = \frac{1}{c_s k} , \quad (4.3)$$

and that of the damping,

$$\tau_2 = \frac{1}{\gamma} . \quad (4.4)$$

Propagation is only possible if

$$\begin{aligned} \tau_1 &\ll \tau_2 \\ &\rightarrow \gamma \ll c_s k \quad \forall k ; \end{aligned} \quad (4.5)$$

in other words, it only takes place provided that the signal reaches the other end of the flock before it gets completely attenuated by the damping. If we require condition (4.5) to be met by the smallest value of k that it is possible to attain in a finite flock of size L , which is

$$k_{min} = \frac{1}{L} , \quad (4.6)$$

then it will be automatically satisfied by all other k 's. The condition for propagation hence becomes

$$\begin{aligned} \frac{c_s}{L} &\gg \gamma \\ &\rightarrow J \gg \frac{1}{n_c \chi} \left(\frac{\eta L}{2a} \right)^2 . \end{aligned} \quad (4.7)$$

As we did in Section 2.3.3, we choose for simplicity

$$\chi = \eta = a = 1 , \quad (4.8)$$

and as long as the system is sitting on the vertices of a cubic lattice, then the number of nearest neighbours is exactly $n_c = 2d = 6$. Typical lattice sides in this work are $L \sim 10 - 15$, hence choosing $J = 10$ will do the job.

The temperature T does not appear in condition (4.7), as we simply assumed the system to be in its deeply ordered phase when we computed c_s using the Spin Wave Approximation; however, we can get quite a precise idea of what this means in practice by observing the phase diagram in Figure 2.2(a), which we realized in Section 2.3.3, although at the time we used the value $J = 1$ for the coupling constant. Indeed, we have to bear in mind that the critical behaviour of the model is controlled not by the temperature T alone, but rather by the combination $\beta J = \frac{J}{T}$ (equation (1.51)): expanding J by a factor 10 (as we do if we choose $J = 10$) is equivalent to adopting a temperature T which is a tenth of the original.

Moreover, we wish our simulated trajectories to look as clean as possible, which wouldn't be the case if the order parameter fluctuated too much. Fluctuations are generally big around the critical temperature T_c , while they grow smaller for both higher and lower temperatures: to see this, it is enough to have a glance at the trend of $\chi(T)$ in Figure 2.2(b), as we know that χ is indeed proportional to the fluctuations of the order parameter. We also recall that fluctuations of the intensive order parameter are of the order

$$\langle \delta \phi^2 \rangle \sim \mathcal{O}\left(\frac{1}{N}\right) , \quad (4.9)$$

which suggests to choose a system that is as big as possible, compatibly with the computational resources at our disposal. All in all, we will mostly adopt the values $T = 0.1$ and $L = 15$ in the simulations *on-lattice* to follow.

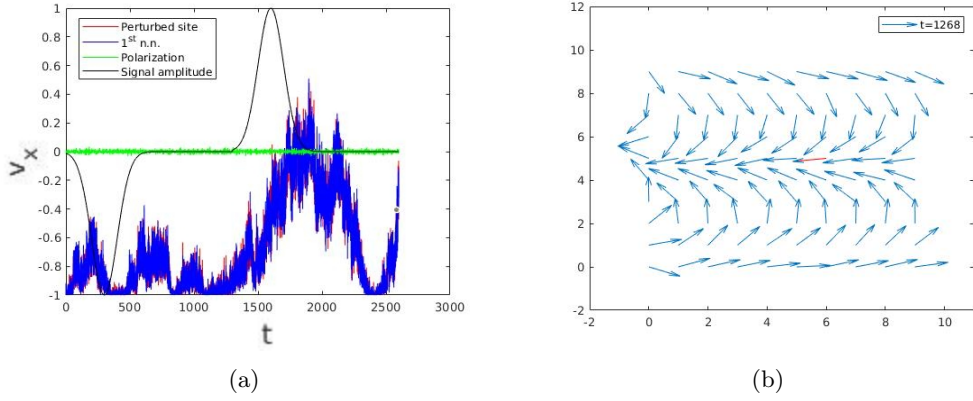


Figure 4.1. Realization of a metastable configuration of the system. The external field is applied in the direction opposite to that of the polarization at time $t_0 = 0$; the perturbed site aligns to it and so do its nearest neighbours, but the rest of the flock maintains its original direction of flight (which is possible, without disrupting the flock, because positions are being kept fixed in time). This configuration does not minimize the energy, but it is indeed an equilibrium state for the system. (a) x -component of the velocity of the perturbed site, of its nearest neighbours and of the polarization Φ , together with the trend of the signal amplitude. (b) A slice of the system around the perturbed site (in red); frame taken at time $t = 1268$ in a simulated footage.

4.1.2 Parameters of the external field

In Chapter 3 we found that an effective way to induce a turn in the system by means of a local field is to couple it to the velocity variable $\mathbf{v}_p$ relative to site p ,

$$\mathcal{H} = \mathcal{H}_0 - \mathbf{H}_p \cdot \mathbf{v}_p. \quad (4.10)$$

This is analogous to immersing a *dipole* $\mathbf{v}_p$ into a uniform field $\mathbf{H}_p$, which produces on the former the force

$$\mathbf{F} = \nabla (\mathbf{v}_p \cdot \mathbf{H}_p); \quad (4.11)$$

this force is null whenever the two vectors $\mathbf{v}_p$ and $\mathbf{H}_p$ are aligned, which represents a minimum for the energy $\mathcal{H}$, but also when $\mathbf{v}_p$ and $\mathbf{H}_p$ are antiparallel, giving rise to a situation of unstable equilibrium.

In the planar ISM, the Hamiltonian reduces to that of equation (3.2), which we rewrite here as

$$\begin{aligned} \mathcal{H} &= \mathcal{H}_0 - H_p(t) \cos(\varphi_p - \alpha_p) \\ \mathcal{H}_0 &= -\frac{J}{2} \sum_{ij} n_{ij} \cos(\varphi_i - \varphi_j) + \sum_i \frac{s_i^2}{2\chi}; \end{aligned} \quad (4.12)$$

in our numerical simulations, we will measure φ_p and α_p with respect to the direction assumed by the global polarization $\mathbf{V}(t)$ at the initial time $t = 0$, as depicted in Figure 3.1.

The first parameter to fix is thus the direction α_p of the field at the initial time t_0 when the perturbation is switched on. We choose to apply the field at right

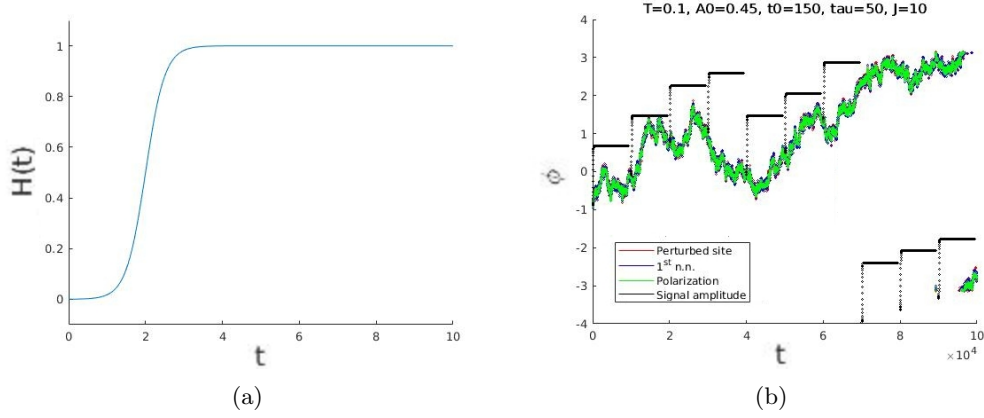


Figure 4.2. (a) Ramp like dependence of the field in equation (4.13), where we made the choice $A_0 = 1$, $t_0 = 2$ and $\tau_0 = 0.5$. (b) Trajectory of the angle $\varphi_p(t)$ of the perturbed site, of its nearest neighbours and of the polarization angle $\Phi(t)$, together with the trend of the signal amplitude. The amplitude of the applied field A_0 is not big enough to induce a collective turn which can be completed in a time interval $\Delta t \ll \tau_w$, hence what we observe is trivially the wandering of the polarization.

angles with respect to the initial direction of flight, that is $\alpha_p = \frac{\pi}{2}$: this way we can control the effective intensity of the field felt by the individual in site p , at least for short times t (such intensity is maximum at $t = 0$). This is also a way to avoid the occurrence of metastable configurations like the one shown in Figure 4.1, which was obtained by choosing $\alpha_p = \pi$.

The next step is to choose a specific shape for the signal $H_p(t)$. Two main parameters are to be set:

- (i) the amplitude of the signal, A_0 , and
- (ii) the duration of the signal, τ .

We will need to devise methods to study their effects separately: this problem is faced in the next two Sections.

4.2 Study of the amplitude: ramp-like perturbation

The simplest way to disentangle the effects of τ from those of A_0 is to send the former to infinity: this is the *ratio* behind the choice of a ramp-like signal in the form

$$H_p(t) = \frac{A_0}{2} \left[1 + \tanh \left(\frac{t - t_0}{\tau_0} \right) \right], \quad (4.13)$$

which is sketched in Figure 4.2(a). Here the focus is not on the ramp itself (hence on its parameters t_0 and τ_0), but rather on its limiting value A_0 ; the steep initial growth is only a way to render the field more realistic than an actual stepfunction $\theta(t)$.

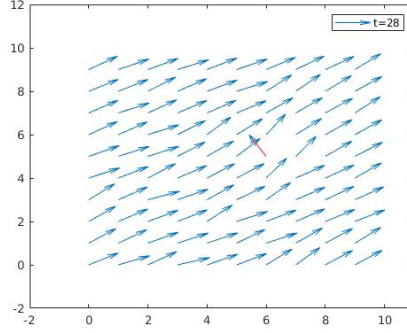


Figure 4.3. A strong field ($A_0 = 100$) is applied locally, so that the perturbed site (in red) suddenly turns by 90° with respect to the direction of its neighbours. Here we show a slice of the system around the perturbed site.

In Section 3.1.3 we have calculated the dynamical response of the system to a field in the form of (4.13): after a rapid transient of $\mathcal{O}(t_0)$, we expect to observe a linear regime where the polarization angle grows like

$$\Phi(t) \sim \frac{A_0 \sin \alpha_p}{\eta N} t \sim \frac{A_0}{N} t, \quad (4.14)$$

where in the last passage we have introduced our particular choice of the parameters α_p and η .

Now we wish to perform several simulations where the value of the amplitude A_0 is varied, with the hope to observe a change in the turning velocity of the flock. But what is a sensible value of A_0 to begin with?

Consider the configuration depicted in Figure 4.3: a strong field is applied on the red site, and the velocity variable coupled to the field immediately aligns to its direction. Now let's keep an eye on the system of equations (3.10) which governs the evolution of the variable $\varphi_p(t)$ subject to the perturbation, which we can translate into the second order equation of motion

$$\begin{aligned} \chi \ddot{\varphi}_p(t) &= -J n_c \sum_j \tilde{\Lambda}_{pj} \varphi_j - 2\gamma s_p - \varphi_p H_p(t) \cos \alpha_p + H_p(t) \sin \alpha_p + \xi_p(t) \\ &= -J n_c \sum_j \tilde{\Lambda}_{pj} \varphi_j - 2\gamma s_p + H_p(t) + \xi_p(t), \end{aligned} \quad (4.15)$$

where we have used $\alpha_p = \frac{\pi}{2}$ in the last passage and where

$$\tilde{\Lambda}_{ij} = -\frac{n_{ij}}{n_c} + \delta_{ij} \quad (4.16)$$

is the Laplacian matrix, here normalized so that $\tilde{\Lambda}_{ij} \xrightarrow{a \rightarrow 0} \nabla^2$.

Some considerations should be made:

- (a) At time $t = 0$, the spin variable $s_p(t = 0) \simeq 0$ because the flock is strongly polarized, so that basically only two forces balance on the right hand side of

equation (4.15): that of the field $H_p(t)$ (which represents a *drift term*, and gets switched on with the external field) and the restoring force that the nearest neighbours exert on the perturbed site. No matter how small the amplitude A_0 of the field is, at every time step Δt we expect $\varphi_p(t)$ to move slightly towards the direction of the field H_p , so that a collective turn can gradually occur as the rest of the flock follows the runaway (Figure 4.6).

- (b) A lower bound on the amplitude of the field is set by the usual *wandering* of the global polarization. Indeed, a really slow collective turn could never be observed in our numerical simulations, because it would get completely covered by wandering effects (as it happens in Figure 4.2(b)). We have seen in Section 3.1.2 that the wandering time is of order

$$\tau_w \sim \frac{\eta N}{2T} , \quad (4.17)$$

while equation (4.14) tells us that, during a collective turn,

$$\Phi(t) \sim \frac{A_0}{\eta N} t . \quad (4.18)$$

Setting $\Phi(t) \sim \mathcal{O}(1)$ for times $t \lesssim \tau_w$ gives the threshold amplitude

$$A_0 \gtrsim 2T \sim 0.2 . \quad (4.19)$$

- (c) The neighbour sites of φ_p do not sense the field $H_p(t)$ directly. In Figure 4.3, the field is strong enough for the perturbed site to align to its new direction straight away; once that happens, and the variable φ_p is at right angles with respect to the rest of the flock, then the variable φ_{p+1} will only realize that one of its nearest neighbours is displaced, no matter how big A_0 was in the first place. Hence, no upper bound is to be expected for the field amplitude A_0 .

To be on the safe side, we choose $A_0 \gtrsim \mathcal{O}(1)$ and we study increasing values of A_0 until the turn becomes too fast to be easily observed and analyzed with our choice of simulation parameters.

Our analysis has been carried out as follows:

- (i) We choose a value of the field amplitude A_0 in the range $[1.5, 30]$.
- (ii) We apply a ramp-like perturbation to a specific spin variable φ_p , chosen well inside the bulk of the cubic lattice of size $L \times L \times L$ that we are simulating.
- (iii) We keep the external field on for a time interval of about $\Delta t \sim 1000$ s, during which the collective turn can take place.
- (iv) We switch the external field off and allow the system to thermalize for a time $T_{term} = 50$ s, during which the polarization of the flock is measured.
- (v) We repeat steps (ii)-(iv) for n_{turns} times: each time, a new field is applied at right angles with respect to the polarization reached by the system after the thermalization interval T_{term} .

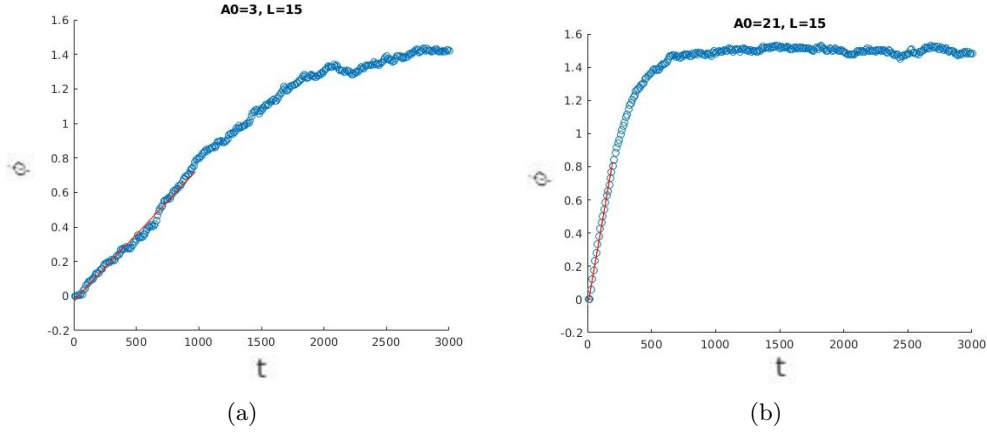


Figure 4.4. A single realization of a turn for two different values of the field amplitude A_0 , when a ramp-like perturbation is applied. The polarization angle $\Phi(t)$ grows linearly in time up until $\Phi \simeq \frac{\pi}{4}$, hence can be fitted by a straight line.

(vi) We change the value of A_0 and we repeat steps (i)-(v).

The results of this analysis are presented in Figure 4.6(a). The first part of these curves ($\Phi(t) \in [0, \frac{\pi}{4}]$) is roughly linear, as predicted by equation (4.14), hence it can be fitted with a straight line (Figure 4.4). The mean value of these n_{turns} slopes is evaluated in correspondence of every amplitude A_0 , and their error is taken as the standard deviation divided by $\sqrt{n_{turns}}$: the results are shown in Figure 4.5, where another linear fit confirms the linear dependence of the slope on A_0 , which again was predicted by (4.14).

To sum up, we found that *tuning the amplitude A_0 of the applied field controls the velocity of the collective turn*. In order for a turn to take place in a time interval $\Delta t \ll \tau_w$, which makes it visible in spite of the wandering of the polarization, we should take the amplitude to be at least of the order of $A_0 \simeq 1.5$.

4.3 Study of the duration: impulse-like perturbation

In order to study the effects of the duration τ of the signal alone, we simply have to set A_0 to a fixed value that we know for sure to induce a neat and readily observable collective turn when a ramp-like field ($\tau = \infty$) of such amplitude is applied; say, $A_0 = 30$. The easiest choice of a time dependence for an impulse-like field of duration τ is the Gaussian

$$H_p(t) = A_0 e^{-\frac{(t-t_0)^2}{2\tau^2}}. \quad (4.20)$$

We have already solved, in Section 3.1.3, for the dynamical response of the polarization angle to such a perturbation; we recall the result

$$\Phi(t) \simeq \frac{A_0 \sin \alpha_p}{\eta N} \tau \sqrt{\frac{\pi}{2}} \left\{ 1 + \operatorname{erf} \left[\frac{1}{\sqrt{2}} \left(\frac{t}{\tau} - 5 \right) \right] \right\}, \quad (4.21)$$

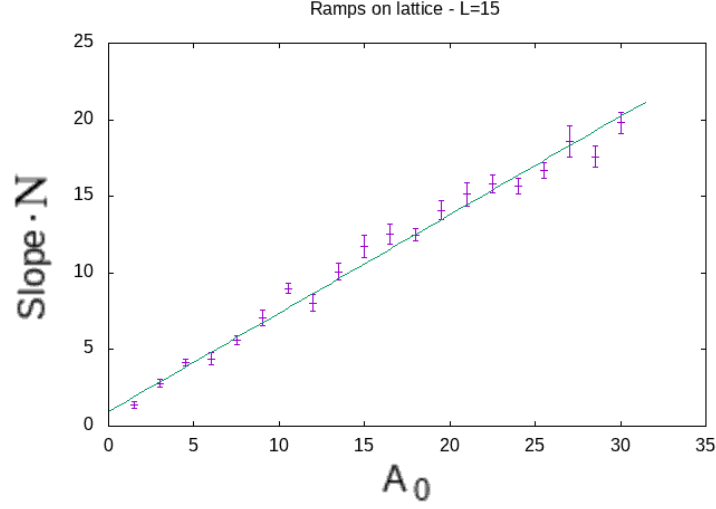


Figure 4.5. Study of the amplitude. For each value of A_0 , the mean value of the n_{turns} slopes of the linear fits like those shown in Figure 4.4 is plotted together with errorbars corresponding to $\frac{\sigma_{slopes}}{\sqrt{n_{turns}}}$; a fit of these data points confirms the prediction of equation (4.14).

where the choice $t_0 = 5\tau$ was adopted just in order to appreciate the whole impulse in the semiaxis $t > 0$ (Figure 3.2(a)). The shape of this function is plotted in Figure 3.2(b); specializing it for $\alpha = \frac{\pi}{2}$, $\eta = 1$ gives

$$\Phi(t) \simeq \frac{A_0\tau}{N} \sqrt{\frac{\pi}{2}} \left\{ 1 + \operatorname{erf} \left[\frac{1}{\sqrt{2}} \left(\frac{t}{\tau} - 5 \right) \right] \right\}. \quad (4.22)$$

We also proved (equation (3.95)) that the slope of the response $\Phi(t)$ around $t \simeq t_0$ should not depend on the choice of τ .

Before we perform several simulations for different values of τ , we need to ask ourselves (as we did for A_0) what is a sensible time scale to begin with. An intuitive argument is the following: surely a collective turn could not take place if the signal starting in a specific site were to be switched off before it had time to reach the other end of the flock. This would imply

$$\tau \geq \frac{L}{c_s} = \frac{15a}{c_s} \simeq 2, \quad (4.23)$$

where we inserted our usual choice of parameters, whence $c_s \simeq 7.75a$.

Now that clarity was made, we begin by applying Gaussian perturbations with τ chosen in the range $[10 - 110]$. Again, as for the study of the amplitude, the same perturbation is applied repeatedly (Figure 4.7(a)), after which the response curves are transposed upon one another and then averaged (Figure 4.7(b)). Quite surprisingly, *the plateau value reached by the polarization angle is observed to grow with τ* . We confirmed the linear nature of this dependence by measuring the plateau value of every single curve: the results are presented in Figure 4.7(c), where we

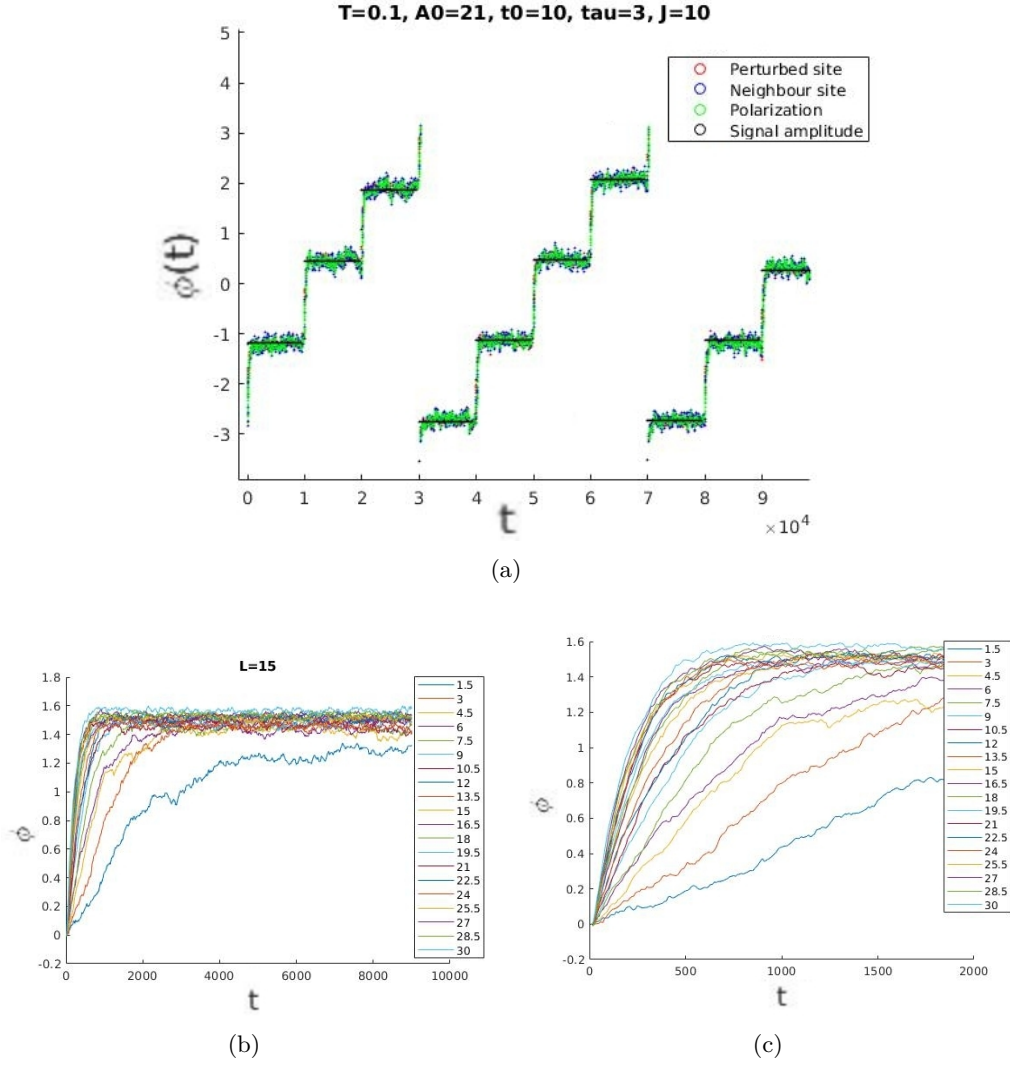


Figure 4.6. Study of the amplitude. Local fields of different amplitudes $A_0 \in [1.5 - 30]$ are repeatedly applied to the system, which behaves as in (a); the trajectory of the perturbed site and of a nearest neighbour are shown together with the polarization angle $\Phi(t)$ and the trend of the applied signal (the black steps). For each value of A_0 , the polarization curves (those colored in green in (a)) are superimposed upon one another and averaged over their n_{turns} realizations: the results are presented in (b) and (c), where the same curves are zoomed in. The slope of the response $\Phi(t)$ is seen to grow with the amplitude A_0 of the applied field.

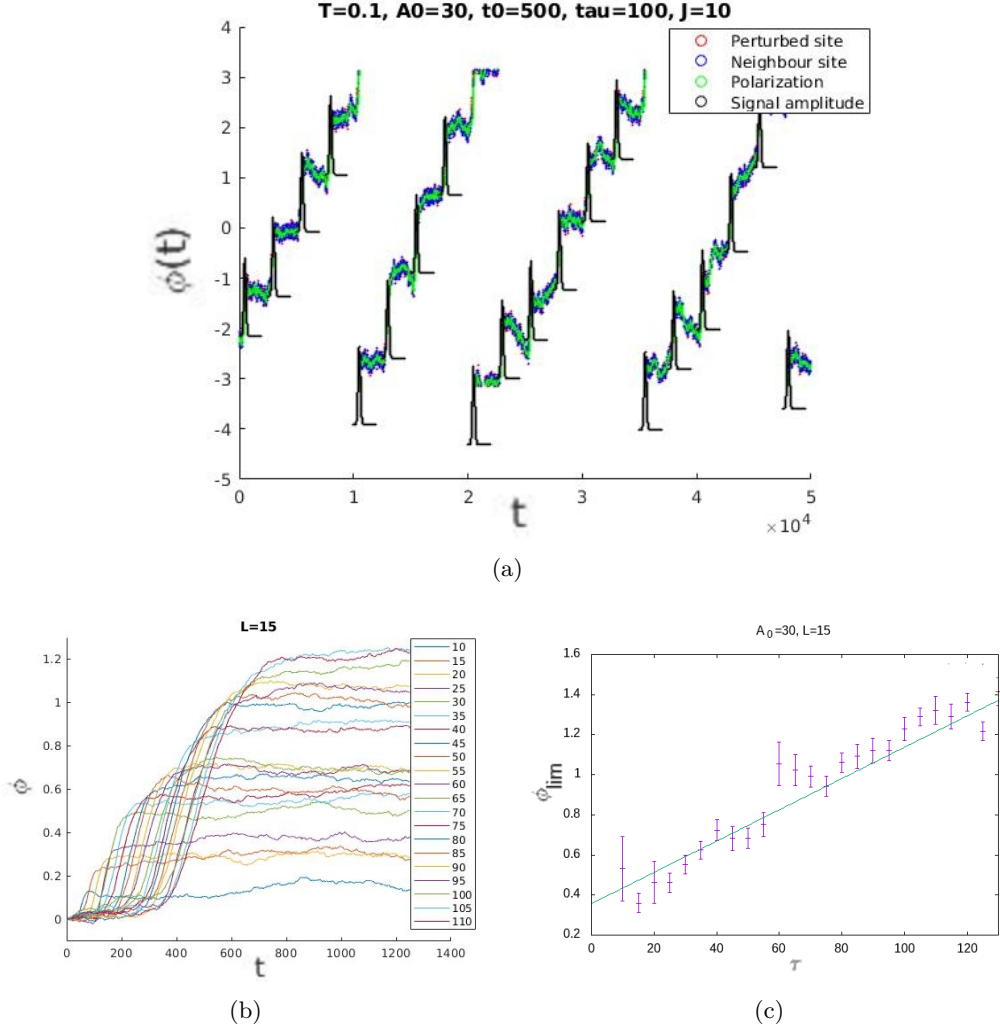


Figure 4.7. Study of the duration. (a) A Gaussian signal of duration τ is applied n_{turns} times to the system, each time pointing at right angles with respect to the direction reached by the polarization after a short thermalization interval which follows each perturbation. (b) Response curves are averaged and here presented for each value of τ . Their plateau value Φ_{lim} appears to grow with τ . (c) The linear dependence of Φ_{lim} on τ is checked by fitting the mean value of the former as measured from the curves above; the error is taken as $\frac{\sigma_{plateau}}{\sqrt{n_{turns}}}$.

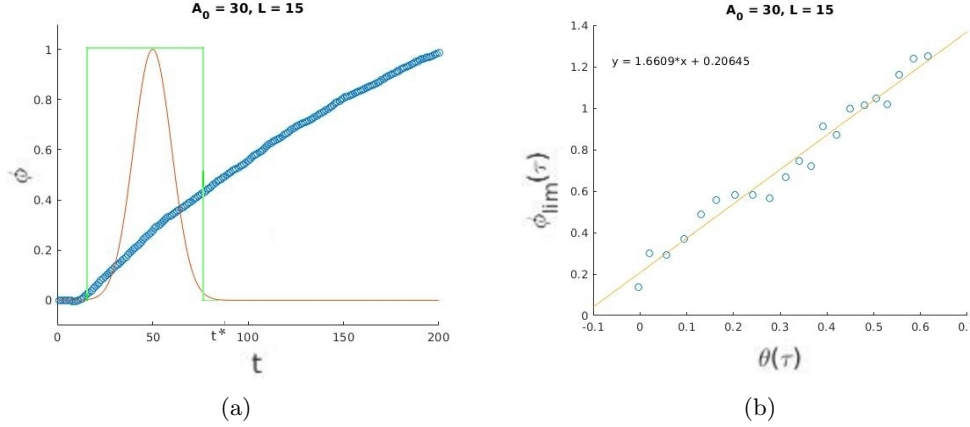


Figure 4.8. Study of the plateau. (a) In *blue*, response of the polarization angle $\Phi_{30}(t)$ to a ramp-like infinite impulse of amplitude $A_0 = 30$. In *red*, Gaussian signal of width $\tau = 10$ (and amplitude $A_0 = 30$, here normalized for graphical purposes), which "ends" at $t^* \simeq 3\tau$; in *green*, its rough rectangular approximation. (b) For each choice of τ , we measured the value $\theta(\tau) \equiv \Phi_{30}(t^*) = \Phi_{30}(3\tau)$ and we plotted it against the plateau $\Phi_{lim}(\tau)$ reached by the polarization angle in correspondence of that specific value of τ (see Figure 4.7(b)). This way we could verify their proportionality.

plotted the mean value of the plateau $\Phi_{lim}(\tau)$ in correspondence of each value of τ , each accompanied by an errorbar corresponding to $\frac{\sigma_{plateau}}{\sqrt{n_{turns}}}$.

A plausibility argument for the observed linear dependence $\Phi_{lim} \sim \tau$ is the following. Consider the response curve $\Phi_{30}(t)$ of the system when it is subject to a ramp-like perturbation of amplitude $A_0 = 30$, just like the ones we analyzed in the last Section; we have already seen that such response is linear in time for small t , as it is also clearly visible in Figure 4.8(a). It is intuitive, by looking at the graph, that the overall effect of a Gaussian-like perturbation of width τ cannot be too dissimilar from that we would obtain if we applied in its place a finite step which lasted for a time interval $T \sim \mathcal{O}(\tau)$. Let $\theta(\tau)$ denote the value reached by the polarization angle $\Phi_{30}(t)$ at a time $t = t^*$ corresponding to the "end" of the Gaussian signal of width τ , that is (practically speaking) $t^* \simeq 3\tau$:

$$\theta(\tau) \equiv \Phi_{30}(t = 3\tau) . \quad (4.24)$$

Plotting the plateau value $\Phi_{lim}(\tau)$ against $\theta(\tau)$, it is possible to check that they are indeed proportional to one another: this result is shown in Figure 4.8(b).

We conclude by checking that the shape of the response $\Phi(t)$ to a Gaussian perturbing field is indeed that predicted by equation (4.22). This can be done by checking that $\Phi(t)$ reaches half of its plateau value Φ_{lim} for $t \sim t_0$ (Figure 4.9(a)), and that the slope of $\Phi(t \simeq t_0)$ does not depend on the choice of τ (Figure 4.9(b)).

All in all, what we found is that when we apply a local field at right angles with respect to the direction of the polarization, *tuning the duration τ of the signal controls the angle of the collective turn* in the range $[0, \frac{\pi}{2}]$. The velocity at which the change of direction occurs is only controlled by the amplitude A_0 of the field.

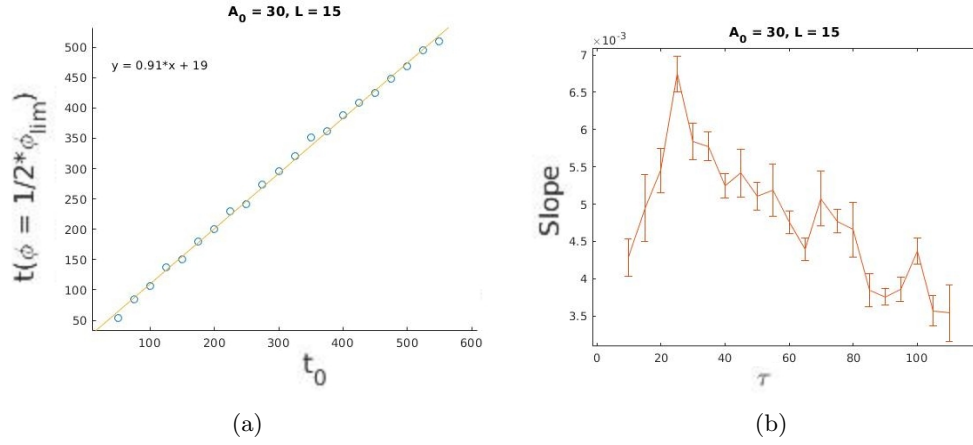


Figure 4.9. Study of the shape of $\Phi(t)$. Equation (4.22) predicts the behaviour of $\Phi(t)$ in response to a Gaussian perturbation like that in (4.20); in particular, $\Phi(t)$ should reach half of its limiting value $\Phi_{lim}(\tau)$ at $t \simeq t_0$, and in that point its slope should be independent of τ . (a) Comparison between t^* s.t. $\Phi(t^*) = \frac{1}{2}\Phi_{lim}(\tau)$ and $t_0(\tau) \equiv 5\tau$. Their dependence is indeed linear. (b) Although we do observe some fluctuations of $\mathcal{O}(10^{-3})$ in its value, the graph shows that no specific trend is associated to the slope of $\Phi(t)|_{t=t_0}$ as we vary τ .

Chapter 5

Study of the response off-lattice

In the last Chapter, we developed numerical protocols which can be used to induce a collective turn in an ISM system *on a regular lattice* by means of an external field coupled to a single velocity variable. Moreover, we measured how the parameters of the field can be tuned to control the dynamical response of the system.

The aim of this Chapter is to understand if and to which extent the same protocols may be applied to control the dynamical response of ISM in its full, *off lattice* version. Here the individual positions $\mathbf{r}_i$ are allowed to evolve in time, following the direction pointed by their respective velocity vector $\mathbf{v}_i$. The relevant parameter we will need to study is $v_0 = \|\mathbf{v}_i\|$: the bigger the speed of the individual particles, the further we expect the behaviour of the system to deviate from that of the *on lattice* case. The reason for this is to be searched in the interplay between two time scales: that of the information propagation, and that of the mutual diffusion of the individuals within their interaction network.

5.0.1 An heuristic argument

Consider two distinct, interacting particles, *i.e* two particles whose relative distance is smaller than the interacting radius r_c in a *metric* interaction picture (see Section 1.2). Let $\delta\varphi$ denote the angular deviation of their flight direction from that of the entire flock, $\mathbf{V}$, as in Figure 5.1. If $\delta\varphi = 0$, they will keep moving parallel to one another and never meet; not so if $\delta\varphi \neq 0$. In fact, it is straightforward to observe, using the Spin Wave Approximation, that $\delta\varphi$ may be related to the polarization ϕ of the flock in its deeply ordered phase by

$$\begin{aligned} \phi &= \frac{1}{v_0} \mathbf{V} \cdot \hat{\mathbf{n}} = \frac{1}{v_0 N} \sum_i v_i^L = \frac{1}{v_0 N} \sum_i v_0 \cos \varphi_i \\ &\simeq \frac{1}{N} \sum_i \left(1 - \frac{\varphi_i^2}{2} \right) = 1 - \frac{1}{2} \langle \varphi^2 \rangle, \end{aligned} \tag{5.1}$$

where as usual $\hat{\mathbf{n}}$ denotes the direction of the polarization vector, v_i^L is the longitudinal component of the i -th velocity vector and the angles φ_i are measured with respect

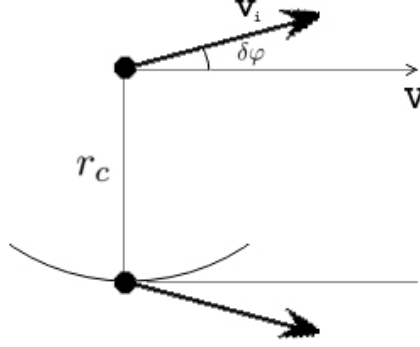


Figure 5.1. Two interacting particles are drifting away from each other due to the fluctuation of their individual velocity $\mathbf{v}_i$ from that of the entire flock, $\mathbf{V}$. Eventually, their distance will overcome the interaction radius r_c and they will stop interacting, which translates into a modification of the connectivity matrix n_{ij} of the system.

to $\hat{\mathbf{n}}$ as in Figure 2.1. We can define the fluctuation $\delta\varphi$ as

$$\begin{aligned}\delta\varphi &\equiv \sqrt{\langle\varphi^2\rangle} \\ &\simeq \sqrt{2(1-\phi)} ,\end{aligned}\tag{5.2}$$

where we used equation (5.1) in the last passage.

In the worst case scenario, the velocity vectors of the two particles will point outwards as in Figure 5.1, so that they will drift away in time from one another. Within a time t_1 defined so that

$$v_0\delta\varphi \cdot t_1 \sim r_c ,\tag{5.3}$$

each particle will be out of the interacting range of the other: hence, we say that the connectivity matrix n_{ij} has rearranged. In our simulations, we will take $r_c = a$, where a is the lattice spacing (as we will explain shortly), so that we can estimate the mutual diffusion time scale t_1 as

$$t_1 \sim \frac{a}{v_0\delta\varphi} .\tag{5.4}$$

On the other hand, when we apply an external field to a single particle p with the hope to induce a collective turn in the system, we have to make sure that the information can reach its nearest neighbours *before* the p -th particle has drifted out of their interaction range. This amounts to comparing the time scale t_1 with another time scale given by

$$t_2 \sim \frac{a}{c_s} ,\tag{5.5}$$

where as usual c_s is the sound speed.

Following this line of reasoning, we expect information propagation to take place in a not too dissimilar way from that of the *on lattice* case as long as

$$\begin{aligned}t_2 &\ll t_1 \\ &\rightarrow v_0 \ll \frac{c_s}{\delta\varphi} .\end{aligned}\tag{5.6}$$

In the following Sections, we will try to test these considerations.

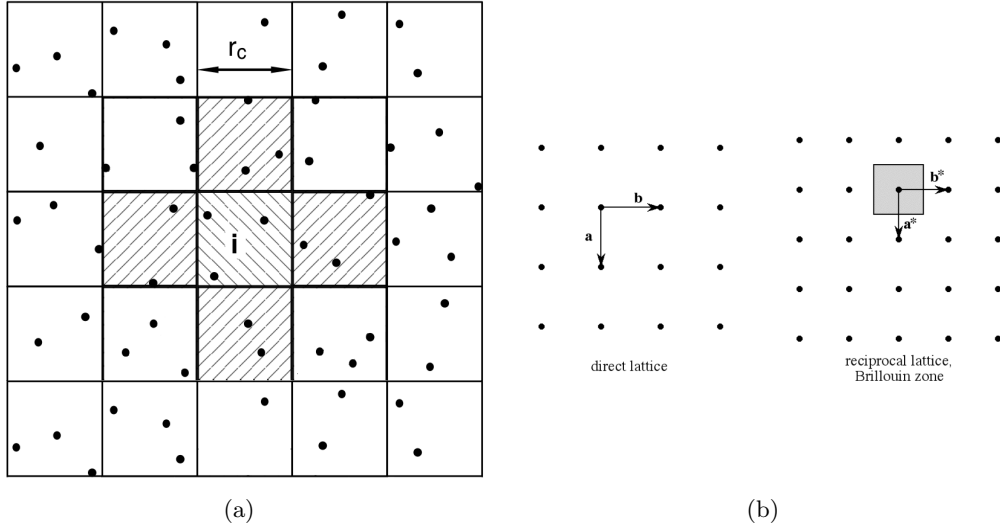


Figure 5.2. A planar scheme of the *mosaic*, which is the pixellation adopted in the numerical code (where, however, the simulated lattice is in 3 dimensions). (a) The physical space is divided into cubic cells of side equal to the radius of interaction r_c . Each particle is allowed to interact with particles in the same cell or in the neighbouring ones. (b) By choosing $r_c = 1$, each vertex of the old lattice can be taken as the center of one of the new cells (which together constitute its *dual* or *reciprocal* lattice).

5.1 Integration scheme

In Section 2.2 we have seen how to integrate numerically the equations of motion (2.9) of the Inertial Spin Model; however, when we finally derived the integration scheme (2.37), we did not specify how the evolution of the connectivity matrix $n_{ij} = n_{ij}(\{\mathbf{r}_i(t)\})$ could be taken into account in our simulations. Indeed, the study of the ISM we have hitherto carried out only concerned the *on lattice* approximation of the model, where the individual positions $\mathbf{r}_i$ are kept fixed in time, hence there was really no need to evolve the matrix n_{ij} at all.

Now we wish to go one step further and study the effects of the relative motion of the components of the system; hence, we proceed as follows. We first divide the original lattice of size $L \times L \times L$ into smaller cubic *cells* of side r_c , as we exemplified in Figure 5.2(a). Choosing $r_c = a = 1$, we can maintain the structure of the original lattice that we have been using in the previous simulations, which however now plays the role of a *reciprocal lattice* (see Figure 5.2(b)): each vertex of the old lattice becomes the center of a *cell*.

In the numerical simulation code, which is fully presented in Appendix B for the *off lattice* case, each active particle in the system is uniquely identified by an index i and so is every n -th cell. Hence, we associate to each cell a *bucket list* containing the indices of the particles which are currently standing inside that specific cell. This is obtained in the code by defining a suitable *struct* such as

```

1 typedef struct{                                // Paradigm for a bucket list
2     int * list;                                // Array list
3     int numData;                               // Number of items it contains
4 } cell;

```

When we update the spins $s_i(t + \Delta t)$ according to (2.37), we need to evaluate the force term

$$f_i(\boldsymbol{\varphi}(t), t) = J \sum_j n_{ij}(t) \sin(\varphi_j(t) - \varphi_i(t)) . \quad (5.7)$$

This can be achieved by letting each particle i interact with those in the same cell or in the neighbouring cells (Figure 5.2(a)). Hence, the connectivity matrix n_{ij} can actually be discarded in favour of

$$f_i(\boldsymbol{\varphi}(t), t) = J \sum_{j \in \mathcal{N}_i(t)} \sin(\varphi_j(t) - \varphi_i(t)) , \quad (5.8)$$

where we indicated with $\mathcal{N}_i(t)$ the *neighborhood* of i , which is made up of the cell where particle i is contained at time t , together with its adjacent cells.

The advantage is that we already know, for each cell n , the indices of all the confining cells, because the cell structure (which was baptized *mosaic* in the code) does not evolve in time.

This makes the algorithm of $\mathcal{O}(N)$ rather than of $\mathcal{O}(N^2)$, because there is no need to cycle over every pair of particles and check if their distance is less than the interaction radius r_c (this class of methods is known as *cell list* or *linked-list* - see, for instance, [23]). Notice also that the self-interaction term $j = i$ in the force (5.8) is null, being the latter proportional to $\sin(\varphi_j - \varphi_i)$, hence there is no need to locate and skip element i in the updating algorithm.

Once all the velocities have been updated using the force term (5.8), then the positions are evolved according to a simple Eulero-Cromer rule

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \mathbf{v}_i(t + \Delta t) \Delta t . \quad (5.9)$$

Notice that, being the velocities two-dimensional in our planar ISM simulations, each particle effectively moves on a plane. If we set the initial state of the system so that the N particles are equally spaced and positioned on the vertices of a regular 3d cubic lattice, then each planar sheet will slide parallel to the other - even though different sheets *do* interact with those adjacent to them.

5.2 Numerical simulations

Now we are ready to use the code presented in Appendix B to perform numerical simulations of the planar ISM in its *off lattice* version.

5.2.1 Choice of parameters

Again, we are faced with the task of choosing the right parameters to feed into our simulation. We begin by taking for simplicity

$$\chi = \eta = a = 1 , \quad (5.10)$$

as we have been doing in all the rest of this work. We choose the number of individuals in the system to be

$$N = (15)^3 , \quad (5.11)$$

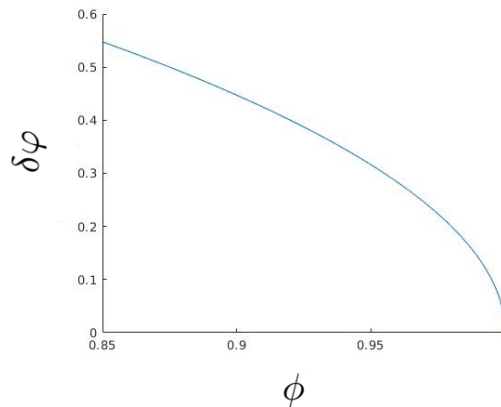


Figure 5.3. Graphical representation of the dependence of the average velocity deviation $\delta\phi$ on the polarization ϕ when the SWA is applied (equation (5.15)).

so that our results can be compared to those of the *on lattice* simulations of Chapter 4, where we had simulated a regular lattice of side $L = 15$.

As we have explained in Section 5.1, in our *off lattice* simulations we recycle the old lattice and use it as a cell structure (the *mosaic*), placing initially a single particle in each cell. If the spatial distribution of the particles remains homogeneous throughout the temporal evolution of their positions, then it is easy to convince oneself, by looking at the cross-shaped interaction domain shown in Figure 5.2(a), that the average number of interacting neighbours per particle should not be too dissimilar from $n_c \simeq 2d = 6$, as it used to be in the *on lattice* simulations with $d = 3$.

Now, as we did in Chapter 4, we need to make sure that the system lies into its underdamped regime, in order to allow for the propagation of information. This is quite a delicate point, because clearly we have derived the very dispersion relation of ISM and the expression of the sound speed

$$c_s = \sqrt{\frac{Jn_c}{\chi}}a \quad (5.12)$$

under the hypothesis of *fixed positions* $\mathbf{r}_i$. As we explained in Section 1.3.2, this is sensible because local bird alignment happens on a much faster timescale than neighbour rearrangement; equivalently, using the nomenclature we introduced at the beginning of this Chapter (Section 5.0.1), we could say that this approximation keeps working in our theoretical model as long as

$$t_2 \ll t_1. \quad (5.13)$$

Looking at condition (5.6), though, we can at least assume c_s to be still given approximately by (5.12) for small values of v_0 . If this is the case, then we can tentatively choose $J = 10$ in analogy with Chapter 4, which gives $c_s \simeq 7.75a$.

Two more parameters are to be fixed, which are the temperature T and the speed v_0 . The former should be chosen in such a way that the system lives in its strongly

ordered phase; but the phase diagram we explored in Section 2.3.3 is not reliable anymore to this purpose. Indeed, the *off lattice* nature of the system is known to affect the polarization diagram: generally speaking, the motion of the network is expected to *help* order propagation, so that $T_c(v_0 \neq 0)$ is generally higher than its fixed-network counterpart. Hence, our first task will be that of exploring the phase diagram $\phi(T)$ for various values of v_0 and locate the critical temperature $T_c(v_0)$.

Once this is set and done, our plan is to perform collective turn simulations following the same perturbation protocols as in Chapter 4, pushing v_0 to higher and higher values until a deviation from the linear response we have encountered in the *on lattice* case can clearly be observed. As we have seen in Section 5.0.1, the threshold value for v_0 should be given by

$$v_0^{lim} \simeq \frac{c_s}{\delta\varphi}, \quad (5.14)$$

where however the average deviation $\delta\varphi$ depends on the system polarization via

$$\delta\varphi \simeq \sqrt{2(1 - \phi)}. \quad (5.15)$$

This last relation is plotted in Figure 5.3 for high values of the polarization, $\phi \in [0.85, 1]$. Clearly, if $\phi = 1$ then the threshold value v_0^{lim} is pushed to infinity: in our heuristic argument, no matter how fast they travel, the particles will move parallel to one another if the flock is perfectly polarized, thus resulting in a trivial rigid displacement of the whole system. Not so if $\phi < 1$: choosing for instance $\phi = 0.9$ yields $\delta\varphi = 0.45$ and, using equation (5.14) with $c_s \simeq 7.75$, we find

$$v_0^{lim} \simeq 16. \quad (5.16)$$

Finally, recall that ϕ is controlled by both T and v_0 . We will thus explore the phase diagrams $\phi(T; v_0)$ with the hope of finding a common value of T which corresponds (almost) to the same polarization ϕ for all the values of v_0 under consideration.

5.2.2 Polarization diagram

The polarization diagram $\phi(T; v_0)$ for several values of the speed v_0 is presented in Figure 5.4. The first consideration to make is that the critical temperature T_c grows with v_0 for small values of the speed (Figure 5.4(a)), confirming our intuition that the motion of the interaction network facilitates order propagation. However, if we push v_0 further (which is important, because we are interested in studying the limit of validity of the model), then T_c is observed to stop growing and it starts instead to decrease steadily until it saturates on $T_c(v_0 \gg 1) \simeq 3.2$.

The reason to this unexpected behaviour can probably be traced back to the *clustering* phenomenon we have observed in numerical simulations of the planar version of ISM; this makes the spatial distribution of particles uneven (Figure 5.5). This pathological effect surely needs further investigation, which is however beyond the scopes of this work (*hic sunt leones*).

An important piece of information we can extract from Figure 5.4 is that the temperature $T = 1$ corresponds virtually to the same polarization $\phi \simeq 0.9$ for all

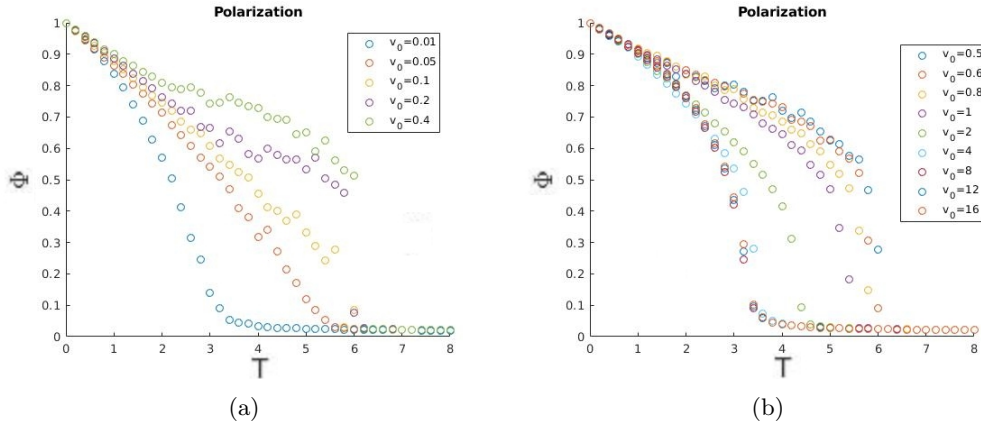


Figure 5.4. Phase diagrams $\phi(T; v_0)$ for several values of v_0 . (a) The critical temperature is observed to grow with v_0 for $v_0 \leq 0.4$, confirming our intuition that the motion of the network helps maintaining collective order in spite of the noise. (b) T_c becomes smaller as we increase v_0 beyond the threshold $v_0 \simeq 0.4$, probably due to clustering effects (see Figure 5.5).

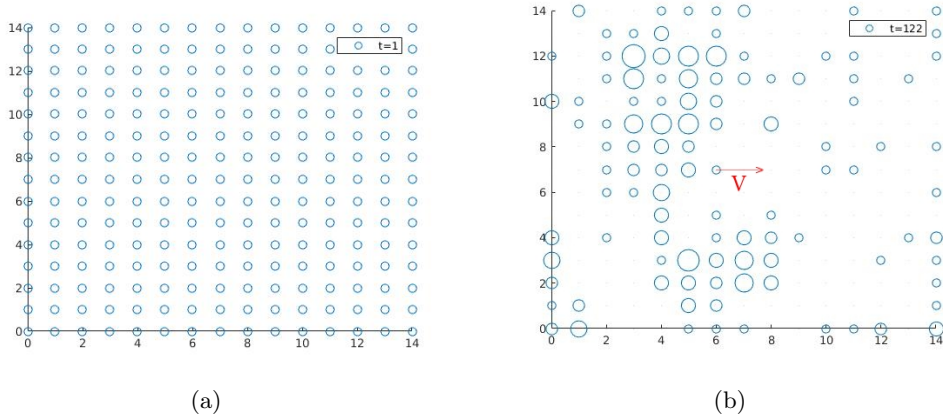


Figure 5.5. Pictorial representation of the *mosaic* for a planar slice of a system with $L = 15$ and $v_0 = 2$ at temperature $T = 1$. A circle is drawn around the center of each cell whose radius is proportional to the number of particles it contains; picturing the mosaic for each time t can be used as a way to monitor the evolution of the spatial distribution of the particles. (a) At $t = 0$, a single particle is placed in the center of each cell. (b) At later times and for high v_0 , the spatial distribution is no longer homogeneous as particles begin to cluster. The global direction of motion is indicated in red (it was inferred by inspecting the adjacent time frames).

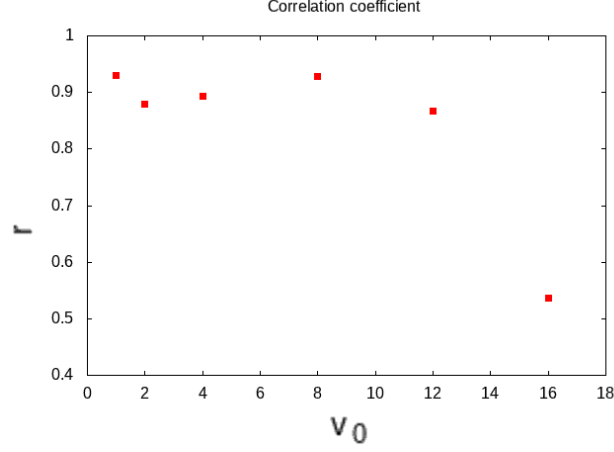


Figure 5.6. Trend of the Pearson correlation coefficient for several values of the particles speed v_0 . An external field with infinite duration and amplitude A_0 is applied to the system, and the response of the polarization angle $\Phi(t)$ is measured, as was extensively explained before. The slope of the linear part of this response can then be plotted as a function of the amplitude A_0 , as in Figure 5.8 (where plots for $v_0 = 1, 2, 4$ are not shown - that in Figure 5.7(b) was obtained by performing a higher number of turns, n_{turns} , so it is not directly comparable). A linear fit of such graphs yields the model $f(x) = ax + b$, and a measure of its goodness is given by the correlation coefficient $r(A_0, f(A_0))$, which can be evaluated as in equation (5.17). We can clearly evince that linearity fails for $v_0 = 16$.

the values of v_0 we are taking into account. This suggests we can choose $T = 1$ in the numerical simulations we are about to perform, and vary v_0 until we observe a deviation from the *on lattice* case. As we noticed in the last Section, this choice of parameters sets the threshold value for the particles speed around $v_0^{lim} \simeq 16$.

5.2.3 Simulated turns

The first, non obvious conclusion we draw in this Section is that collective turns *do* occur in our simulated system when the same perturbation protocols as in Chapter 4 are applied, at least for small values of the particles speed v_0 . Figure 5.7 shows the response of the polarization angle of an *off lattice* system with $v_0 = 1$ both to ramp-like and impulse-like perturbations: the result is completely analogous to those of the *on lattice* case.

Now, as announced, we wish to explore the behaviour of the system for higher values of v_0 . In Section 3.3 we summarized the two perturbation protocols we have been applying on the fixed-network system in Chapter 4. In the following, we will adopt the ramp-like protocol (*i.e.* long lasting signals for various field amplitudes A_0) because we know it to induce a simple, linear growth of the polarization angle $\Phi(t)$ as a dynamical response (equation (3.132)); it should thus be relatively easy to detect a deviation from this behaviour.

Indeed, such deviation is clearly visible in Figure 5.8, where we applied the perturbation protocol to systems with $v_0 = 8$ and higher. In particular, the

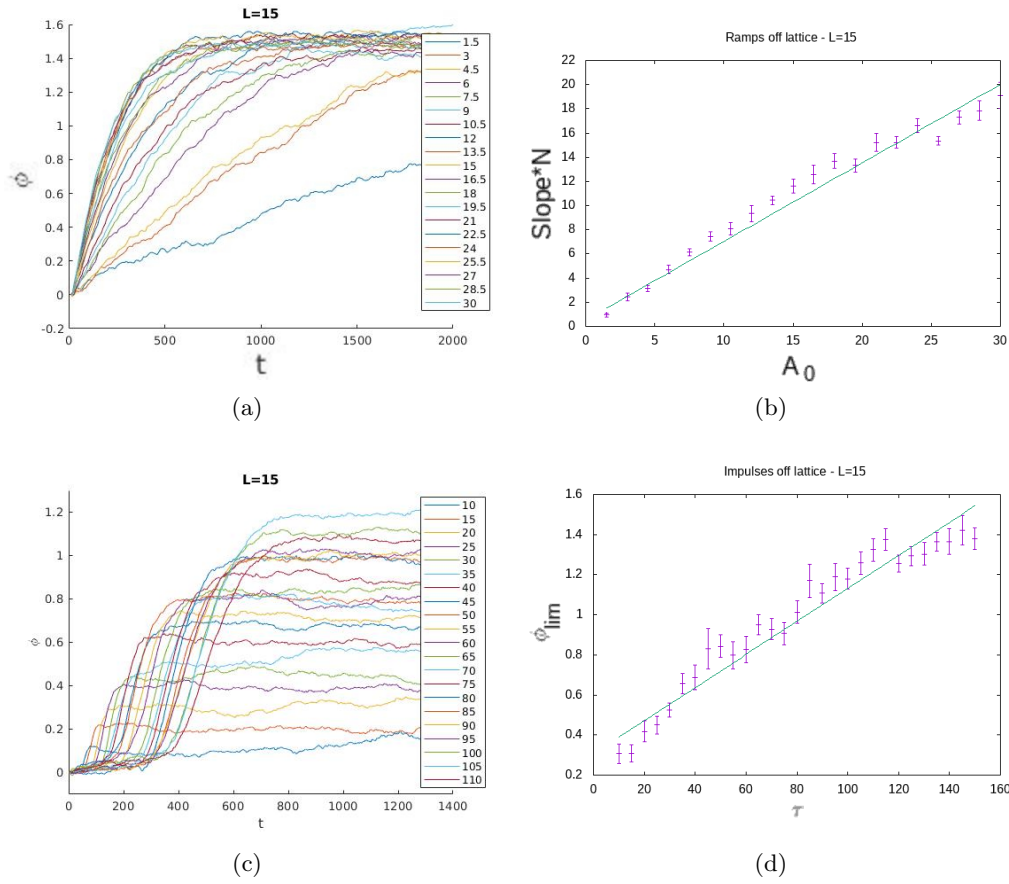


Figure 5.7. Realization of collective turns in the *off lattice* version of the system ($v_0 = 1$), using the same perturbation protocols as in Chapter 4. (a) Response of the polarization angle to ramp-like perturbations of various amplitudes A_0 . Compare with Figure 4.6(c). (b) Slope of the same response curves in the linear regime for short times t . Compare with Figure 4.5. (c) Response of the polarization angle to impulse-like perturbations of fixed amplitude and various durations τ . Compare with Figure 4.7(b). (d) Limiting values reached by the polarization angle after the application of the same impulse-like perturbations. Compare with Figure 4.7(c).

behaviour of the system is significantly far from the linear case for the highest value of v_0 we have tested, which is $v_0 = 16$.

A more quantitative way to check this is by inspecting the Pearson correlation coefficient, which is defined as

$$r(X, Y) \equiv \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} \quad (5.17)$$

for the variables X and Y in terms of their covariance and standard deviations. The quantity $r(A_0, f(A_0))$ is plotted in Figure 5.6 for some selected values of v_0 : here we indicated with A_0 the set of chosen amplitudes (on the x -axis) and with $f()$ the linear regression model obtained by fitting graphs like that in Figure 5.8(b). Again, the linearity hypothesis is observed to fail for $v_0 = 16$.

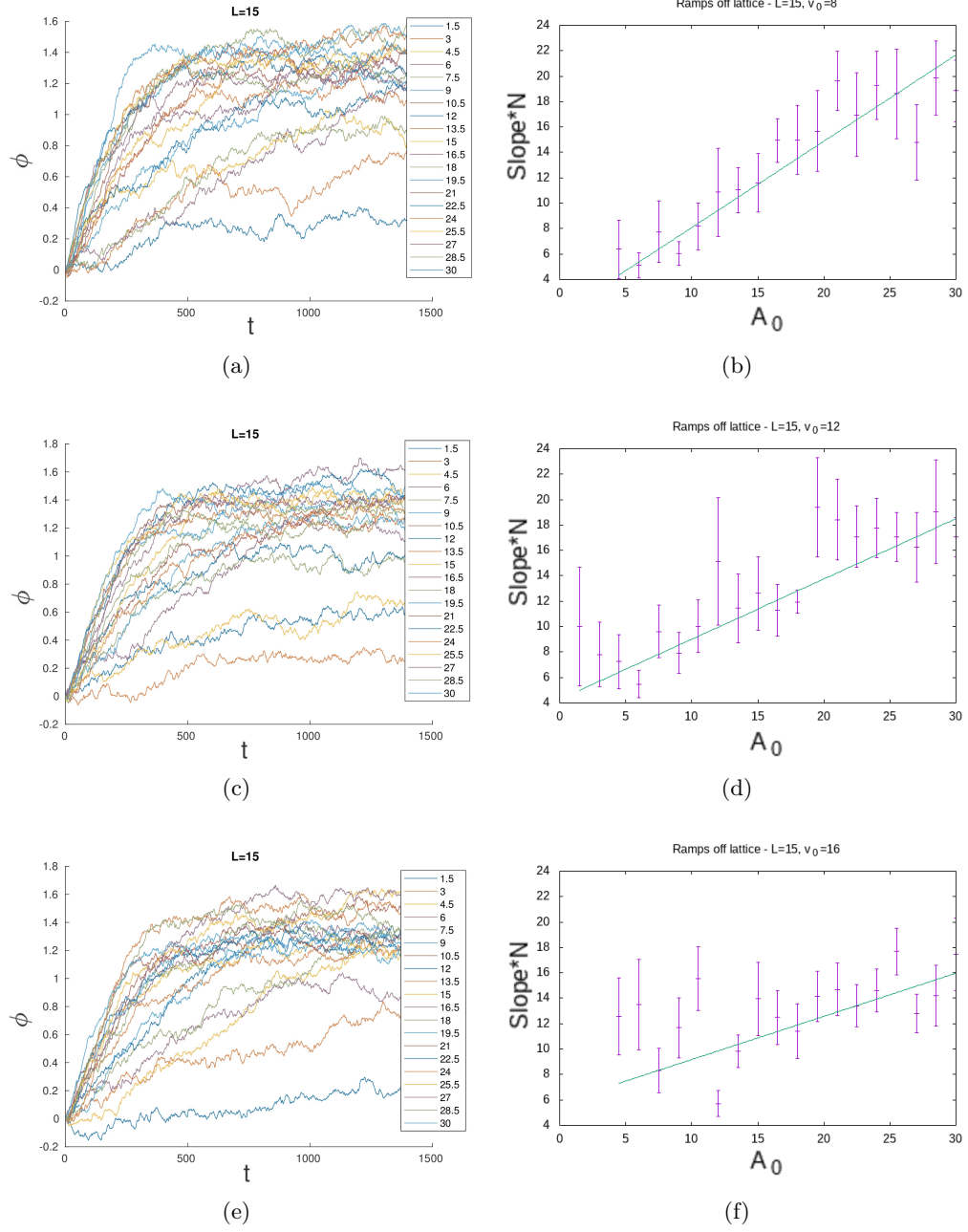


Figure 5.8. Realization of collective turns in the *off lattice* version of the system, using the ramp-like perturbation protocol described in Chapter 4 (see Summary on page 61). The results for several values of v_0 are presented here: $v_0 = 8$ (a-b), $v_0 = 12$ (c-d), $v_0 = 16$ (e-f). For each of them, we show the response of the polarization angle to ramp-like perturbations of various amplitudes A_0 and the slope of the same response curves in the linear regime for short times t . A significant deviation from the *on lattice* case is observed for $v_0 = 16$, as we could also clearly evince by inspecting its Pearson correlation coefficient in Figure 5.6.

Conclusions

The purpose of this work has been that of understanding how a local external field can affect the global order of a finite size system of active particles in its low temperature, ordered phase.

After briefly describing the Inertial Spin Model, which was devised to correctly account for the dynamics of biological many-body systems, we introduced its planar version (*i.e.* with a 2-dimensional order parameter) and we studied its static properties on a regular lattice using Langevin dynamics numerical simulations.

Having thus set up the stage, we presented a low temperature expansion of the planar model on a lattice (the Spin Wave Approximation) and performed a full derivation of the response of the system at first order in Linear Response Theory. To do so, we modeled the external perturbation using a field $\mathbf{H}_i(t)$ coupled to a single velocity variable $\mathbf{v}_i$ in the microscopic Hamiltonian; we furthermore showed that coupling a field to its conjugated momentum $\mathbf{s}_i$, a choice which is equally plausible in principle, is in practice ineffective insofar as changing the direction of the order parameter $\mathbf{V}$ (the global velocity of the system) is desired.

In the context of collective behaviour in biological systems, this amounts to relating a collective turn (here represented as a change in the direction of the flock's total velocity $\mathbf{V}$) to the perturbation which caused it, and which was measured to always present a well localized spatial origin in experiments.

By simulating such collective turns numerically, we linked their dynamical properties (*i.e.* their angular extent and their rotational speed) to the parameters of the external perturbing field, which are its amplitude and its time dependence. In particular we found that, when a finite-time local field is applied at right angles with respect to the direction of the polarization,

- (i) tuning the duration τ of the signal controls the angle of the collective turn in the range $[0, \frac{\pi}{2}]$, and
- (ii) tuning the amplitude A_0 of the applied field controls the velocity at which the change of direction occurs.

When an infinite-time signal is applied, on the other hand, a collective turn is always observed to take place, in which the order parameter $\mathbf{V}$ is forced to align to the direction singled out by the external field $\mathbf{H}$.

The last part of this work consisted in applying the same perturbation protocols to the full, off-lattice system, where the individual particles in positions $\mathbf{r}_i$ are

allowed to move according to their velocity vectors $\mathbf{v}_i$, instead of standing still on the vertices of a fixed lattice. We found that the same qualitative considerations we made in the on-lattice case about the dynamical response of the system still apply to the off-lattice case, provided that the particles speed v_0 is taken to be smaller than a threshold value $v_0^{lim} \simeq 16$. This represents approximately the velocity at which an individual particle drifts out of the interaction range of its turning neighbour before the turning signal (which travels with the sound speed c_s) is able to reach it, thus impeding optimal information propagation.

The natural continuation of this work should include a detailed analysis of the following aspects:

- (a) As we pointed out in Section 5.1, the particles effectively move on parallel planar sheets in our *off lattice* simulations of the planar ISM. This limitation could be simply overcome by adding some noise in the evolution equations for the $\hat{z}$ component of the positions $\mathbf{r}_i$ (*i.e.* the direction perpendicular to that of the planes). It is still unclear, however, what other side effects this random motion could cause and how the intensity of the noise should be chosen.
- (b) In Section 5.2.2 we observed an anomaly in the polarization diagram of the *off lattice* model: for high values of the particles speed ($v_0 > 1$), the critical temperature T_c decreases as we increase v_0 , thus contradicting our intuition that the motion of the interacting network should help maintaining long range order. We also noticed on that occasion (see Figure 5.5) that a clustering phenomenon occurs in our numerical simulations, which makes the spatial distribution of particles no longer homogeneous. That the former be a consequence of the latter is a possibility which should be carefully examined; at the same time, an inquiry over the origins of the clustering process is strongly advisable.
- (c) Once a perturbation protocol has been defined for the planar Inertial Spin Model, then its generalization to the full 3D model should be quite straightforward. However, much more attention should be paid to the numerical simulations in that case: as the number of degrees of freedom of the system grows in comparison to the simple planar case, new expedients should be devised that make the code execution much faster.

Appendices

A Computation of the propagator

Here we derive the propagator for the ISM dynamics, to be used in Section 3.2.

A.1 Linear response

To fix the notation, let's first revise some of the tools of Linear Response Theory; we will follow the steps of [13]. Let $\varphi(\mathbf{x}, t)$ be a field and call $\varphi_0(\mathbf{x})$ its stationary value; if we apply a localized perturbation in the form of a scalar field $h(\mathbf{x}, t)$, then we can define the *impulse response function*

$$R(\mathbf{x}, t; \mathbf{x}', t') = \left. \frac{\delta \varphi(\mathbf{x}, t)}{\delta h(\mathbf{x}', t')} \right|_{h=0}, \quad (\text{A.1})$$

so that the response of the field to the external perturbation can be given in terms of the convolution

$$\varphi(\mathbf{x}, t) = \varphi_0(\mathbf{x}) + \int d\mathbf{x}' dt' R(\mathbf{x}, t; \mathbf{x}', t') h(\mathbf{x}', t') + \mathcal{O}(h^2). \quad (\text{A.2})$$

If we choose the initial condition on the field φ so that $\varphi(\mathbf{x}, t=0) = \varphi_0(\mathbf{x})$, *i.e.* equal to its stationary value, then the response function cannot depend on the specific time t when it is applied, but only upon the difference

$$R(\mathbf{x}, t; \mathbf{x}', t') = R(\mathbf{x}, \mathbf{x}'; t - t'). \quad (\text{A.3})$$

In order to preserve causality, we may also assume

$$R(\mathbf{x}, \mathbf{x}'; \tau < 0) = 0, \quad (\text{A.4})$$

so that the integration in dt' in equation (A.2) actually runs over the past of t only, $\int_{-\infty}^t dt'$. Taking the perturbing field $h(\mathbf{x}, t)$ to be null before $t = 0$, which can be achieved for instance by choosing

$$h(\mathbf{x}, t) = \hat{h}(\mathbf{x}) \theta(t), \quad (\text{A.5})$$

equation (A.2) finally takes the form

$$\varphi(\mathbf{x}, t) = \varphi_0(\mathbf{x}) + \int d\mathbf{x}' \int_0^t dt' R(\mathbf{x}, \mathbf{x}'; t - t') \hat{h}(\mathbf{x}'). \quad (\text{A.6})$$

The *dynamic susceptibility* or *integrated response* of the system can be defined as

$$\chi(\mathbf{x}, \mathbf{x}', t) = \left. \frac{\delta \varphi(\mathbf{x}, t)}{\delta \hat{h}(\mathbf{x}')} \right|_{\hat{h}=0} = \int_0^t R(\mathbf{x}, \mathbf{x}'; t') dt' ; \quad (\text{A.7})$$

taking its limit for $t \rightarrow \infty$ yields the *static susceptibility*

$$\chi(\mathbf{x}, \mathbf{x}') = \lim_{t \rightarrow \infty} \chi(\mathbf{x}, \mathbf{x}'; t) = \int_0^\infty R(\mathbf{x}, \mathbf{x}'; t') dt' . \quad (\text{A.8})$$

Relation with the Green's function

Consider a partial differential equation in the space and time variables

$$\mathcal{D}_{\mathbf{x},t} \varphi(\mathbf{x}, t) = f(\mathbf{x}, t) , \quad (\text{A.9})$$

where the field $\varphi(\mathbf{x}, t)$ is defined over a set $\mathbf{x} \in \mathcal{R} \subset \mathbb{R}^d$, $t \in [0, \infty)$ and subject to some boundary and initial conditions, so that the uniqueness of the solution is ensured. The Green's function, or *fundamental solution* associated to (A.9) is defined as

$$\begin{cases} \mathcal{D}_{\mathbf{x},t} G(\mathbf{x}, t; \mathbf{x}', t') = \delta(\mathbf{x} - \mathbf{x}') \delta(t - t') \\ \lim_{t \rightarrow 0^+} G(\mathbf{x}, t; \mathbf{x}', t') = 1 \end{cases} \quad (\text{A.10})$$

and it can be used to build a particular solution of (A.9) for any given $f(\mathbf{x}, t)$ via

$$\varphi(\mathbf{x}, t) = \varphi_{hom}(\mathbf{x}) + \int_{\mathcal{R}} d\mathbf{x}' \int_0^\infty dt' G(\mathbf{x}, t; \mathbf{x}', t') f(\mathbf{x}', t') , \quad (\text{A.11})$$

where $\varphi_{hom}(\mathbf{x})$ is a solution for the homogeneous problem obtained by setting $f(\mathbf{x}, t) = 0$. Notice the analogy with the linear response function defined in equation (A.2): in fact, it is straightforward to prove that

$$\mathcal{D}_{\mathbf{x},t} R(\mathbf{x}, t; \mathbf{x}', t') = \frac{\delta \mathcal{D}_{\mathbf{x},t} \varphi(\mathbf{x}, t)}{\delta f(\mathbf{x}', t')} = \frac{\delta f(\mathbf{x}, t)}{\delta f(\mathbf{x}', t')} = \delta(\mathbf{x} - \mathbf{x}') \delta(t - t') , \quad (\text{A.12})$$

i.e. the response function itself is a solution of the equation (A.10) which defines the Green's function.

A.2 Propagator for an inertial dynamics

As a prototype of our dynamical equation, which is of the second order in time, we consider the differential operator

$$\mathcal{D}_{\mathbf{x},t} = \chi \partial_{tt}^2 + \eta \partial_t - L_{\mathbf{x}} , \quad (\text{A.13})$$

where $L_{\mathbf{x}}$ is a linear and Hermitian operator. In our system $L_{\mathbf{x}}$ acts as a restoring force, *i.e.* it is negative definite: this implies $L_{\mathbf{x}}$ only involves even space derivatives, which preserve the $\mathbf{x} \rightarrow -\mathbf{x}$ symmetry. Let's consider its eigenfunction basis

$$\begin{cases} -L_{\mathbf{x}} u_n(\mathbf{x}) = \lambda_n u_n(\mathbf{x}) \\ \lambda_n \geq 0 \quad \forall n \end{cases} \quad (\text{A.14})$$

and formally expand the Green's function (after having set $t' = 0$ for brevity)

$$\begin{cases} G(\mathbf{x}, t; \mathbf{x}') = \sum_n G_n(t; \mathbf{x}') u_n(\mathbf{x}) \\ G_n(t; \mathbf{x}') = (u_n, G) , \end{cases} \quad (\text{A.15})$$

where we introduced the scalar product

$$(f, g) = \int_R f^*(\mathbf{x}) g(\mathbf{x}) d^d x . \quad (\text{A.16})$$

Projecting equation (A.10) onto $u_n(\mathbf{x})$ gives an equation for the $G_n(t; \mathbf{x}')$ component: using the fact that $L_{\mathbf{x}}$ is Hermitian, we obtain

$$\begin{aligned} \left[\chi \partial_{tt}^2 + \eta \partial_t \right] (u_n(\mathbf{x}), G(\mathbf{x}, t; \mathbf{x}')) - (L_{\mathbf{x}} u_n(\mathbf{x}), G(\mathbf{x}, t; \mathbf{x}')) &= (u_n(\mathbf{x}), \delta(\mathbf{x} - \mathbf{x}')) \delta(t) \\ \left[\chi \partial_{tt}^2 + \eta \partial_t \right] G_n(t; \mathbf{x}') + \lambda_n G_n(t; \mathbf{x}') &= u_n^*(\mathbf{x}') \delta(t) , \end{aligned} \quad (\text{A.17})$$

which is still subject to the initial condition

$$G_n(t \rightarrow 0^+; \mathbf{x}') = 0 . \quad (\text{A.18})$$

The solution can be easily found by going into Fourier space,

$$\begin{aligned} G_n(\omega; \mathbf{x}') &= \int_{-\infty}^{+\infty} dt e^{i\omega t} G_n(t; \mathbf{x}') \\ G_n(t; \mathbf{x}') &= \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} e^{-i\omega t} G_n(\omega; \mathbf{x}') ; \end{aligned} \quad (\text{A.19})$$

we soon obtain

$$G_n(\omega; \mathbf{x}') = \frac{u_n^*(\mathbf{x}')}{-\chi \omega^2 - i\eta \omega + \lambda_n} , \quad (\text{A.20})$$

whose poles are in

$$\begin{aligned} \omega_{\pm} &= -i\gamma \pm \omega_n \\ \gamma &\equiv \frac{\eta}{2\chi} , \quad \omega_n \equiv \sqrt{\frac{\lambda_n}{\chi} - \gamma^2} , \end{aligned} \quad (\text{A.21})$$

to be compared with the spectrum of the propagator in the discrete case, equation (3.29). We notice that both $\omega_{\pm}$ lie in the lower half of the complex ω plane provided that

$$\gamma > \sqrt{\frac{\lambda_n}{\chi}} , \quad (\text{A.22})$$

being $\eta, \chi, \lambda_n \geq 0$.

Let's now perform the integration in $d\omega$ and compute the propagator in real space, $G_n(t; \mathbf{x}')$. Looking at the $e^{-i\omega t}$ factor in the Fourier transform (A.19), we notice that, for $t < 0$, the integration path in the complex plane should be chosen as in Figure A.1 in order for the integrand to go to zero as $r \rightarrow \infty$. This way no poles are enclosed within the path and the result of the integration is zero, which ensures

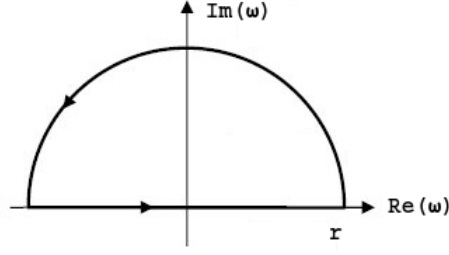


Figure A.1. Integration path in the complex ω plane for the evaluation of the propagator (A.26) in the case $t < 0$.

causality. On the other hand, if $t \geq 0$ the semi-circle should be closed in the lower half of the complex plane, $\text{Im}\{\omega\} < 0$: this way both poles are enclosed within the integration path and we find

$$\begin{aligned} G_n(t; \mathbf{x}') &= \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} e^{-i\omega t} \frac{u_n^*(\mathbf{x}')}{-\chi(\omega - \omega_+)(\omega - \omega_-)} \\ &= \theta(t) \frac{u_n^*(\mathbf{x}')}{\chi} e^{-\gamma t} \sin(\omega_n t). \end{aligned} \quad (\text{A.23})$$

Finally, we can use this result to compute the eigenfunction expansion of the Green's function via equation (A.15), for $t > 0$:

$$G(\mathbf{x}, t; \mathbf{x}', t') = \sum_n u_n(\mathbf{x}) u_n^*(\mathbf{x}') e^{-\gamma(t-t')} \frac{\sin \omega_n(t-t')}{\chi \omega_n}. \quad (\text{A.24})$$

If the region R where $L_{\mathbf{x}}$ is defined coincides with the whole $\mathbb{R}^d$, then an eigenfunction basis is simply given by the plane waves

$$u_{\mathbf{k}}(\mathbf{x}) = e^{i\mathbf{k} \cdot \mathbf{x}}, \quad (\text{A.25})$$

where the wavenumber $\mathbf{k}$ plays the role of the eigenfunction index n ; the propagators in real and in Fourier space thus take the form

$$\begin{aligned} G(\mathbf{x}, t; \mathbf{x}', t') &= \frac{1}{(2\pi)^d} \int_{-\infty}^{+\infty} d^d k e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')} e^{-\gamma(t-t')} \frac{\sin \omega_{\mathbf{k}}(t-t')}{\chi \omega_{\mathbf{k}}} \\ G(\mathbf{k}, \omega) &= \frac{1}{-\chi \omega^2 - i\eta \omega + \lambda_{\mathbf{k}}}, \end{aligned} \quad (\text{A.26})$$

where the dispersion relation, which can be found by setting $G^{-1}(\mathbf{k}, \omega) \equiv 0$, reads

$$\begin{aligned} \omega_{\pm}(\mathbf{k}) &= -i\gamma \pm \omega_{\mathbf{k}} \\ \omega_{\mathbf{k}} &= \sqrt{\frac{\lambda_{\mathbf{k}}}{\chi} - \gamma^2}. \end{aligned} \quad (\text{A.27})$$

Due to the absence of borders, the propagator can only depend upon the difference $(\mathbf{x} - \mathbf{x}')$ and not on their specific choice, as we expected for translational symmetry reasons.

Going back to equation (A.11), we see now that the convolution which gives the response to an external perturbation $f(\mathbf{x}, t)$ is more easily performed in Fourier space, where it simply becomes a product:

$$\varphi(\mathbf{x}, t) = \frac{1}{(2\pi)^{d+1}} \int d^d k d\omega e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} G(\mathbf{k}, \omega) \hat{f}(\mathbf{k}, \omega) . \quad (\text{A.28})$$

If $\hat{f}(\mathbf{k}, \omega)$ has no poles in the half plane $\text{Im}\{\omega\} < 0$ (which happens whenever $f(\mathbf{x}, t)$ is not exponentially divergent in t), then the integration in $d\omega$ can be performed in the same way as we did in equation (A.23) and we obtain

$$\varphi(\mathbf{x}, t) = \frac{1}{(2\pi)^d} \int d^d k e^{i\mathbf{k} \cdot \mathbf{x}} \frac{e^{-\gamma t}}{i\chi\omega_{\mathbf{k}}} \left[e^{-i\omega_{\mathbf{k}} t} \hat{f}(\mathbf{k}, \omega_+(\mathbf{k})) - e^{i\omega_{\mathbf{k}} t} \hat{f}(\mathbf{k}, \omega_-(\mathbf{k})) \right] . \quad (\text{A.29})$$

B Code employed in numerical simulations

Here we present an extract from the original code written to perform the numerical studies exposed in this work. The simpler *on lattice* version of this program and the data analysis routines were not included.

```

2 // Numerical simulation of ISM off-lattice, with external perturbations.
3 // Roma, 14.03.2019. Davide Venturelli
4 // Last update 05.08.19
5 /* COMMENTS
6 - Variables: r(t) position, o(t) polar angle, u(t) angular velocity.
7 - Variables that have to be updated by external functions at every time step are local,
8   but their pointers are global.
9 - Eulero-Cromer method on position update, can easily be improved to Midpoint.
10 */
11
12 // SIMULATION PARAMETERS
13
14 #define L 15 // Lattice side
15 #define N (L*L*L) // Total number of items
16 #define N_ext 1 // Number of sites subject to the external field
17 #define t0 150 // Initial time of the perturbation
18 #define tau 50 // Duration of the perturbation
19 #define A0 30 // Amplitude of the perturbation
20 #define v0 2 // Velocity of the active particles
21
22 // LIBRARIES, TYPES
23
24 #include <stdio.h>
25 #include <stdlib.h>
26 #include <math.h>
27 #include <string.h>
28
29 typedef struct{
30     double tmax; // Total duration
31     double dt; // Integration time step
32     double K; // Chi, inertial coefficient
33     double E; // Eta, viscosity coefficient
34     double J; // Alignment strenght coefficient
35     double T; // Temperature (noise) coefficient
36 } data;
37
38 typedef struct{
39     double X; // Paradigm for a 3D vector
40     double Y;
41     double Z;
42     int tile; // Location on the mosaic
43 } vector;
44
45 typedef struct{
46     int * list; // Paradigm for a bucket list
47     int numData; // Array list
48 } cell; // Number of items it contains
49
50
51 // FUNCTION PROTOTYPES
52
53 data cmd_input(int argc, char **argv); // User input
54 void cmd_error(); // Error message, instructions and interruption
55 void init(); // Initializes all lattice variables (random)
56 void initUP(); // Initializes all lattice variables (aligned)
57 void startOnLattice(); // Places particles initially on lattice vertices
58 void evolve(double *p, cell * mosaic); // Temporal evolution
59 void boxmuller(double * address); // Generates two gaussian random numbers
60 void neighborhood(int * vic); // Fills the nearest neighbors list
61 void hybrisSwap(double **a, double **b); // Sends x_(n+1) into x_n
62 void updateMosaic(cell * mosaic); // Updates the mosaic
63 vector polarization(); // Evaluates the polarization
64 double energy(cell * mosaic, data par); // Evaluates the energy
65 double modular(double a, double b, double m); // Distance in modular arithmetic
66 void push(cell * bucket, int item); // Adds an element to the list
67 void pop(cell * bucket, int index); // Removes an element from the list
68 void flush(cell * bucket); // Empties the list
69 int x(int i); // Converters: vector components <-> list indices
70 int y(int i);
71 int z(int i);
72 int ind(int ix, int iy, int iz);
73 // Temporal evolution in the presence of external fields
74 void evolve_field(double *p, cell *mos, double t, double shift);
75 // Applies the same perturbation back and forth
76 void shake(double * factor, cell * mosaic, data par);
77 // Repeated turn of a specified angle
78 void turn_angle(double *fac, cell *mos, data par, double angle);
79 // Integrates and prints the trajectory
80 void trajectory(double * factor, cell * mosaic, data par);

```

```

82 // Integrates and prints observables evaluated in real time
void realtime(double * factor, cell * mosaic, data par);

84

86 // GLOBAL VARIABLES

88     int site[] = {555}; // Look-up table for sites subject to perturbations
    double *op, *Op, *up, *Up; // Pointers to trajectory arrays
    vector *rp; // Pointer to position array
    double gauss[2*N]; // Stores gaussian variables
    double cost=1/(2*sqrt(3)); // Coefficient of noise in the integrator
    double f_n[N]; // Stores f(x_n) for evolve()
    double A_n[N]; // Stores A_n for evolve()
    double f_n1; // Stores f(x_{n+1}) for evolve()
    int near[6*N]; // List of adjacent cells

98

100 // EXTERNAL PERTURBATIONS

102 // Definition of external perturbations. They have to be as many as N_ext.
// Fields coupled to the spin
double h_1(double t){return 0;}
double (*h_ext[])(double t) = {h_1}; // Function pointer (h_1...h_N_ext)
// Fields coupled to the velocity
double H_1(double t){return A0/2*(tanh((t-t0)/tau)+1);}
double (*H_ext[])(double t) = {H_1}; // Function pointer (H_1...H_N_ext)

108

110 // MAIN PROGRAM

112 int main(int argc, char ** argv){
    // User input acquisition and one-time operations on input
    data par;
    par = cmd_input(argc, argv); // Reads input parameters

    double factor[10]; // Integration factors

    factor[9] = v0; // v0
    factor[8] = 1/par.K; // 1/K
    factor[7] = par.J/par.K; // J/K
    factor[6] = par.E/par.K; // gamma
    factor[5] = sqrt(2*par.T*par.E)/(par.K)*sqrt(par.dt); // sigma*sqrt(dt)
    factor[4] = par.dt*par.E/par.K; // dt*gamma
    factor[3] = 0.5*par.dt; // 0.5*dt
    factor[2] = factor[5]*par.dt; // sigma*dt^(3/2)
    factor[1] = 0.5*par.dt*par.dt; // 0.5*dt^2
    factor[0] = par.dt; // dt

    // Seed of random function
    int seed, i;
    FILE * devran;
    devran = fopen("/dev/random", "r");
    fread(&seed, 4, 1, devran);
    fclose(devran);
    srand48(seed);

    // Creation of trajectory arrays
    double o[N]; // Polar angle
    double O[N]; // (t-1, t)
    double u[N] = {0}; // Angular velocity
    double U[N]; // (t-1, t)
    vector r[N]; // Position vector
    op = o; // Matches arrays to their global pointers
    Op = O;
    up = u;
    Up = U;
    rp = r;

    // Initializes lattice
    startOnLattice(); // Places particles on lattice sites
    neighborhood(near); // Fills in the list of nearest neighbors
    //init(); // Initializes velocities with random values
    initUP(); // Alternative, initial velocities aligned

    // Creates mosaic.
    // Each cell contains an array of integers (list) and a counter (numData).
    cell mosaic[N];
    for(i=0; i<N; i++){
        mosaic[i].list = (int *) malloc(L*L * sizeof(int));
    }
    updateMosaic(mosaic); // Initializes mosaic

    // Integration
    for(i=0; i<(int)(100/par.dt); i++){ // Warmup cycle
        evolve(factor, mosaic);
    }
    //turn_angle(factor, mosaic, par, M_PI/2); // Repeated turns of the specified angle
    //trajectory(factor, mosaic, par); // Prints trajectory
    realtime(factor, mosaic, par); // Prints polarization and energy
}

```

```

172 // Free memory and exit
173 for(i=0; i<N; i++){
174     free(mosaic[i].list);
175 }
176 exit(EXIT_SUCCESS);
177 }
178
179 // DEFINITION OF FUNCTIONS
180
181 // Update of the mosaic
182 void updateMosaic(cell * mosaic){
183     int i,j;
184     for(j=0; j<N; j++){
185         flush(mosaic+j); // Cycles over mosaic tiles
186         // Empties the bucket
187     }
188     for(i=0; i<N; i++){
189         j = ind( (int)((rp+i)->X), (int)((rp+i)->Y), (int)((rp+i)->Z) ); // Cycles over particles and locates them
190         push(mosaic+j,i); // Adds particle index to j-th bucket
191         (rp+i)->tile = j; // Saves tile index
192     }
193 }
194
195 // Time evolution using Vanden-Eijnden and Ciccotti's algorithm.
196 // Its arguments are the list of integration factors and the mosaic.
197 void evolve(double * p, cell * mosaic){
198     int i,j,k;
199     double tmp;
200
201     // Updates angles and positions
202     for(i=0; i<N; i++){
203         // Extracts two gaussian numbers
204         boxmuller(gauss+2*i);
205         // Evaluates f(x_n)
206         f_n[i]=0;
207         for(j=0; j<6; j++){ // Cycles over adjacent tiles
208             k=0;
209             while(k< mosaic[near[6*((rp+i)->tile)+j]].numData){
210                 f_n[i] += sin( *(op+mosaic[near[6*((rp+i)->tile)+j]].list[k]) - *(op+i) );
211                 k++;
212             }
213         }
214         k=0;
215         // Cycles over same tile (self-interaction is null!)
216         while(k < mosaic[(rp+i)->tile].numData){
217             f_n[i] += sin( *(op+mosaic[(rp+i)->tile].list[k]) - *(op+i) );
218             k++;
219         }
220         f_n[i] *= p[7];
221
222         // Evaluates A_n
223         A_n[i] = p[1]*(f_n[i]-p[6]*(*(up+i))) + p[2]*(0.5*gauss[2*i]+cost*gauss[2*i+1]);
224         // Evaluates x_(n+1)
225         *(Op+i) = *(op+i) + p[0]*(*(up+i)) + A_n[i];
226         /**(Op+i) = *(op+i); //stops the velocity update
227
228         // Updates positions (Euler-Cromer)
229         // Castings ensure the boundary conditions.
230         // 100 is arbitrary (must be an integer bigger than the maximum increment).
231         tmp = (rp+i)->X + cos(*(Op+i))*p[0]*p[9];
232         (rp+i)->X = ((int)(tmp+L))%L + tmp - (int)(tmp+100) +100;
233         tmp = (rp+i)->Y + sin(*(Op+i))*p[0]*p[9];
234         (rp+i)->Y = ((int)(tmp+L))%L + tmp - (int)(tmp+100) +100;
235         (rp+i)->Z += NOISE;
236     }
237 }
238
239 // Updates the mosaic
240 updateMosaic(mosaic);
241
242 // Updates angular velocities
243 for(i=0; i<N; i++){
244     // Evaluates f(x_(n+1))
245     f_n1 = 0;
246     for(j=0; j<6; j++){
247         k=0;
248         while(k< mosaic[near[6*((rp+i)->tile)+j]].numData){
249             f_n1 += sin( *(Op + mosaic[near[6*((rp+i)->tile)+j]].list[k]) - *(Op+i) );
250             k++;
251         }
252     }
253     k=0;
254     while(k < mosaic[(rp+i)->tile].numData){
255         f_n1 += sin( *(Op+mosaic[(rp+i)->tile].list[k]) - *(Op+i) );
256         k++;
257     }
258     f_n1 *= p[7];
259
260     // Evaluates u_(n+1)
261     *(Up+i) = (1-p[4])*(*(up+i)) + p[3]*(f_n1 + f_n[i]) + p[5]*gauss[2*i] - p[6]*A_n[i];

```

```

262     }
263
264     // Sends x_n(n+1) into x_n
265     hybrisSwap(&op, &Op);
266     hybrisSwap(&up, &Up);
267 }
268
269 // Time evolution in the presence of external perturbing fields.
270 // "shift" is the phaseshift added to the angle.
271 void evolve_field(double * p, cell * mosaic, double t, double shift){
272     int i,j,k;
273     double tmp, x_n, x_n1;
274
275     // Updates angles
276     for(i=0; i<N; i++){
277         // Extracts two gaussian numbers
278         boxmuller(gauss+2*i);
279         // Evaluates f(x_n)
280         f_n[i]=0;
281         for(j=0; j<6; j++){
282             k=0;
283             while(k< mosaic[near[6*((rp+i)->tile)+j]].numData){
284                 f_n[i] += sin( *(op+mosaic[near[6*((rp+i)->tile)+j]].list[k]) - *(op+i) );
285                 k++;
286             }
287             k=0;
288             // Cycles over same tile (self-interaction is null!)
289             while(k < mosaic[(rp+i)->tile].numData){
290                 f_n[i] += sin( *(op+mosaic[(rp+i)->tile].list[k]) - *(op+i) );
291                 k++;
292             }
293             f_n[i] *= p[7];
294
295             // Evaluates A_n
296             A_n[i] = p[1]*(f_n[i]-p[6]*(*(up+i))) + p[2]*(0.5*gauss[2*i]+cost*gauss[2*i+1]);
297             // Evaluates x_n(n+1)
298             *(Op+i) = *(op+i) + p[0]*(*(up+i)) + A_n[i];
299             //*(Op+i) = *(op+i); // stops the velocity update
300         }
301     }
302
303     // Adds perturbations to the angles
304     for(i=0; i<N_ext; i++){
305         x_n = *(op+site[i]) - shift;
306         *(Op+site[i]) += p[3]*( (p[4]-1)*(h_ext[i])(t) - p[8]*p[0]*(H_ext[i])(t)*sin(x_n)
307             - (h_ext[i])(t+p[0]) );
308     }
309
310     // Updates positions (Euler-Cromer)
311     for(i=0; i<N; i++){
312         // Castings ensure the boundary conditions.
313         // 100 is arbitrary (must be an integer bigger than the maximum increment).
314         tmp = (rp+i)->X + cos(*(Op+i))*p[0]*p[9];
315         (rp+i)->X = ((int)(tmp+L))%L + tmp - (int)(tmp+100) + 100;
316         tmp = (rp+i)->Y + sin(*(Op+i))*p[0]*p[9];
317         (rp+i)->Y = ((int)(tmp+L))%L + tmp - (int)(tmp+100) + 100;
318         (rp+i)->Z += NOISE;
319     }
320
321     // Updates the mosaic
322     updateMosaic(mosaic);
323
324     // Updates angular velocities
325     for(i=0; i<N; i++){
326         // Evaluates f(x_n(n+1))
327         f_n1 = 0;
328         for(j=0; j<6; j++){
329             k=0;
330             while(k< mosaic[near[6*((rp+i)->tile)+j]].numData){
331                 f_n1 += sin( *(Op + mosaic[near[6*((rp+i)->tile)+j]].list[k]) - *(Op+i) );
332                 k++;
333             }
334             k=0;
335             while(k < mosaic[(rp+i)->tile].numData){
336                 f_n1 += sin( *(Op+mosaic[(rp+i)->tile].list[k]) - *(Op+i) );
337                 k++;
338             }
339             f_n1 *= p[7];
340
341             // Evaluates u_n(n+1)
342             *(Up+i) = (1-p[4])**(up+i) + p[3]*(f_n1 + f_n[i]) + p[5]*gauss[2*i] - p[6]*A_n[i];
343         }
344     }
345
346     // Adds perturbations to the angular velocities
347     for(i=0; i<N_ext; i++){
348         x_n = *(op+site[i]) - shift;
349         x_n1 = *(Op+site[i]) - shift;
350         *(Up+site[i]) += p[3]*( (1-p[4])*((1-p[4])*((h_ext[i])(t) - p[8]*(H_ext[i])(t)*sin(x_n)
351             + p[6]*(h_ext[i])(t+p[0]) - p[8]*(H_ext[i])(t+p[0])*sin(x_n1) );

```



```

352 }
353
354 // Sends x_(n+1) into x_n
355 hybrisSwap(&op, &Op);
356 hybrisSwap(&up, &Up);
357 }
358
359 // Integrates and prints the trajectory in phasespace for every time step.
360 // The perturbation is applied repeatedly, each time reversed.
361 void shake(double * factor, cell * mosaic, data par){
362     int i, j, k, imax, kmax, side;
363     double window;
364
365     window = t0+10*tau; // Duration of each turn
366     kmax = (int)(par.tmax/window); // Total number of turns
367     imax = (int)(window/par.dt); // Number of steps in every window
368
369     for(k=0; k<kmax; k++){
370         side = (-2)*(k%2)+1; // Direction of the field (0/PI)
371         for(i=0; i<imax; i++){ // Time evolution
372             if(i%10==0){ // Prints time, angle
373                 printf("%.8lf", (k*imax+i)*par.dt);
374                 for(j=0; j<N; j++) printf("\t%.8lf", *(op+j));
375                 printf("\t%.8lf", (*h_ext[0])(i*par.dt)/A0*side); // Prints signal amplitude h(t)
376                 printf("\t%.8lf", (*H_ext[0])(i*par.dt)/A0*side); // Prints signal amplitude H(t)
377                 printf("\n");
378             }
379             // Updates all lattice sites
380             evolve_field(factor, mosaic, i*par.dt, (k%2)*M_PI);
381         }
382     }
383 }
384
385 // Integrates and prints the trajectory in phasespace for every time step.
386 // A turn of "angle" wrt to the current polarization is applied repeatedly.
387 void turn_angle(double * factor, cell * mosaic, data par, double angle){
388     int i, j, k, imax, ifree, kmax, count;
389     double window, free, shift, phi, phi_loc, cosines;
390     vector pol;
391
392     // Prints polarization data
393     FILE * out;
394     out = fopen("angles.txt", "a");
395     fprintf(out, "\nA0=%f\tN=%d\n", A0, N);
396     fprintf(out, "Signal\tFinal\tDifference\t1-<cos(phi-shift)>\n");
397
398     // Cycle preparation
399     window = 10000; // Duration of each turn
400     free = 500; // Relaxation time after the turn
401     kmax = (int)(par.tmax/window); // Total number of turns
402     imax = (int)((window-free)/par.dt); // Number of steps in every window
403     ifree = (int)(free/par.dt); // Number of relaxation steps
404     pol = polarization(); // Initial polarization (approx.)
405     phi = atan2(pol.Y, pol.X);
406
407     // Perturbation cycle
408     for(k=0; k<kmax; k++){
409         shift = phi + angle; // Direction of the perturbation
410         shift = atan2(sin(shift), cos(shift)); // Puts it in the right quadrant
411         fprintf(out, "%.8lf\t", shift);
412
413         for(i=0; i<imax; i++){ // Time evolution
414             if(i%1000==0){ // Prints time, angles
415                 printf("%.8lf", (k*(imax+ifree)+i)*par.dt);
416                 for(j=0; j<N; j++) printf("\t%.8lf", *(op+j));
417                 // Prints signal amplitude h(t)
418                 printf("\t%.8lf", (*h_ext[0])(i*par.dt)/A0*angle+shift-angle);
419                 // Prints signal amplitude H(t)
420                 printf("\t%.8lf", (*H_ext[0])(i*par.dt)/A0*angle+shift-angle);
421                 printf("\n");
422             }
423             // Updates all lattice sites
424             evolve_field(factor, mosaic, i*par.dt, shift);
425         }
426
427         // Relaxation cycle
428         count = 0;
429         phi = 0;
430         cosines = 0;
431
432         for(i=0; i<ifree; i++){ // Polarization after thermalization
433             if(i%100==0){
434                 pol = polarization();
435                 phi_loc = atan2(pol.Y, pol.X); // Current angle
436                 phi += phi_loc; // Average angle
437                 cosines += cos(phi_loc-shift); // Average h dot phi
438                 count += 1;
439             }
440             if(i%1000==0){ // Prints time, angles

```

```

442         printf("%.8lf", (k*(imax+ifree)+imax+i)*par.dt);
443         for(j=0; j<N; j++) printf("\t%.8lf", *(op+j));
444         // Prints signal amplitude h(t) (zero)
445         printf("\tNaN");
446         // Prints signal amplitude H(t) (zero)
447         printf("\tNaN");
448         printf("\n");
449     }
450 }
451
452 // Updates all lattice sites
453 evolve(factor, mosaic);
454 }
455
456 phi /= count;
457 cosines /= count;
458 fprintf(out, "%.8lf\t%.8lf\t%.8lf\n", phi, modular(shift,phi,2*M_PI), 1-cosines);
459 }
460 fclose(out);
461 }
462
463 // Integrates and prints the trajectory in phasespace for every time step
464 void trajectory(double * factor, cell * mosaic, data par){
465     int i, j, imax;
466     imax = (int)(par.tmax/par.dt);
467
468     for(i=0; i<imax; i++){
469         // Time evolution
470         if(i%100==0){
471             // Prints time, angle
472             printf("%.8lf", i*par.dt);
473             for(j=0; j<N; j++) printf("\t%.8lf", *(op+j));
474             for(j=0; j<N; j++) printf("\t%.8lf", (rp+j)->X);
475             for(j=0; j<N; j++) printf("\t%.8lf", (rp+j)->Y);
476             for(j=0; j<N; j++) printf("\t%.8lf", (rp+j)->Z);
477             printf("\n");
478         }
479         // Updates all lattice sites
480         evolve(factor, mosaic);
481     }
482 }
483
484 // Integrates and prints the polarization in real time
485 void realtime(double * factor, cell * mosaic, data par){
486     printf("# Time\t\tphiX\t\tphiY\t\tPolarization\n");
487     // Prints headline
488     int i, imax;
489     vector vec;
490     imax = (int)(par.tmax/par.dt);
491     for(i=0; i<imax; i++){
492         // Time evolution
493         if(i%100 == 0){
494             // Prints time and observables
495             vec = polarization();
496             printf("%.8lf\t%.8lf\t%.8lf\t%.8lf\n", i*par.dt, vec.X, vec.Y, vec.Z);
497         }
498         // Updates all lattice sites
499         evolve(factor, mosaic);
500     }
501 }
502
503 // Evaluates the vector polarization (i.e. the order parameter) and its modulus
504 // (i.e. the scalar polarization, stored in the z-component).
505 // phi = v_0/N * sqrt( (sum_i cos(theta_i))^2 + (sum_i sin(theta_i))^2 )
506 vector polarization(){
507     int i;
508     double c,s;
509     // Sum of cosines and sines
510     vector pol;
511     c=0;
512     s=0;
513
514     for(i=0; i<N; i++){
515         c += cos(*(op+i));
516         s += sin(*(op+i));
517     }
518
519     pol.X = c/N;
520     pol.Y = s/N;
521     pol.Z = sqrt(c*c+s*s)/N;
522     return pol;
523 }
524
525 // Evaluates the energy
526 // E = -J/2 sum_{i,j} n_{ij} v_i dot v_j + k*u^2/2
527 double energy(cell * mosaic, data par){
528     double en=0;
529     double cin=0;
530     int i,j,k;
531
532     for(i=0; i<N; i++){
533         for(j=0; j<6; j++){
534             // Adjacent cells
535             k=0;
536             while(k< mosaic[near[6*((rp+i)->tile)+j]].numData){
537                 en += cos( *(op+i) - *(op + mosaic[near[6*((rp+i)->tile)+j]].list[k]) );
538             }
539         }
540     }
541     return en;
542 }

```

```

532         k++;
533     }
534     k=0;
535     while(k < mosaic[(rp+i)->tile].numData){           // Same cell
536         en += cos( *(op+i) - *(op + mosaic[(rp+i)->tile].list[k]) );
537         k++;
538     }
539     en = en-1;                                           // Subtracts self-energy
540     cin += (*(up+i))*(*(up+i));
541 }
542
543 //return - (par.J)/2*en + 1/(par.K*2)*cin;
544 return - (par.J)/2*en;                                 // No kinetic energy
545 }
546
547 // Translates from a list index (i) to 3d lattice indices (ix,iy,iz), and viceversa
548 inline int x(int i) {return i/L;}
549 inline int y(int i) {return ((int)(i/L))%L;}
550 inline int z(int i) {return (int)(i/(L*L));}
551 inline int ind(int ix, int iy, int iz) {return ix + L*iy + L*L*iz;}
552
553 // Generates two normally distributed random numbers via Box-Muller transform,
554 // then stores them in the position addressed by "address" and the following.
555 void boxmuller(double * address){
556     double u1, u2;
557
558     u1 = drand48();
559     u2 = drand48();
560     *(address) = sqrt(-2*log(u1))*cos(2*M_PI*u2);
561     *(address+1) = sqrt(-2*log(u1))*sin(2*M_PI*u2);
562 }
563
564 // Sends x_(n+1) into x_n.
565 // Recall (double ** a) is a pointer to an array of doubles.
566 void hybrisSwap(double **a, double **b){
567     // The content of 'a' (i.e. the array, i.e. the address of its first element)
568     // is assigned to the pointer tmp.
569     double *tmp = *a;
570     // The array to which 'b' points is assigned to the one to which 'a' points
571     *a = *b;
572     *b = tmp;
573 }
574
575 // Acquires input data from user.
576 data cmd_input(int argc, char** argv){
577     if (argc != 7) cmd_error();
578
579     data val;
580     val.tmax = atof(argv[1]);
581     val.dt = atof(argv[2]);
582     val.K = atof(argv[3]);
583     val.E = atof(argv[4]);
584     val.J = atof(argv[5]);
585     val.T = atof(argv[6]);
586
587     return val;
588 }
589
590 // Initializes lattice variables (random)
591 void init(){
592     int i;
593     for(i=0; i<N; i++){
594         *(op + i) = 2*M_PI*drand48();
595     }
596 }
597
598 // Initializes lattice variables (everyone UP)
599 void initUP(){
600     int i;
601     for(i=0; i<N; i++){
602         *(op + i) = 0;
603     }
604 }
605
606 // Places particles initially on lattice vertices
607 void startOnLattice(){
608     int i;
609     for(i=0; i<N; i++){
610         (rp+i)->X = x(i);
611         (rp+i)->Y = y(i);
612         (rp+i)->Z = z(i);
613     }
614 }
615
616 // Returns the distance mod(m)
617 double modular(double a, double b, double m){
618     return fabs(m/2 - fmod(3*m/2 + a - b, m));
619 }
620 }

```

```

622 // Data storage functions for a bucket list
623 void push(cell * bucket, int item){
624     bucket->list[(bucket->numData)++] = item;
625 }
626 void pop(cell * bucket, int index){
627     bucket->list[index] = bucket->list[--(bucket->numData)];
628 }
629 void flush(cell * bucket){
630     bucket->numData = 0;
631 }
632
633 // Fills the nearest neighbours list (3d with periodic boundary conditions).
634 // Positions [6*i,6*(i+1)] are the list indices of its 6 nearest neighbours on the 3d lattice.
635 void neighborhood(int * vic){
636     int k, jx, jy, jz;
637
638     for(k=0; k<N; k++){
639
640         jx = (x(k) + 1)%L; // Finds x of x-right neighbour
641         vic[6*k] = ind(jx, y(k), z(k)); // Stores its index
642
643         jx = (x(k) + L - 1)%L; // Finds x of x-left
644         vic[6*k+1] = ind(jx, y(k), z(k));
645
646         jy = (y(k) + 1)%L; // y-up
647         vic[6*k+2] = ind(x(k), jy, z(k));
648
649         jy = (y(k) + L - 1)%L; // y-down
650         vic[6*k+3] = ind(x(k), jy, z(k));
651
652         jz = (z(k) + 1)%L; // z-up
653         vic[6*k+4] = ind(x(k), y(k), jz);
654
655         jz = (z(k) + L - 1)%L; // z-down
656         vic[6*k+5] = ind(x(k), y(k), jz);
657     }
658 }
659
660 // Prints an error message and the instructions, then kills the program.
661 void cmd_error(){
662
663     fprintf(stderr, "\n\tError: wrong input format.\n\n\t
664     NUMERICAL SIMULATION OF ISM (Inertial Spin Model)\n\n\t
665     Insert the following values, separated by a single space: <tmax><dt><chi><eta><J><T>\n\n\t
666     i.e. total duration, time step and the parameters of the model. \n\n\t
667     \n\n");
668     exit(EXIT_FAILURE);
669 }
670

```


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